



OPERATOR'S AND OWNER'S MANUAL

WARNING

- *Make sure these important instructions are available for the operator.
Keep this manual in the pocket on the seat back.*
- *Do not start the truck unless you understand these instructions.*

8FGU15·18·20·25·30·32

8FDU15·18·20·25·30·32

8FGCU15·18·20·25·30·32

8FGCSU20

CONTENTS

I. OWNER'S SECTION	1
● OSHA/ANSI Laws and Standards	
● Employer Responsibility: Environment, Training, Work Rules, Workplace Design	
● Selection of Lift Trucks and Accessories	
● Lift Truck Maintenance and Modifications	
● Information Resources	
II. SAFETY INFORMATION.....	7
● Training	
● OSHA/ANSI Safety Rules	
● Operational Hazards and Avoidance	
III. KNOW YOUR TRUCK.....	17
● General Description, ANSI Designation	
● Intended Uses, Typical Applications	
● Prohibited Uses	
● Expected Work Environment	
● Expected Skill Levels of Operators/Mechanics	
● Maintenance Requirements	
● Warning and Informational Decals	
● Controls, Switches, Instruments, Restraints	
● Data Plate/Capacities	
IV. DAILY OPERATION	67
● Daily Inspection (Pre-Operation Checks)	
● Starting Engine (Gas, LP, Diesel)	
● Brakes	
● Lifting and Lowering Controls	
● Direction Controls	
● Lifting Loads	
● Traveling	
● Stacking Loads	
● Attachments	
● Parking	
● Unexpected Emergencies	

V. OPERATOR MAINTENANCE.....	81
● Fueling (Gas, LP, Diesel)	
● Tires/ Wheels	
● Antifreeze	
● Fuses	
● Battery	
● Parking Brake Adjustment	
● Tag Out	
VI. EMERGENCY PROCEDURES	89
● Dead Battery, Jump Starting	
● Lifting	
● Transporting	
● Towing	
● Jacking	
VII. PLANNED MAINTENANCE.....	95
● Training	
● Manuals, Parts	
● Planned Maintenance, Dealer Service	
● Storage	
● Weekly Maintenance	
● Maintenance Schedules	
● Replacement Schedules	
● Service Data	
● Lubrication Chart	

I**II****III****IV****V****VI****VII**

I. OWNER'S SECTION

CONTENTS

- OSHA/ANSI Laws and Standards
- Employer Responsibility: Environment, Training, Work Rules, Workplace Design
- Selection of Lift Trucks and Accessories
- Lift Truck Maintenance and Modifications
- Information Resources

AN IMPORTANT MESSAGE FOR THE OWNER/ EMPLOYER	1
SELECTION OF LIFT TRUCKS AND ACCESSORIES	1
CHOOSING OPTION AND ATTACHMENTS	2
LIFT TRUCK OPERATIONS	3
WORKPLACE ENVIRONMENT	3
LIFT TRUCK MAINTENANCE.....	4
LIFT TRUCK MODIFICATIONS	5
INFORMATION RESOURCES.....	5

AN IMPORTANT MESSAGE FOR THE OWNER/EMPLOYER

The most expensive part of a material handling operation is not the equipment, but the people who operate it. Toyota has designed this lift truck to work safely and efficiently with the operator, but you have the primary responsibility for the well-being of your employees.

Training of operators and the responsibility for providing a safe workplace are required by the Occupational Safety and Health Act (OSHA). Both OSHA and ANSI (American National Standards Institute) provide operational and maintenance rules. The last page of this section provides sources for these and other publications that can help you.

This section of the manual will help you identify ways you can carry out your responsibilities, and the sections relating to safety information point out hazards that you can eliminate from the workplace as well as ideas that can be incorporated into your training programs and work rules.

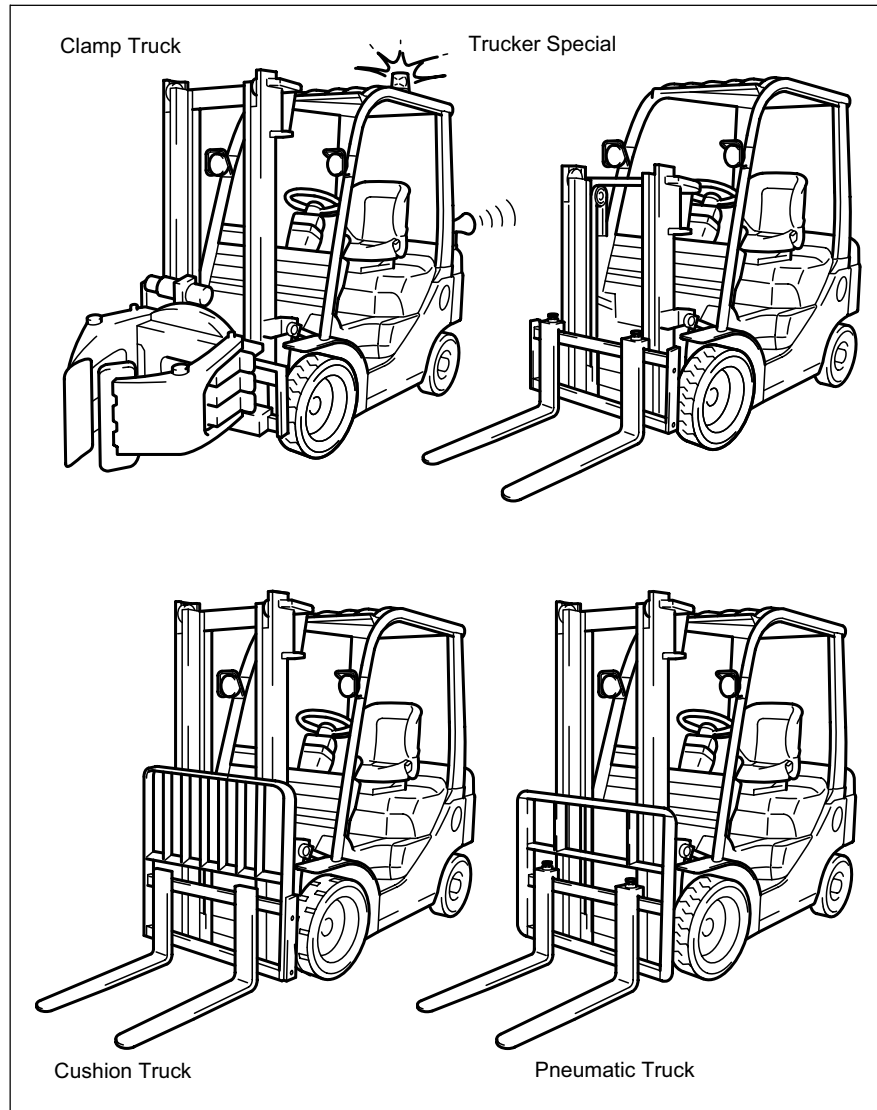
SELECTION OF LIFT TRUCKS AND ACCESSORIES

The Occupational Safety and Health Act (OSHA) and the American National Standards Institute (ANSI) have established rigorous requirements for the design and operation of industrial trucks. Every Toyota industrial truck has been designed to meet and exceed those requirements. It is important, however, that your Toyota truck be selected to meet the needs and demands of your particular work environment. Consult with your Toyota dealer regarding proper lift truck selection.

Once the proper Toyota truck for your work environment has been selected, both OSHA and ANSI require that you evaluate and determine whether your lift truck should be equipped with additional accessories. For information regarding additional accessories, please consult the Accessories Brochure provided with every new Toyota truck or the Toyota Industrial Equipment website at www.toyotaforklift.com.

Toyota strongly suggests that you consult with a safety professional familiar with your type of industrial setting when assessing your specific work place safety requirements.

I. OWNER'S SECTION



CHOOSING OPTION AND ATTACHMENTS

Now that you have selected the right truck for your business, spend time with your Toyota dealer to discuss how options and attachments can help make your employees' jobs safer and more efficient. Sideshifters can speed pallet movement for little additional cost; clamps and rotators are more sophisticated, but if you handle paper rolls or transfer liquids from barrels to larger containers, the payback in increased efficiency and reduced product damage will continue as long as you own your truck.

Many other options are available which cannot all be listed here. Your Toyota dealer is an expert in solving material handling problems so take them to him or invite him in to help you build the truck that is right for you.

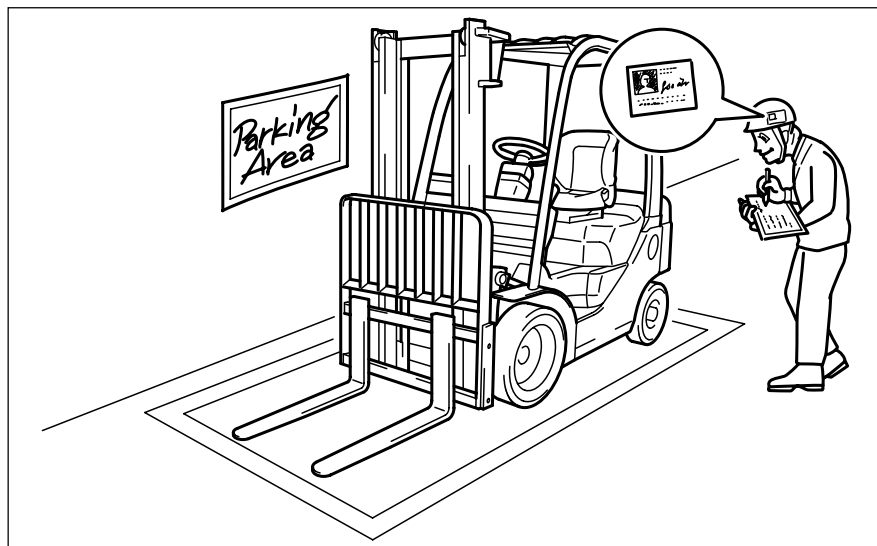
Clamp Truck with Audible and Visual Alarms: This highly specialized truck works in close spaces with a high noise level and many pedestrians. The owner chooses audible and visual alarms tailored to his workplace.

Trucker Special: This lift truck has special equipment that lets it work easily in loading and unloading trailers.

Cushion Truck: The solid tires on this truck provide resistance to damage or punctures, but their small diameter and the under-clearance of the truck make it suitable only for use indoors on clear, smooth, dry floors.

Pneumatic Truck: Larger air filled tires allow this truck to operate on less smooth surfaces or outside, but care must still be taken to provide a workplace free of obstacles. Outside use must be restricted to improved roadways, no dirt or gravel.

LIFT TRUCK OPERATIONS

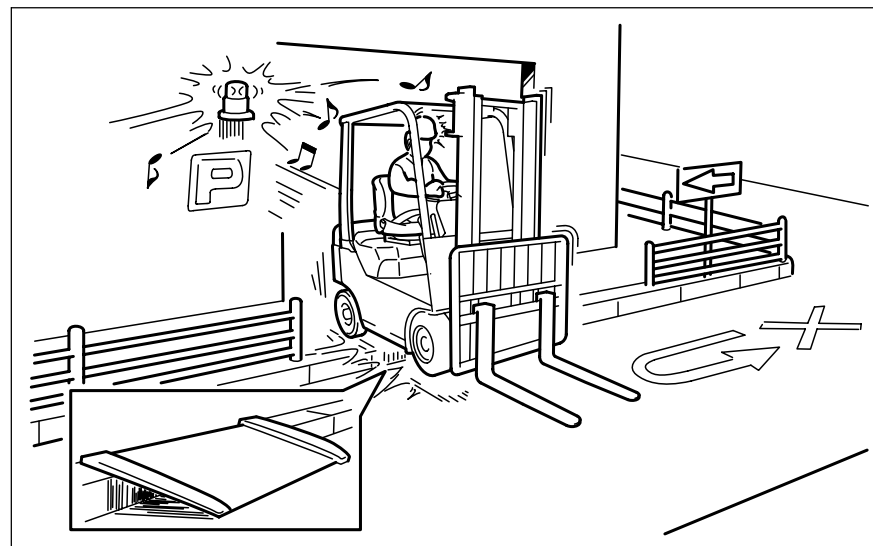


ANSI Standard B56.1 requires operator qualification prior to selection based on visual, auditory, physical and mental ability relative to the specific requirements of the employer. Similarly, the training required by OSHA and ANSI must be tailored to the workplace. Unique operations can present hazards even to an experienced operator if the work is different.

Training of new operators, refresher training for experienced operators, plus appropriate supervision, will provide the basics for safe lift truck operation. One of the most important aspects of training is the experience necessary for the daily truck inspections required by OSHA. Your Toyota dealer can provide training aids to help you meet all of your employee obligations.

All training must include study of the appropriate Toyota materials and OSHA and ANSI operating rules; but most importantly, the workplace rules specific to your operation. This book and the National Safety Council publications listed at the end of this section will help you develop your rules.

WORKPLACE ENVIRONMENT



Many injuries related to lift truck operation are directly traced to the work environment and injuries are most easily avoided by removing hazards presented by plant layout, by improved housekeeping, or instituting and enforcing work rules for truck operators and nearby workers.

Clean, smooth lift truck aisles separated from pedestrian walkways, large well-placed intersection mirrors, work rules requiring horn use at blind intersections, and pedestrians trained to be alert to mobile equipment, will be effective in lowering the incidence of pedestrian injuries.

Dock related injuries can be reduced or eliminated by the use of trailer locks, proper edge guarding and well designed dock levelers. To eliminate creeping, never allow your forklifts to enter trailers that are not properly parked. Chock the wheels and use jacks to support the front and rear when necessary.

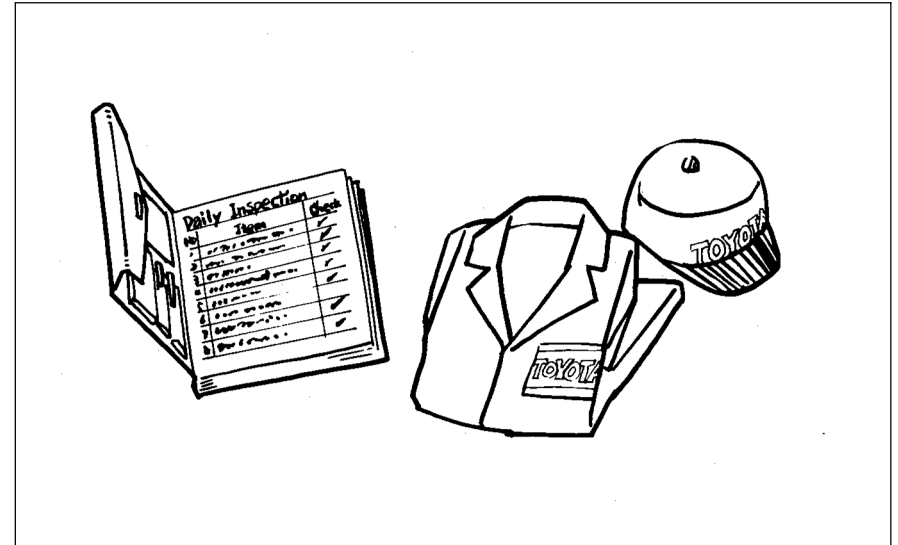
I. OWNER'S SECTION

Lift trucks are versatile industrial machines, but don't use them when other equipment is needed. Man-lifts and mobile work platforms are designed to lift people; lift trucks are not. Ladders or scaffolding may require more time to reach high areas than climbing up a lift truck mast, but the extra safety is worth the wait.

Lift truck tipovers can be the source of serious injuries. You can eliminate hazards which cause them by training your operators, enforcing work rules against overloading and high speeds, and promoting the use of appropriate operator restraints. Provide a hazard free environment for your people so that they can work more efficiently.

Please read the section on workplace hazards and eliminate them so your operators will not have to recognize and avoid them while also concentrating on their job. The section on operator rules is for you also. It will help you set up work rules suited to your workplace.

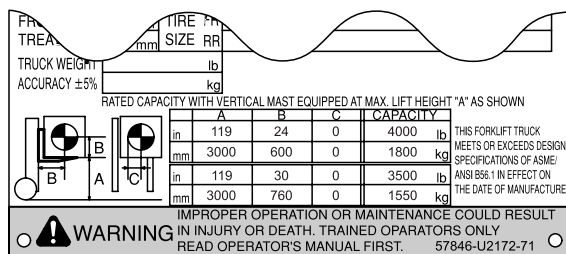
LIFT TRUCK MAINTENANCE



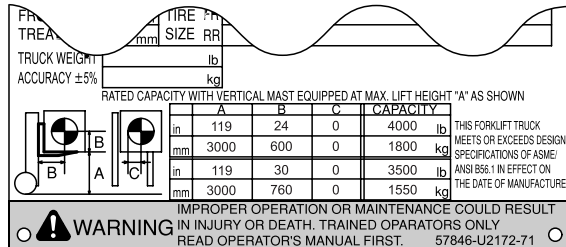
Planned maintenance provides not only the least expensive way to operate your Toyota trucks, but also the safest. OSHA requires inspection of lift trucks daily or at the start of every shift. This daily check is an important aspect of equipment safety and availability, and you should provide the time, training materials, and discipline necessary to make it work for you.

OSHA and ANSI require that authorized mechanics perform maintenance and repairs, not operators. Toyota dealers can provide planned maintenance service, daily operator check lists and genuine Toyota replacement parts to help you comply with lift truck maintenance standards.

LIFT TRUCK MODIFICATION



Standard mast



**FSV mast
(Full-Free Lift
Three-Stage mast)**

The diagrams show the mast configurations and their corresponding capacity tables. Both tables list dimensions in inches and millimeters, and capacities in pounds and kilograms. A warning label is present on each diagram, stating: "IMPROPER OPERATION OR MAINTENANCE COULD RESULT IN INJURY OR DEATH. TRAINED OPERATORS ONLY. READ OPERATOR'S MANUAL FIRST. 57846-U2172-71".

The capacity rating of your Toyota lift truck is determined by extensive testing to provide you with a safe, efficient material handling tool. Changes or additions to the truck, such as installation of a sideshifter or changing a mast, change the load capacity. OSHA and ANSI require that any changes made to a lift truck that potentially affect its safety must be approved by the manufacturer in writing. Please protect your employees by contacting your Toyota dealer before making any change to your truck.

The examples of adding a sideshifter or changing a mast or counterweight are some of the many changes that require approval and revision of a machine's nameplate. Other changes to the steering or braking system, overhead guards, wheels and tires, can all affect safety. If you are in doubt, call your Toyota dealer and ask.

INFORMATION RESOURCES

You should become familiar with the operating, maintenance and safety instructions contained in these publications:

General Industry Standards, OSHA 2206: "OSHA Safety and Health Standards (29 CFR 1910)", "Subpart N-Materials Handling and Storage", Section 1910.178 "Powered Industrial Truck". For sale by: Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

ANSI/ITSDF B56.1: "Safety Standard for Low Lift and High Lift Trucks" (Safety Code or Powered Industrial Trucks). Published by: Industrial Truck Standards Development Foundation, 1750 K Street NW, Suite 460, Washington DC 20009.

NFPA 505: "Fire Safety Standard for Powered Industrial Trucks; Type Designations, Areas of Use, Maintenance and Operation".

NFPA 58: "Storage and Handling Liquefied Petroleum Gases". Available from: National Fire Protection Association, Inc., Batterymarch Park, Quincy, MA 02269.

National Safety Council Data Sheets: "I-664- Writing and Publishing Employee Safety Regulations"; "479-Liquefied Petroleum Gases for Industrial Trucks"; "317-Powered Hand Trucks"; "653-Powered Industrial Lift Trucks"; "195.71-The 'In-Plant' Pedestrian". Available from: National Safety Council, 444 North Michigan Avenue, Chicago, IL 60611.

Accident Prevention Manual for Industrial Operations: Two volumes: "Administration and Programs", and "Engineering and Technology", Available from: National Safety Council, 444 North Michigan Avenue, Chicago, IL 60611.

Publications Concerning Safe Handling and Storage of LP Gas: Contact: National LP Gas Association, 1301 West 22nd Street, Oakbrook, IL 60521.

If you have any questions about the application of this information to your workplace, contact your Worker's Compensation Insurance Company or Toyota Material Handling, U.S.A., Inc. 1 Park Plaza, Suite 1000, P.O.Box 17419, Irvine, CA 92623-7419.

I. OWNER'S SECTION

II. SAFETY INFORMATION

CONTENTS

- Training
- OSHA/ANSI Safety Rules
- Operational Hazards and Avoidance

SAFETY AND OPERATION INFORMATION.....	7
OPERATOR AND OWNER SAFETY INFORMATION	8
WORKPLACE RULES.....	9
HAZARDS	13

SAFETY AND OPERATION INFORMATION

Don't just read over safety information, it could save your life. Make sure you understand it. If you don't understand something, ask your supervisor before you take a chance of being wrong.

 Every time you see this safety alert symbol in this book or on your truck, there is important safety information for you. It means **PAY ATTENTION! LOOK OUT! YOUR SAFETY IS INVOLVED!**

The word after the  symbol will tell you about the hazard or importance of the rule.

 **DANGER** -- An extreme hazard that will result in death or cause very serious injury every time if you don't follow the rules.

 **WARNING** -- Not as serious as DANGER, but a hazard that may cause serious injury or death if you don't follow the rules.

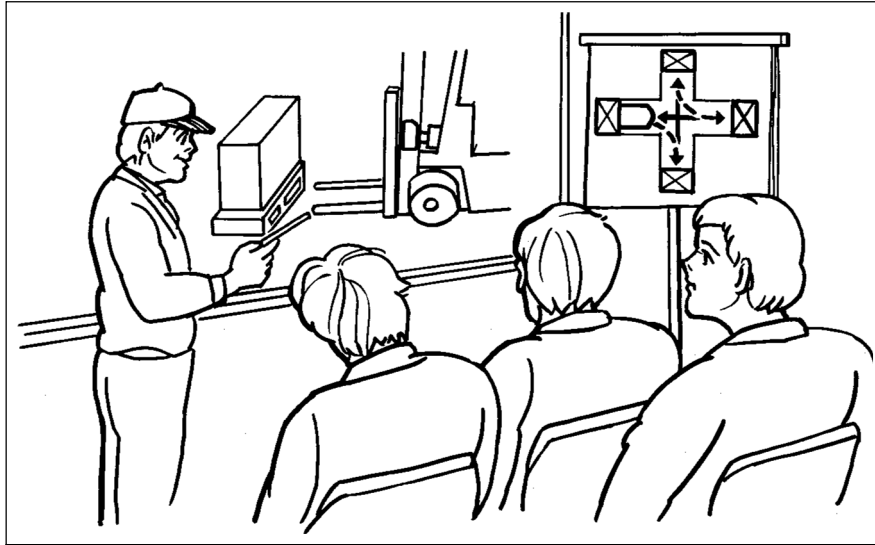
Two other words will be used to help you find better ways to run your truck:

Caution -- When you see this word, there will be a message to help you avoid minor injury or damage to your truck.

Note -- Ways to do your job right and easier and to keep your truck running well, follow this word.

II. SAFETY INFORMATION

OPERATOR AND OWNER SAFETY INFORMATION



TRAINING

Even simple hand tools can't be used efficiently or safely by people not trained or experienced with them. Lift trucks are complicated industrial machines that can be dangerous if used by people who are not trained. Lift trucks are not all the same. Even if you are trained on one type it does not mean you can operate any machine or do any job.

OSHA and ANSI require that formal training, specific to the place where you work, must be given to lift truck operators. On-the-job training must be a part of any training program, but it must really be training with close supervision, not just driving around or being assigned to a potentially dangerous job without gradual increases in difficulty.

Increased safety and fewer accidents are major benefits of proper training, but so are decreased costs due to better efficiency, decreased maintenance costs, and lower product damage rate.

When people successfully complete the training course, give them recognition for it. Issue authorization cards or jacket patches that show they are authorized operators, and if possible assign them to one truck that they have the responsibility for.

Contact your Toyota dealer for help in putting together a training course suited to your operation and think about the following areas as they apply to the work you do.

- How and where to perform daily truck inspections (OSHA regulation for every shift in multiple shift operations).
- Specialized trucks or attachments that you use (for example, roll clamp, ASME LP tank).
- Particular areas where hazards to pedestrians are created (for example, lunch room, entrance doors).
- Hazards that can't be eliminated (for example, outside ramp with ice in shadow, low overhead clearance).
- Loading docks, rail cars or containers (for example, loading container with portable ramp).
- Plant traffic rules (for example, speed zones, right-of-way, one-way aisles).
- Procedures for reporting truck maintenance or repair problems (for example, forms, out-of-service tags).
- Special refueling procedures (for example, LP tank storage, "Diesel Only" sign on truck).
- Classified hazard areas are restricted to certain types of lift trucks (for example, acetone barrels, cotton warehouse).

The examples above are intended to be thought joggers, not necessarily all of the things that need to be taught to your operators. Before putting your training course together, walk around the plant, look at your operations, and ask people who work there if they know of ways to teach operators how to become safer and more efficient.



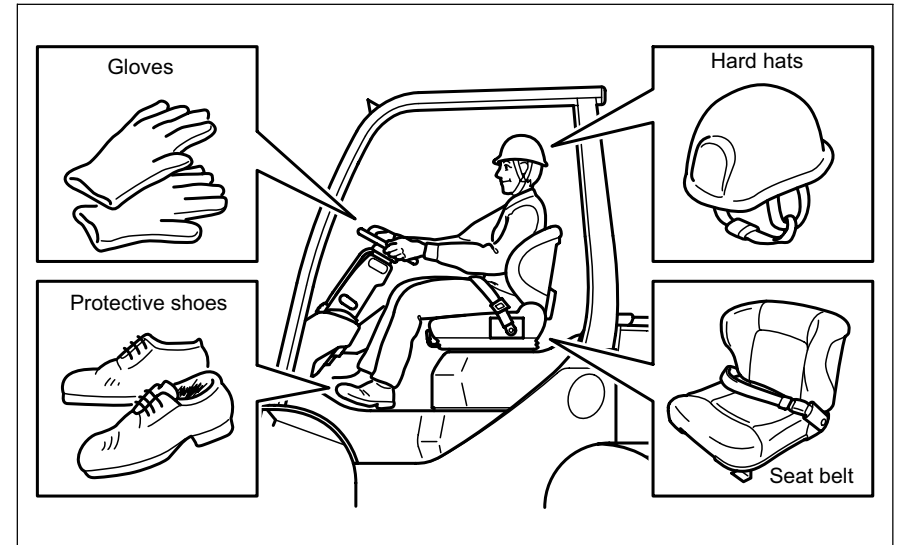
WORKPLACE RULES

Lift truck operators need to remember these rules for safety while they are working. Some may be more important in your workplace and you may know of some others that are only necessary where you work. In either case, supervisors need to know all the rules to help operators stay safe.

Remember, this is not a training manual for new operators. Lift trucks are so much different than cars that even experienced automobile drivers need formal training, and OSHA requires it!

Many of these rules are in the OSHA, ANSI, National Safety Council, and other publications at the end of the first section of this book. We can't put all of that information in this book, so please study them when setting your work rules.

Use this section of this book as a starter to develop workplace rules that keep your plant safe. Call meetings with your people to explain your rules -- post them where they can be read and enforce them.



OPERATOR

- **Training:** Must be formally trained on specific equipment before being authorized to operate fork lift truck.
- **Drugs/Alcohol Mentality:** Mixing emotional problems, drugs or alcohol with a heavy industrial machine is a dangerous situation.
- **Personal Protective Equipment, Clothing:** Proper protection may be necessary in some operating environments. Use hard hats, seat belts, protective shoes, gloves, etc. Mental ability is affected by comfort and fatigue level.
- **Smoking:** Never smoke when refueling, in other hazardous areas of the plant, or when operating the truck. Both hands are needed for safe operation.
- **Operator Restraints:** Restraints must be used to reduce the possibility of injury from overturns or other accidents.

II. SAFETY INFORMATION

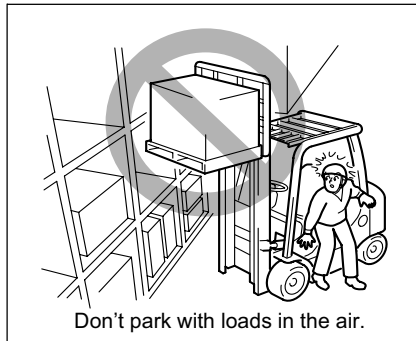


TRUCK

- **Daily Check:** OSHA requires daily or each shift checks of major equipment. Don't operate faulty equipment. Report problems and tag machine so others will not use it.
- **Capacity:** Capacity is based on load and length of load. Either condition in excess can cause loads to fall or tip a truck over.
- **Maintenance:** Operators are to perform only authorized simple maintenance such as refueling or checking fluid levels and tire pressures. Authorized, specially trained mechanics are needed to repair trucks. Park a truck that needs to be repaired. Don't risk injury by trying to fix it.
- **Truck Application:** Know the ways your truck can operate safely. Don't try to operate on rough ground with a cushion truck, or push a rail car with the carriage.

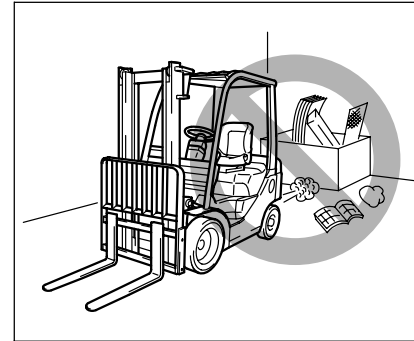


- **Ramps/ Inclines:** Travel loaded with load uphill at all times. Never turn or park on an incline.
- **Stay Protected:** Don't travel with legs, arms or head outside the overhead guard. Watch for long objects that could come into the compartment as you travel. Never put hands or feet through the mast.
- **Rules of the road:** Unless directed otherwise by your employer, signs, etc., keep right; no passing at intersections or other dangerous places. Use horn only when necessary for signaling. Give pedestrians right-of-way.



PARKING

- **Hydraulic Drift:** Don't park with loads in the air. Even well maintained hydraulic systems can change position over time and cause a load to drop.
- **Safety Exits/Equipment:** Don't park where emergency exits are blocked, heavy traffic must go, or where emergency equipment such as fire extinguishers are stored.
- **Forks Down:** Lower forks or attachments to help keep the truck from rolling and to avoid trip hazards.
- **Direction Control:** Shift control to NEUTRAL even if parked for a short time.
- **Parking Brake:** Make sure the parking brake works and use it every time you stop. Chock wheels if there is any question about the truck moving.
- **Key Switch:** Stop the engine when you park. For LP engines, turn off the tank valve, run engine out of fuel, then turn off the key.



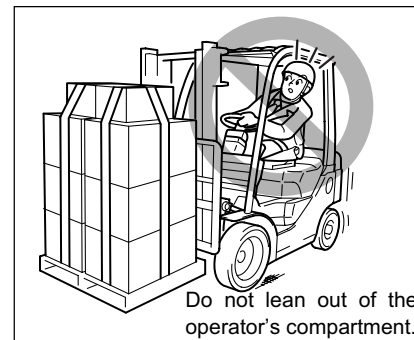
=====

⚠ WARNING

Flammable and/or combustible materials can be damaged, and in some cases ignited, by a hot exhaust system or hot exhaust gases. To minimize the possibility for such damage or fire, the operator must obey the following recommended practices:

- Do not operate lift truck over or near flammable and/or combustible materials, including dried grass and paper scraps;
- Park lift truck with rear end at least 12" away from lumber, veneer board, paper products and other similar materials to avoid discoloration, deformation or combustion of those materials.

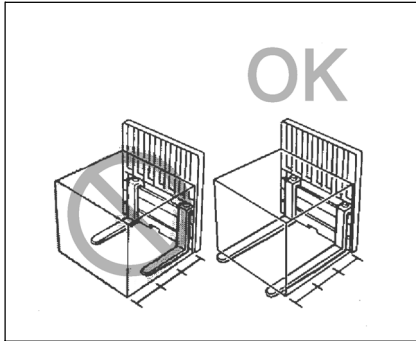
=====



LIFTING

- **Operating Position:** Operate the truck only when in the normal operating position -- never standing on the floor or leaning outside the operator's compartment.

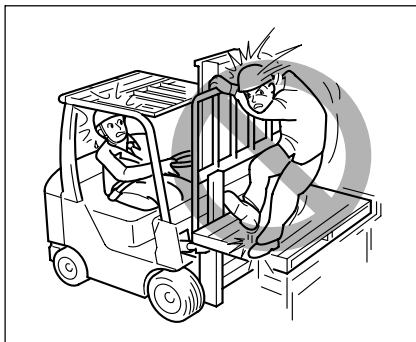
II. SAFETY INFORMATION



- **Forks:** Forks must be long enough to go at least 2/3 of the way under a load. Spread forks as far as possible and lock them to the carriage.



- **Loose Loads:** Don't carry loose or unstable loads; restack if necessary. Carry all loads lifted just high enough to clear the floor and tilted back to cradle them.



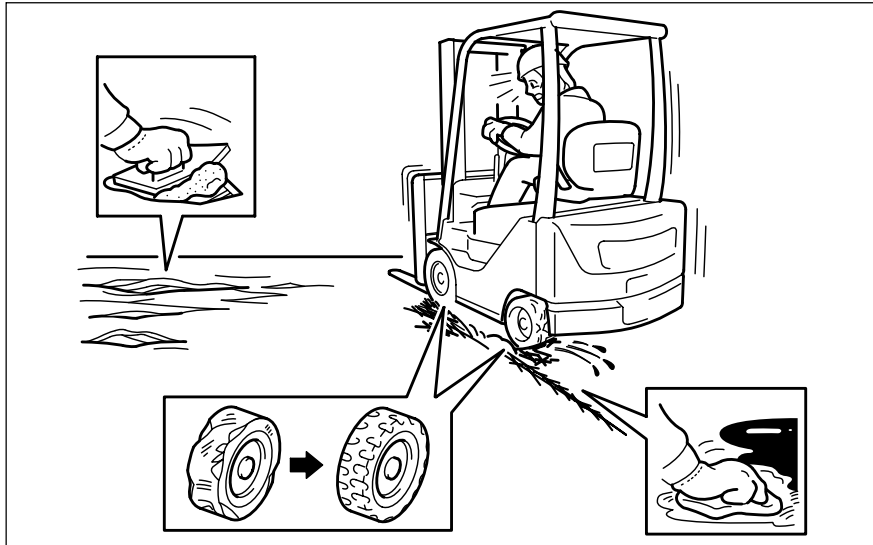
- **People:** Don't use a lift truck to raise people -- use a work platform or ladder. Don't use the mast or overhead guard to reach high points. Don't let anyone stand under raised forks.



TRAVELING

- **Pedestrians:** Give right-of-way to pedestrians; especially watch tail swing -- pedestrians may not know about it. Never take riders. Use the horn to make sure pedestrians see you. Flashing lights and beepers are not always effective.
- **Speed:** Smoothness is important for safety and to avoid loads falling or being damaged. Turn, stop and start gently.
- **Go Backwards:** REVERSE direction is to be used if vision is blocked by attachments or load. Always make sure your intended path is clear. Watch for people.
- **Mirrors:** Never depend on mirrors for traveling in reverse, look where the truck will be going. Use mirrors for extra side vision and to check tail swing when going forward.
- **Weight:** A loaded 6000lb truck can weigh over 8 tons (17600lb). Elevators, upper floors, ramps or highway trucks may not be able to support this weight. Ask your Toyota dealer if you are not sure what your truck weighs.

HAZARDS



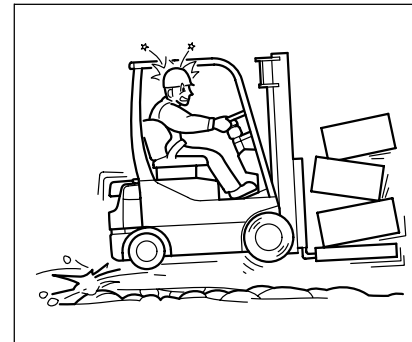
WORKPLACE HAZARDS

We can't know all the hazards in your work area, but here are three very good rules to follow to help recognize and avoid some hazards:

1. If you see one of these conditions that can cause an accident or injury, eliminate it. Guard drop offs and fill pot holes. Don't allow maintenance of your truck to fall behind. Train your people about the unique design of the lift truck.
2. If a hazard occurs, avoid it. For example, stay clear of slippery spots and make sure they are cleaned up. Change or repair a low tire; don't keep driving.
3. Look for other hazards that may only be in your workplace; things that could cause injury that only you know about and that only you can eliminate or avoid.

LIFT TRUCK HAZARDS

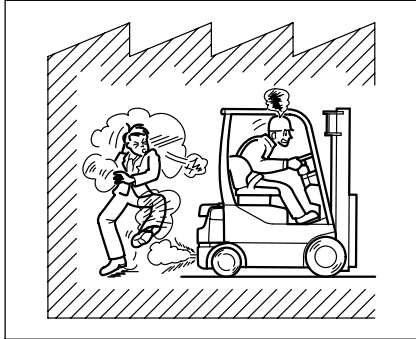
- **Truck vs Car:** Lift trucks are not cars. They have small tires, no springs, and are very heavy. They can pick up a heavy weight that raises the center of gravity and they steer from the rear very sharply.
- **Rear Steer:** Rear steer is necessary for doing its job. Combined with sharper turning, results in tail swing that can surprise and injure pedestrians or damage product.



- **No Suspension:** Lift trucks cannot go over bumps or through pot holes because springs can't be used when the weight and center of gravity change due to picking up loads. Avoid these hazards.

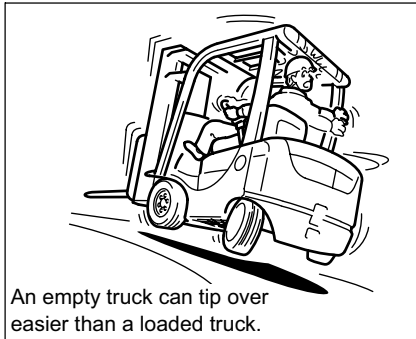
- **Overturn:** Lift trucks can overturn if improperly operated or if hazards are not avoided. Empty trucks can overturn at high speed; loaded trucks can overturn while stopped.
- **Attachments:** Attachments that hold onto loads or even sidershifters can cause dynamic loadings that can upset a truck if smooth operation of controls or brakes is not used.
- **Slack Chains:** When slack chains occur, something is wrong with the mast. Operate the lift control to take up the slack before you move the truck. Forks or rails may be caught or jammed. Never put your hands in the mast. Ask your Toyota dealer to inspect and repair.

II. SAFETY INFORMATION



- **Carbon Monoxide:** All engines produce carbon monoxide, even LP and diesel. Don't operate without good ventilation. Don't let trucks idle unnecessarily. Stop truck and get fresh air if you feel drowsy, sick or have a headache from exhaust fumes.

- **Maintenance:** Don't drive a truck that is not mechanically sound. Park it and tag it out so others won't be hurt.

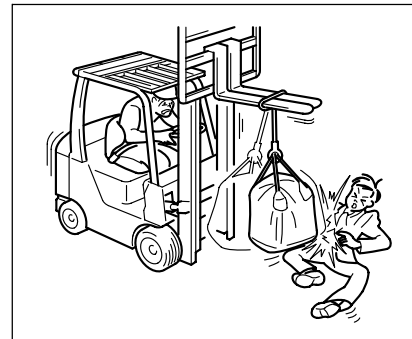


- **Empty Trucks:** A large counterweight on the rear balances the load in front and you know that high speed or quick turns with that load could cause an overturn or spilling of the load. When you drop off the load, that does not give you the green light to forget about safety. The heavy counterweight is still with you, so when empty, the truck can be more unbalanced than when loaded. This large weight can cause a truck to tip over when the rear wheels are being straightened when coming out of a curve. Steering tighter again will restore balance until the truck can be slowed, but it is always best to turn, stop and start smoothly to avoid tipovers.

- **Refueling:** Gasoline, diesel, LP and CNG all can cause serious injuries if not handled properly. Avoid spills, see the Operator Maintenance section for more important information.

LOAD HAZARDS

- **People:** Lift trucks are designed ruggedly to carry and stack heavy loads. If problems occur in a mast, it is often when it is lightly loaded. Stacking operations rarely require a mast to go to full height as is often done when trying to reach a high area. Don't use your truck to lift people.
- **High, Wide, Unstable:** These loads can fall off and be damaged, interfere with a doorway or overhead beam, and tip a truck over or fall onto the operator or a bystander. Sometimes wide loads must be carried high to be lifted over other stacked material. If so, minimum load, height and speed must be used.

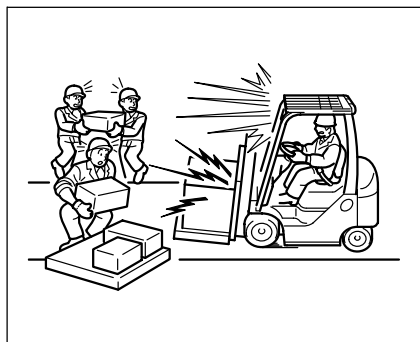


- **Suspended Loads:** Carrying a load on a chain or cable can introduce dynamics due to swinging that can unbalance a truck or strike a bystander.

- **Overhead Guard, Load Backrest Extensions:** These design features are incorporated to protect the operators from falling loads. Make sure they are in place and in good repair.
- **Inclined Storage:** Stacking loads on an incline can result in damaged goods, falling loads, injury to bystanders, or upsetting of the truck.
- **Mast Tilt:** Getting it just right takes training and experience. Too far forward could cause tip up; too far rearward could contribute to lateral overturn.

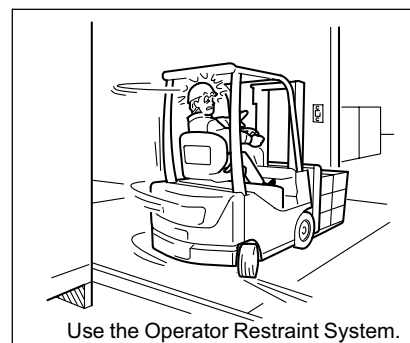
ENVIRONMENTAL HAZARDS

- **Slippery Surface:** Ice, water, oil, snow, sand, etc., can cause loss of control. Avoid these hazards-get them removed.
- **Rough Terrain:** Pot holes, bumps and pavement edges can damage or upset a truck. Go around them and see that they are fixed.
- **Railroad Tracks:** If you must cross them, go at an angle so one wheel crosses at a time. Trucks can be damaged, loads lost, or the truck upset.
- **Poor Lighting/ Noise:** Loud noises can make it hard to concentrate or damage your hearing. Noise and hearing protectors make it harder for pedestrians to notice you. If you need light to be seen or to work properly, your Toyota dealer can supply a truck mounted light to fit operating conditions.



- **Pedestrians:** People are a big part of a work area. They have their own jobs and are concentrating on them, not lift trucks. Your responsibility is safe operation. Don't depend on flashing lights or beepers to avoid injuring people. Watch for them and make sure they know your are near. Use your horn and make eye contact.

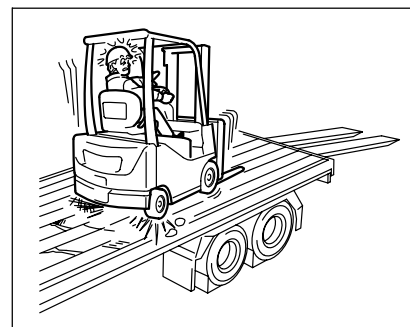
- **Explosions/Fire:** Chemical vapors and even some types of dust can explode if you drive into classified hazard areas requiring certain types of lift trucks. Reference OSHA 29 CFR 1910.178. Fibers such as cotton may collect on the truck and have to be removed from the radiator screen or exhaust area. Make sure your truck is the right one for the job and that it stays clean.
- **Floor Hazards:** Watch for loose materials, open or loose drainage channel covers, posts or gratings. Lift trucks need smooth, level floors for safe operation.



LOADING DOCK HAZARDS

- **Falls:** Rail or highway docks present drop offs, many of which are not guarded. Use the Operator Restraint System and stay with the truck if it falls. Watch rear steering near edges, back away, don't go forward.

- **Trailers:** Unrestrained trailers can creep due to a lift truck going in or out. Chock wheels if dock restraints are not used; follow signals if they are. If the trailer moves forward, the dock board or lift truck can fall. Trailers without a tractor must be supported with jacks at front and rear when necessary.



- **Floors:** Make sure floors in rail cars, highway trucks, and trailers are strong enough for your truck. Don't fall through a weak area.

- **Dock Boards:** Must be properly sized for the job and firmly in place. Watch for any shifting or movement.

- **Portable Ramps:** If using a portable ramp to load a trailer, make sure it is secured to the trailer, that the lip remains in place and that all wheels are off supporting surfaces. Keep driving surface clean and dry. Operate lift truck slowly and smoothly.
- **Truck Drivers:** Don't become distracted by truck drivers, make sure they are not in your operating area, but in a safe zone away from the loading/unloading traffic. Don't rely on them to chock wheels, make sure it is done properly. If a trailer is not held up by a tractor, it must be supported with jacks at front and rear when necessary.

II. SAFETY INFORMATION

III. KNOW YOUR TRUCK
CONTENTS

- General Description, ANSI Designation
- Intended Uses, Typical Applications
- Prohibited Uses
- Expected Work Environment
- Expected Skill Levels of Operators/Mechanics
- Maintenance Requirements
- Warning and Informational Decals
- Controls, Switches, Instruments, Restraints
- Nameplate/Capacities

MAIN COMPONENTS.....18

LIFT TRUCK DESCRIPTION19

PRECAUTIONS TO BE TAKEN WHEN USING SAS MODELS20

OPS SYSTEM23

RETURN-TO-NEUTRAL24

TRAVEL AND LOAD HANDLING CONTROL (OPTION).....25

WARNING AND INFORMATIONAL DECALS26

OPERATING CONTROLS.....30

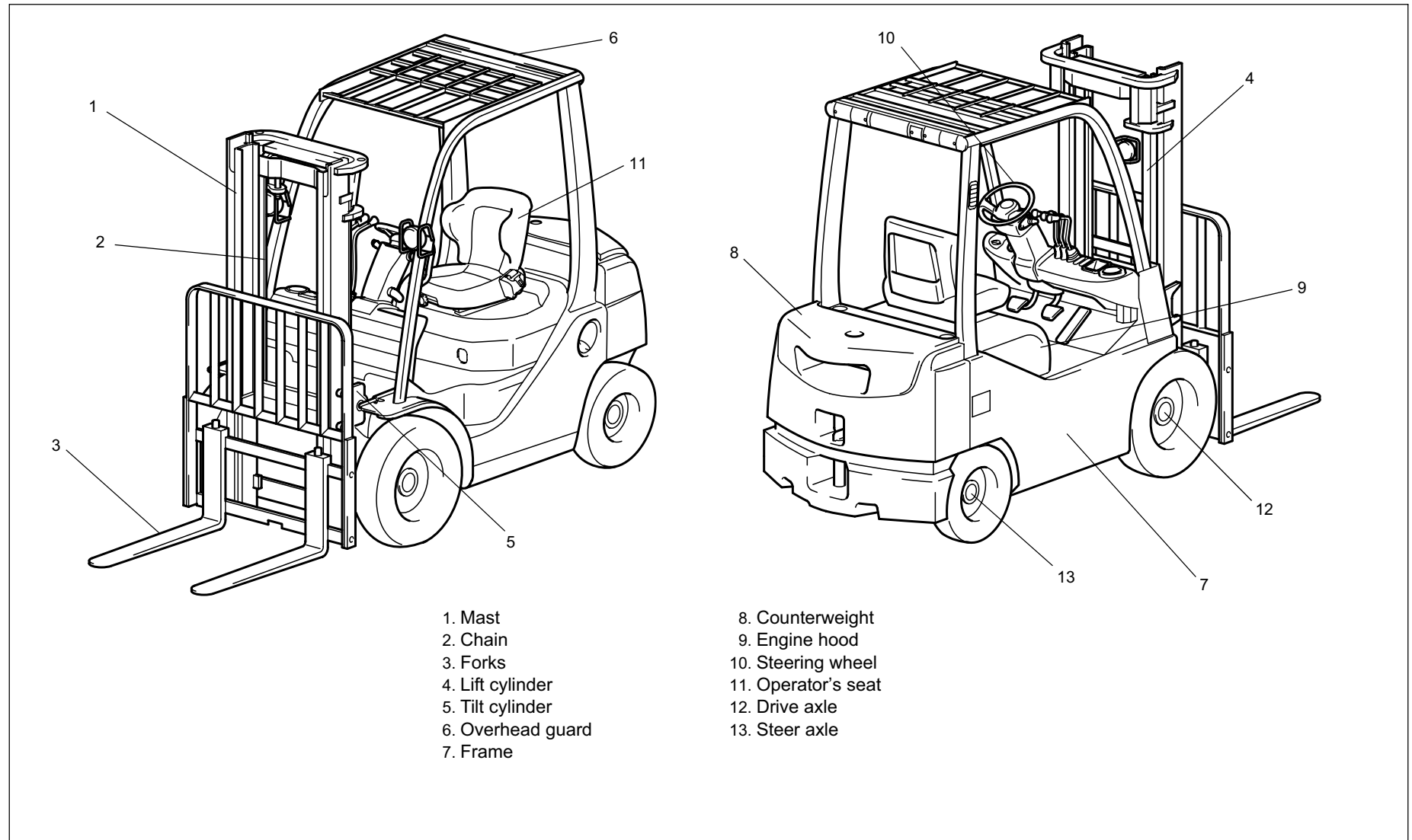
INSTRUMENTS.....31

MULTIFUNCTION DISPLAY DX (OPTION).....38

SWITCHES AND LEVERS.....43

BODY COMPONENTS.....57

MAIN COMPONENTS



If you want to perform your own maintenance and repairs, do not attempt it without asking your dealer for the proper instructions and manuals. OSHA requires it.

TOYOTA FORKTRUCK						
MODEL				SERIAL NO.		
MAST	BACK TILT			ATTACH		
TYPE						
FRONT TREAD	In mm		TIRE FR SIZE RR			
TRUCK WEIGHT	lb					
ACCURACY ±5%	kg					
RATED CAPACITY WITH VERTICAL MAST EQUIPPED AT MAX. LIFT HEIGHT "A" AS SHOWN						
		A	B	C	CAPACITY	
		In	119	24	0	4000 lb
		mm	3000	600	0	1800 kg
		In	119	30	0	3500 lb
mm	3000	760	0	1550 kg		
THIS FORKTRUCK MEETS OR EXCEEDS DESIGN SPECIFICATIONS OF ASME / ANSI B56.1 IN EFFECT ON THE DATE OF MANUFACTURE						
		WARNING IMPROPER OPERATION OR MAINTENANCE COULD RESULT IN INJURY OR DEATH. TRAINED OPERATORS ONLY READ OPERATOR'S MANUAL FIRST. 57846-U217-71				

⚠ WARNING

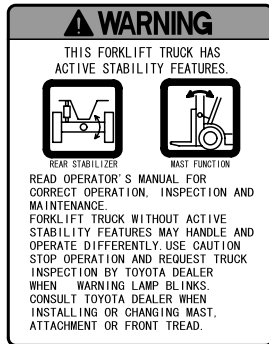
This warning is not meant to substitute for training, but is here as a reminder of the more important areas covered in a training course. Use it to refresh your memory often.

III. KNOW YOUR TRUCK

PRECAUTIONS TO BE TAKEN WHEN USING SAS MODELS (SAS: System of Active Stability)

Whenever you use a Toyota forklift equipped with SAS, check the warning plate to determine which SAS feature(s) your truck has been equipped with. Do not operate your truck if any SAS feature is not operating properly.

[Figure 1]



The following are examples of the warning plates which will be affixed to SAS equipped trucks and identify the SAS features installed on that specific truck.

Figure 1.

This label will be used on SINGLE TIRE Toyota trucks equipped with SAS. As the pictorial symbols on the label indicate, this truck is equipped with the following SAS features:

- Swing lock control on the rear axle;
- Mast forward tilt angle control.

[Figure 2]

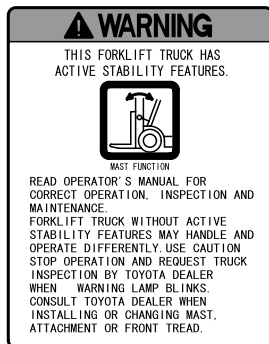
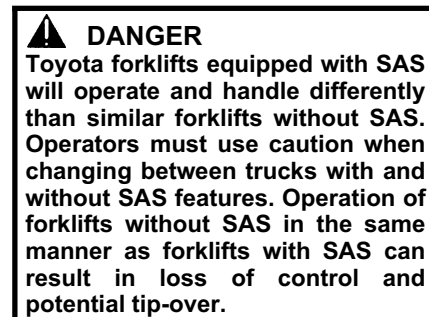


Figure 2.

This label will be used on DUAL TIRE Toyota trucks equipped with SAS. As the pictorial symbols on the label indicate, this truck is equipped with the following SAS feature:

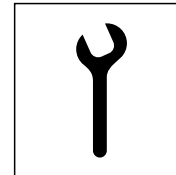
- Mast forward tilt angle control.



Caution

Before operating a truck equipped with SAS, familiarize yourself with the SAS feature or features and be sure that the SAS warning lamp is not illuminated. Either condition indicates a malfunction in the SAS system and the truck should not be operated until it has been repaired.

- While operating a truck equipped with SAS, should the diagnosis lamp illuminate, or an error code be indicated by a warning lamp or hour-meter, park the truck in a safe location and have the truck inspected by a Toyota dealer.
- The SAS system is electronically controlled. The system may need to be initialized after completion of maintenance.
- Do not remove or modify the SAS features. If inspection of the SAS system is required, contact your Toyota dealer.
- When washing your truck, take caution to prevent water from being directly applied to the electronics used in the SAS system. These electronics include the controller, sensors and switches.



This warning lamp will illuminate in the event of a SAS malfunction.

DESCRIPTION OF FEATURES AVAILABLE ON SAS MODELS:

Active Control Rear Stabilizer:

This feature is designed to temporarily lock and prevent the movement / articulation of the rear axle on your truck in order to increase the stability of your truck during right and left turns. This feature is intended to enhance the stability of your forklift under certain conditions, but cannot prevent your forklift from tipping over under all circumstances. You must continue to use caution when operating your forklift.

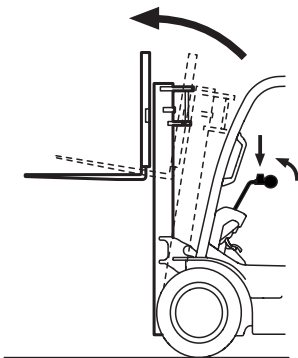
AUTOMATIC FORKS LEVELING CONTROL

- With tilting the mast forward while pressing the tilt lever knob switch will cause the fork to automatically stop at its horizontal position (the mast vertically positioned).

Motion of Mast when changing Tilt from Backward to Forward, with Tilt Lever Knob Switch Pressed:

	Not Loaded	Loaded
High Lift (Over 6 feet)	Forks will stop at the horizontal position (with mast vertically positioned)	The mast will not tilt forward
Low Lift	Forks will stop at the horizontal position (with mast vertically positioned)	

- After stopping the forks at the horizontal position with the tilt lever knob switch pressed, you may want to tilt the forks further forward. Return the tilt lever to the neutral position once. Then, after releasing the tilt lever knob switch, operate the tilt lever again.



When tilting the mast forward while pressing the tilt lever knob switch, the mast will stop automatically at the vertical position so that the upper surfaces of the forks will be horizontal.

Caution

- When tilting the mast forward with a heavy load at high lift, pressing the tilt lever knob switch will cause the mast to suddenly stop moving. Avoid this operation. Under some conditions, suddenly stopping the mast while tilting forward may cause the truck to tip forward.
- Some specialty models onto which a heavy attachment is mounted may not be equipped with the automatic forks leveling control. Confirm with a Toyota dealer in advance.

Note

- The mast will not move if it is tilted forward while pressing the tilt lever knob switch with a high load at a high lift (more than 6 feet).
- When tilting backward, the forks will not stop at the horizontal position even if the tilt lever knob switch is pressed.

ACTIVE MAST FRONT TILT ANGLE CONTROL

According to the lift height and load weight, the angle at which the mast can be tilted forward is automatically controlled within the range of angles illustrated below.

	Light Load (no load)	Medium Load	Heavy Load
High Lift (Over 6 feet)	Forward tilt angle not controlled	Angle controlled between 1° and 5° degrees forward tilt	Forward tilt limited to 1° degree
Low Lift	Forward tilt angle not controlled		

Caution

- If a load needs to be raised while the forks are tilted forward at low lift, you can exceed the recommended forward tilt level at high lift and potentially tip the truck over. Use caution.
- With a heavy load at high lift, never tilt the mast forward beyond vertical in order to avoid the potential for truck tip over.

III. KNOW YOUR TRUCK

Caution

- Some specialty models onto which a heavy attachment is mounted may not be equipped with the mast forward tilt control. Confirm with a Toyota dealer in advance.
- Once you have mounted or replaced any attachment on a SAS equipped fork lift, ask a Toyota dealer to inspect the truck to verify proper SAS calibration.
- If you use two or more removable attachments alternately, the heaviest one should be used to carry out matching (SAS setting). Contact your Toyota dealer to request compatibility matching.
- When attaching forks or attachments to a forkless model, the attachments must be compatible with the model. Contact your Toyota dealer to request compatibility matching.

Note

When the forks are elevated to the maximum height, a high pressure (relief pressure) may remain in the lift cylinder. This high pressure causes the truck to judge that it has a high load even if there is no load. As a result, the mast will be stopped from tilting forward. In this case, lower the forks slightly (to release the pressure) and the mast may be tilted forward.

ACTIVE MAST REAR TILT SPEED CONTROL

- At a high lift height, the mast has a backward tilt speed controller (slow down) irrespective of the load. When lowering from a high lift height to a lower lift height while tilting the mast backward, the control speed will not change.
- At a low lift height, the mast can be tilted at full speed even if there is a load. If the mast is tilted backward at a low lift height with the tilt knob switch depressed, the mast has the backward tilt speed controlled (slowed down) as long as the tilt lever knob switch is depressed.
- If the low lift height is then changed to a high lift height while tilting the mast backward, the control speed will not change as long as the tilt lever knob switch is pressed. The mast may be tilted backward at the full speed so long as the tilt lever knob switch is not depressed.

KEY-LIFT INTERLOCK

When the ignition switch is positioned to OFF, the forks will not lower even if the lift lever is operated. However, if you sit in the seat and turn on the ignition switch, you can lower the forks.

SHOULD A PROBLEM WITH THE SAS SYSTEM OCCUR:

The SAS system consists of a controller, a sensor and various actuators. Should one or more of these components experience a problem, the following may occur:

- The automatic forks leveling control, active mast front tilt angle control and/or active mast rear tilt speed control may become inoperable.
- The swing lock may not unlock.

Should any of the above occur, you can expect the following:

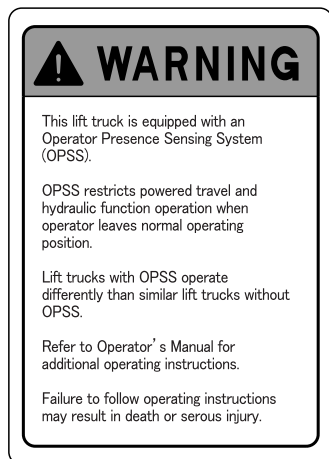
- An error code will be displayed in the hour meter.
- The diagnosis lamp will light up or blink.

Should one or more of these conditions occur, move the truck to a safe location and contact a Toyota dealer for repair.

Active steering synchronizer (Option)

If the steering wheel knob is not angularly matched with the steer tires, such out-of-position will be automatically corrected while turning the steering wheel. Thus, the optional knob is kept at a constant position relative to the steer tires.

If an error occurs within the OPSS, the diagnosis lamp will blink to inform the operator of the error. This indicates that there is a malfunction in the OPSS. Have the truck inspected by your Toyota Dealer.



⚠ WARNING

OPSS activation is shown by  lamp appearing on the instrument panel when operator is not in the normal operating position - operator's seat.

Stop operation and contact your Toyota dealer for inspection if any of the following conditions are present:

- Lamp is not turned on when operator is out of the normal operating position.
- Lamp is not turned off when operator is in the normal operating position. (Note: Lamp may stay on during brief engine warm-up, if so, this is not an OPSS malfunction.)
- Lamp is flashing, and does not go off when operator returns to normal operating position the second time.

This truck is equipped with the OPSS. Confirm the OPSS is functioning properly before operating truck.

If the operator leaves the seat while the truck is traveling, the OPSS lamp will come on and powered travel will be stopped after 2 seconds. However, the OPSS will not apply the brakes to stop coasting or prevent rolling back on inclines. (The Operator can continue moving the truck if he/she returns to the seat within 2 seconds.)

To resume powered travel, release the accelerator pedal, return the direction control lever to its neutral position and sit in the seat. Then move the direction control lever to the desired direction and press on the accelerator pedal.

Load-Handling OPSS Functions

If the operator leaves the seat for 2 seconds or more while conducting load-handling operations the OPSS lamp will come on and load-handling operations will be stopped. (The operator can continue lowering operations if he/she returns to the seat within 2 seconds.) If the operator leaves the seat while using load-handling control levers, operation can be continued for 2 to 4 seconds.

If load-handling OPSS is activated when the lift lever is in the lowering position, move the lever to a position other than the lowering position and return to the normal seated position to deactivate load handling OPSS. If load-handling OPSS is activated when the lift lever is in a position other than the lowering position, load handling OPSS is deactivated 1 second after the operator returns to the normal seated position.

** Warning:**

Insure the load handling control levers are in their neutral positions before returning to the operator's seat. If not, lowering functions will resume 1 second after the operator returns to the seat.

Forklifts with a mini lever / joy stick lever (Option)

If the operator leaves the seat during load handling operations, the OPSS lamp will light up, and two seconds later, load handling operations will be stopped. (If the operator returns to the seat within 2 seconds, load handling operations can be continued.)

To resume load handling functions, return to the seat and return all of the levers to the neutral position.

III. KNOW YOUR TRUCK

OPSS Operation Functions

If the operator leaves the seat, the OPSS lamp will come on to inform the operator that the OPSS is going to be activated. Returning to the seat will turn off the OPSS lamp.

OPSS/SAS Controller Abnormality Warning

If an error occurs within the OPSS, the diagnosis lamp will blink to inform the operator and the diagnosis error content will be displayed in the hour meter display area. This indicates that there is a malfunction in the OPSS. Park the truck in a safe location and have it inspected by your Toyota dealer.

Note

If the OPSS registers an abnormality, the diagnosis lamp will blink to inform the operator. If the diagnosis lamp starts blinking, the OPSS may have malfunctioned. Park the truck at a safe location and request an inspection by your Toyota dealer. In the following cases, park the truck in a safe location and have the truck inspected by your Toyota dealer.

- The OPSS lamp does not light up even if the operator leaves the seat.
- The OPSS lamp does not turn off even if the operator is re-seated.

Caution

If the operator remains seated in the seat for a long period with the ignition switch turned OFF, the next time the ignition switch is turned to ON, the diagnosis lamp may start blinking. If this occurs, turn the ignition switch to OFF, return to the normal seated position and then turn the ignition switch back to ON. The diagnosis lamp will then turn off.

Return-to-Neutral Warning

When the traveling OPSS function is activated, even if the operator is re-seated without returning the control lever to the neutral position, the traveling OPSS function cannot be reactivated. Please return the control lever to the neutral position.

Forklifts with a standard lever

When the load handling OPSS function is activated, even if the operator is re-seated while the lift lever is still set in the lowering position, the lowering OPSS function cannot be reactivated. Please return the lift lever to the neutral position.

Forklifts with a Mini lever / joy stick lever (Option)

When the load handling OPSS function is activated, even if the operator is re-seated without returning all of the levers to the neutral position, the load handling OPSS functions cannot be reactivated. Please return all of the levers to the neutral position.

Travel and Load handling Control Functions (Option)

Caution

- The optional Travel and Load Handling Control Functions limit maximum travel speed and high speed acceleration relative to load height and weight and reduces the chances of tip-overs. However, this feature will not prevent all tip-overs in all circumstances.
- Depending on road surface conditions and loading operations, temporary changes in speed limits and sense of acceleration may be experienced.
- During operation using the inching and brake pedal, temporary changes in speed limitations and sense of acceleration may be experienced.
- If traveling is commenced while raising the forks, the traveling speed may accelerate.

Speed limits after load lifting height and weight sensing

(Maximum speed limiter)

This function restricts maximum speed depending on the load height, to reduce truck instability if a sudden stop is required.

Note

- Although the maximum speed limiter will be deactivated when the height of the load is lowered, sudden accelerations will remain limited until the accelerator pedal is re-engaged.
- If the load height is raised when the speed limit is exceeded, the speed will be gradually reduced until the set speed limit is reached.

(Travel and load handling control indicator)

While the speed limiting load height and weight sensors are active, the operator will be notified by the travel control indicator on the multi function display.

Load height and weight sensors to prevent sudden start ups

(Acceleration limiter)

When the load is in an elevated position, this function limits sudden acceleration of the truck in accordance with the weight of the load.

(Preventing sudden starts)

When the load is in an elevated position at a high engine RPM, this function will suppress the sudden acceleration of the truck and resultant loss of the load even if the truck is operated erroneously such as by abruptly releasing the brake pedal or the inching and braking pedal while the accelerator pedal is engaged, or by setting the control lever in forward or reverse direction.

(Loading priority function)

The load height and load weight sensors will be activated by certain load height and weight conditions to prevent sudden acceleration. These sensors will be deactivated when the inching/brake pedal is engaged or the directional control lever is set to neutral. Sensor deactivation will continue when the accelerator is re-engaged as long as the truck travel speed is less than the set travel speed.

(Travel and load handling control indicator)

When the load height and weight sensor are active, the operator will be notified by the travel control indicator on the multi function display.

III. KNOW YOUR TRUCK

Low speed setting

When the low speed setting switch is pressed, traveling will not be possible at the preset speed or greater.

Pressing the low speed setting switch again will deactivate this function.

Maximum speeds can be set between approximately 5 - 9 mph.

Note

- Depending on the truck weight, the set speeds may not be attained while traveling uphill. Similarly, the set speed may be exceeded during downhill travelling, but will resume at the set speed when it is reached after going down the hill.
- When a maximum speed limit has also been set, the lower speed setting will take precedence.
- When setting the maximum speed, consult your supervisor or a Toyota dealer.

Maximum speed limit

This function prevents the truck from traveling at a faster speed from that speed preset by your supervisor or company. The maximum speed can be set between approximately 5 - 9 mph.

Note

- Depending on the truck weight, the set speed may not be attained while traveling uphill. Similarly, the set speed may be exceeded during downhill traveling, but will resume at the set speed when it is reached after going down the hill.
- When setting the maximum speed, consult your supervisor or a Toyota dealer.

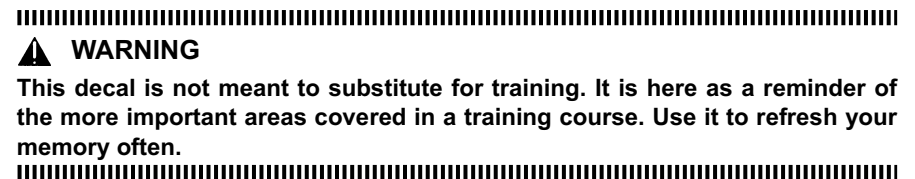
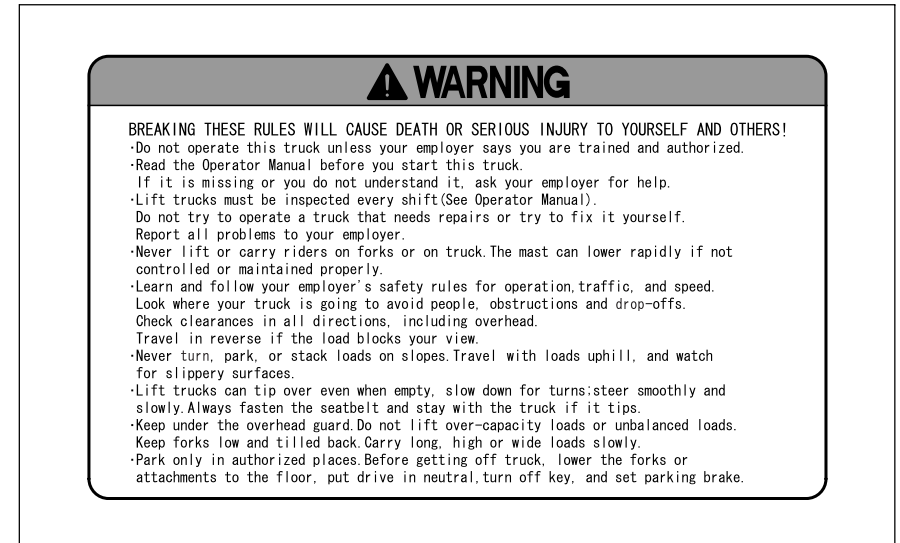
Increasing the idling lifting speed

When the lift lever is raised, the forks can be raised at a steady speed without pressing on the accelerator pedal to increase the engine RPM.

Note

- There are circumstances where the idle-up of engine increases the truck speed when the lift lever is operated.
- The lifting speed will differ depending on the truck model, specifications, and load conditions.

WARNING AND INFORMATIONAL DECALS

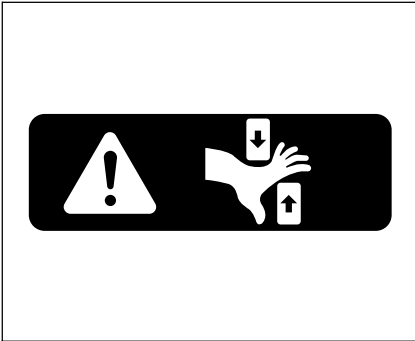


⚠ WARNING

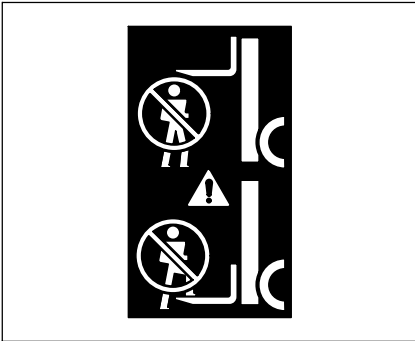
⚠ WARNING

BREAKING THESE RULES WILL CAUSE DEATH OR SERIOUS INJURY TO YOURSELF AND OTHERS!

- Do not operate this truck unless your employer says you are trained and authorized.
- Read the Operator's and Owner's Manual before you start this truck. If it is missing or you do not understand it, ask your employer for help.
- Lift trucks must be inspected every shift (See Operator's and Owner's Manual). Do not try to operate a truck that needs repairs or try to fix it yourself. Report all problems to your employer.
- Never lift or carry riders on the forks or the truck. The mast can lower rapidly if not controlled or maintained properly.
- Learn and follow your employer's safety rules for operation, traffic, and speed.
Look where your truck is going to avoid people, obstructions and drop-offs.
Check clearances in all directions, including overhead. Travel in REVERSE if the load blocks your view.
- Never turn, park or stack loads on slopes. Travel with the loads uphill, and watch for slippery surfaces.
- Lift trucks can tip over even when empty. Slow down for turns; steer smoothly and slowly. Always fasten the seat belt and stay with the truck if it tips.
- Keep under the overhead guard. Do not lift over-capacity loads or unbalanced loads.
- Park only in authorized place. Lower the forks or attachments to the floor. Put drive in NEUTRAL, turn off key, and set parking brake.



This decal may be placed in several places on the truck, but look for it, especially on the mast and on attachments. When you see it, stay clear.

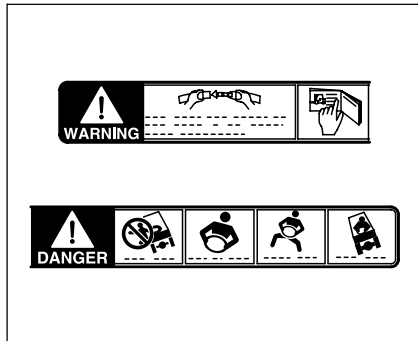


Never lift people with your truck or allow anyone under the load. Safety of people around you is your responsibility.

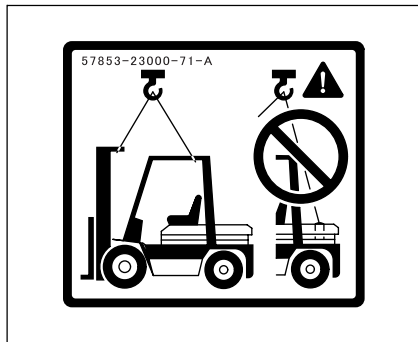


If you fill the fuel tank, make sure you use the proper fuel. A mistake could cause very costly repairs.

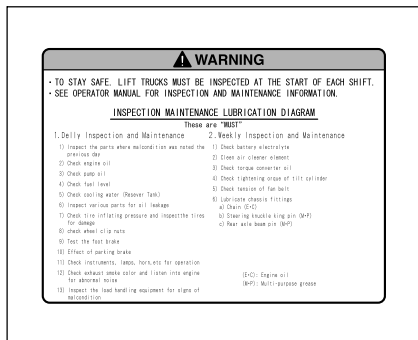
III. KNOW YOUR TRUCK



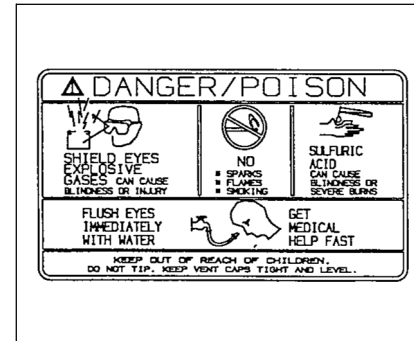
Tipovers can cause serious injury. Learn to see and avoid the hazards that cause them and use the seat belt. If your truck tips, stay in the seat, lean away from the fall, hang on, and brace your feet.



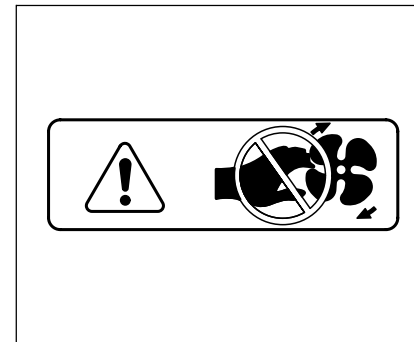
Don't risk dropping the truck, never use the counterweight to pick it up.



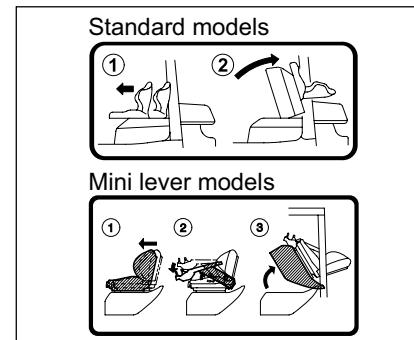
This is a reminder of inspections that are required by OSHA for all lift trucks.



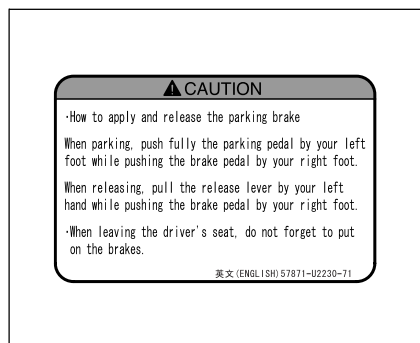
Batteries can be very dangerous if you do not follow these rules.



The fan is white to make it easier to see when it is spinning. You still need to watch out for this decal and stay clear.



Don't damage your truck, adjust the seat forward first.

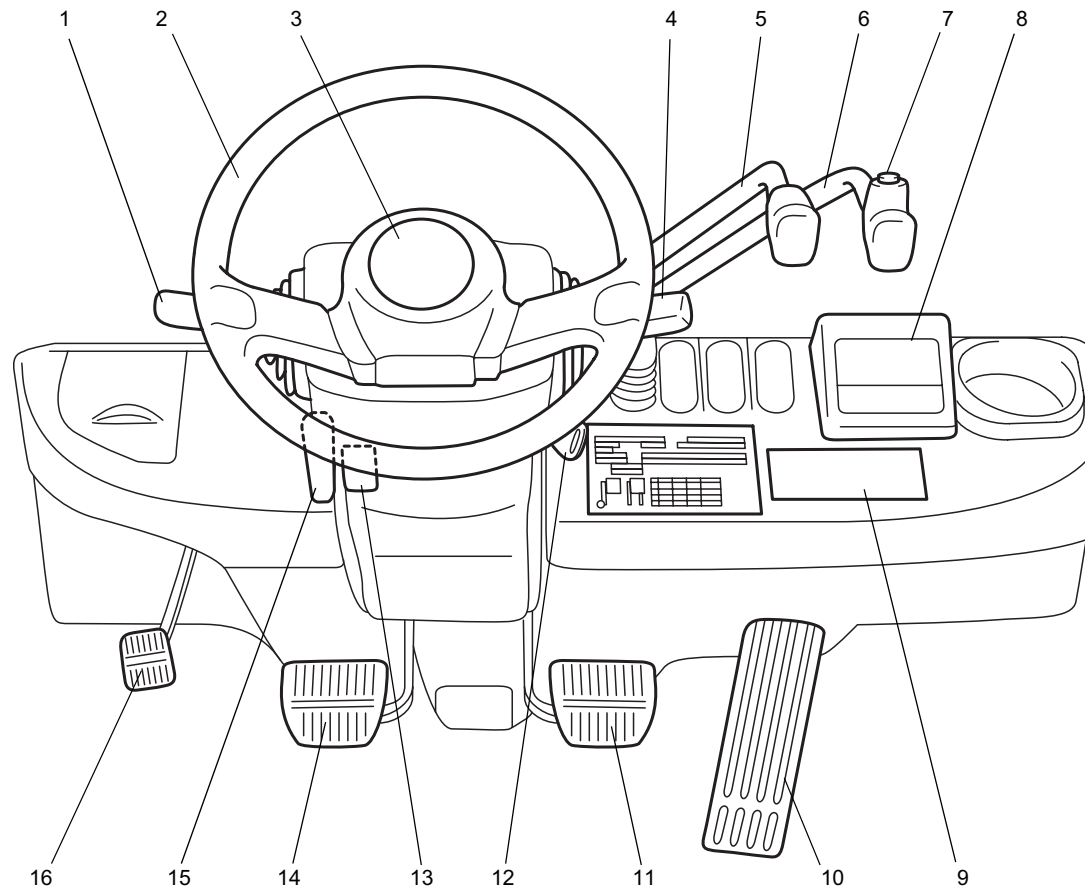


Apply the parking brake according to the content of caution label.

Caution

Trucks equipped with the wet brake option require a warm up period before starting load handling operations when temperature is below -10°C (14°F). The warm up method consists of driving the truck without load for approximately 200 meters (600ft).

OPERATING CONTROLS



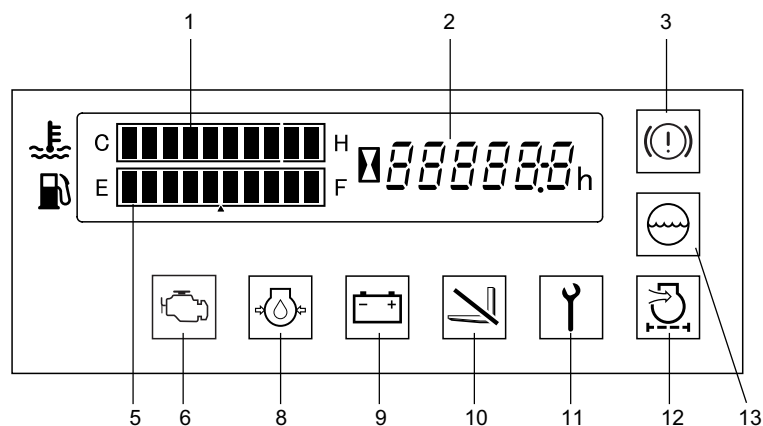
1. Direction control lever
2. Steering wheel
3. Horn button
4. Light control and turn signal switch

5. Lift lever
6. Tilt lever
7. Tilt lever knob switch
8. Multi function display (Option)

9. Combination meter
10. Accelerator pedal
11. Brake pedal
12. Ignition switch

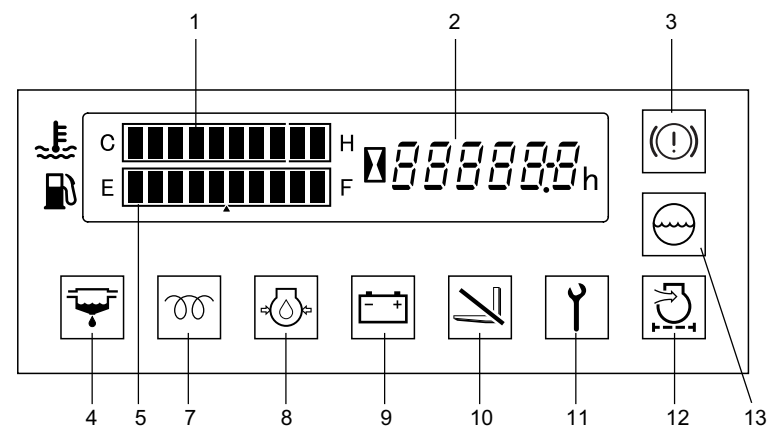
13. Parking brake release lever
14. Inching and brake pedal
15. Tilt steering adjust lever
16. Parking brake pedal

INSTRUMENTS



Gasoline engine models

1. Water temperature gauge
2. Hour meter
3. Brake warning lamp (OK monitor :Option)
4. Sediment warning lamp (Diesel engine models)
5. Fuel gauge
6. Engine check lamp (Gasoline engine models)
7. Glow indicator lamp (Diesel engine models)



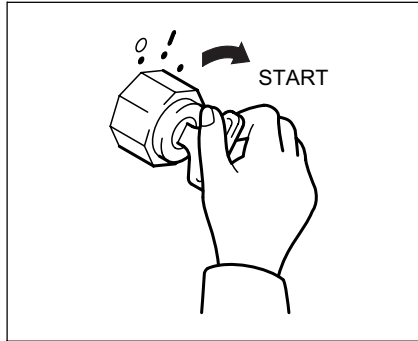
Diesel engine models

8. Engine oil pressure warning lamp
9. Charge warning lamp
10. OPSS lamp
11. Diagnosis lamp
12. Air cleaner warning lamp (OK monitor :Option)
13. Engine coolant level warning lamp (OK monitor :Option)

Integrated monitoring center

Meter illumination lamps are provided for easy meter reading at night. They come on when the light control switch is set to ON.

III. KNOW YOUR TRUCK



Warning lamp check method

1. Please check if all warning lamps come on when the ignition switch is turned to ON.
2. If any lamp does not come on, the lamp may be burned out. Inspect the lamp.

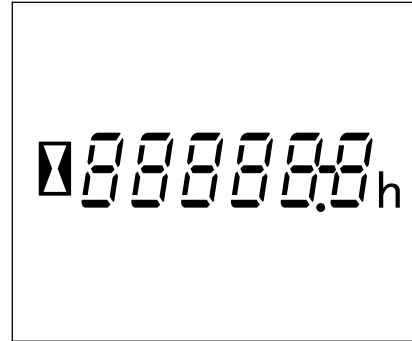
Note

Use the light control switch to check the meter-lighting lamp.

Caution

The glow indicator lamp (1DZ-II engine models) is only on for 2 seconds when the engine coolant temperature exceeds 50°C.

If a lamp does not light up, contact your Toyota dealer to request an inspection.



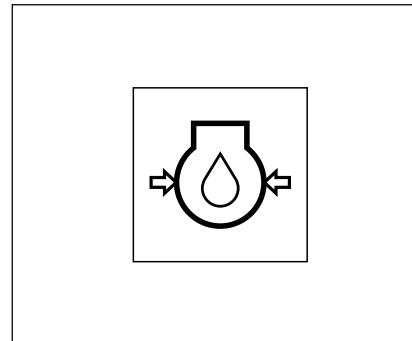
Hour meter also serving as diagnosis indicator

The hour meter operates when the ignition switch is turned to ON. It indicates the total number of truck operating hours. The unit of the right most digit is 1/10 hour.

Please use this meter for the timing of periodic maintenance and recording the operating hours. When an abnormality occurs to the truck (diagnosis lamp lights up or blinks) the error code and hour meter will be alternately displayed.

Caution

Should an error code be displayed, park the truck in a safe location and contact your Toyota dealer to request an inspection.

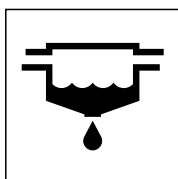


Engine oil pressure warning lamp

1. Comes on to indicate low engine oil pressure while the engine is running.
2. If normal, the lamp comes on when the ignition switch is turned to ON and goes off when the engine starts.
3. If the lamp comes on while the engine is running, either the engine oil is low or the lubrication system is faulty. Stop the operation immediately and ask your Toyota dealer to inspect and repair.

Note

The "engine oil pressure warning lamp" does not indicate the oil level. Check the oil level with the oil level gauge before starting work.



Sediment warning lamp (Diesel engine models)

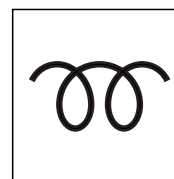
The sedimenter is a device for separating water from the fuel:

1. The warning lamp comes on to indicate water in the sedimenter exceeds the predetermined level while the engine is running.
2. If normal, the lamp comes on when the ignition switch is turned to ON, and goes off when the engine starts.
3. If the lamp comes on while the engine is running, drain water immediately.

(See the Self-Service Section for the draining instructions.)

Caution

Continued operation with the lamp on may cause seizure of the injection pump and pump damage. If the warning lamp lights up, always make sure to drain the water.



Glow indicator lamp (Diesel engine models)

Indicates heating of the glow plugs. When the ignition switch is turned to ON, the lamp comes on and glow plug heating begins. The lamp goes off automatically when glow plug heating is complete.

The engine will start easily once the glow plugs are heated.

Note

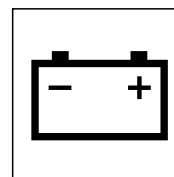
The glow indicator lamp is on for 2 seconds when the engine coolant temperature exceeds 50°C (122°F).

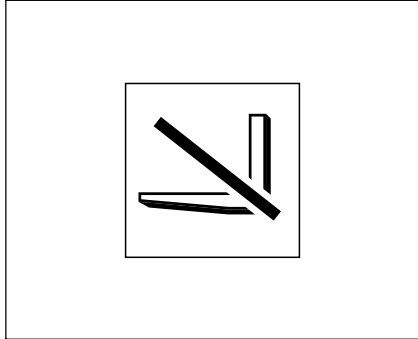
Caution

If the indicator lamp does not go off the glow plugs may be defective. Please ask your Toyota dealer to inspect and repair.

Charge warning lamp

1. This lamp comes on to indicate a problem in the charging system while the engine is running.
2. If normal, the lamp comes on when the ignition switch is turned to ON and goes off when the engine starts.
3. If the lamp comes on while the engine is running, stop immediately, park the truck in a safe location, stop the engine and after the engine has cooled down inspect the fan belt for cuts or looseness, adjust it, and restart the engine. If lamp does not go off, the electrical system may be faulty. Please ask a Toyota dealer immediately for inspection and repair.





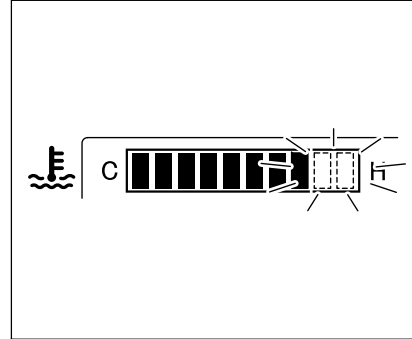
OPSS lamp

This lamp comes on when the operator leaves the seat, in order to inform the operator that the OPSS is going to be activated. If this lamp comes on, and OPSS is activated, return the direction control lever, lift and tilt levers and accelerator pedal to their neutral positions and return to the seat.

Caution

In the following cases, a malfunction may have occurred to the OPSS. Park the truck in a safe location and contact your Toyota dealer to request an inspection.

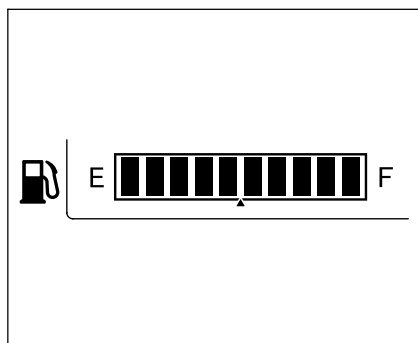
- If the operator leaves the seat, the OPSS lamp does not light up.
- Even when the operator is re-seated, the OPSS lamp does not turn off.



Water temperature gauge

Indicates the temperature of the engine coolant.

1. This gauge functions with the ignition switch ON, and displays the coolant water temperature from left to right in a 10-stage gradation scale.
2. The operator will be notified when the water temperature is 115°C (239°F) or over (above 8th stage), as the final two stages at far right begin blinking. Again, when the engine protection function activates (for trucks with multifunction display: option), the entire gauge will start blinking to inform the operator.
3. Temporary overheating may be caused by coolant leakage, low engine coolant level, loosened fan belt, or other problem in the cooling system. Contact your Toyota dealer to request an inspection.



Fuel gauge

(Excluding LPG and CNG models)

Indicates the amount of remaining fuel in the fuel tank in terms of a 10-stage gradation scale. The operator will be notified that remaining fuel level is low when the two stages at far left start blinking.

It takes some time for the indicator to stabilize after refuelling and the ignition switch is turned to ON.

Note

- If the operating area is not level, attention must be paid because the correct level may not be indicated.
- When the gauge begins blinking, refuel as soon as possible.
- In case of diesel engine, be sure to refuel it before it runs out because when this occurs it becomes necessary to bleed air from the fuel supply system.

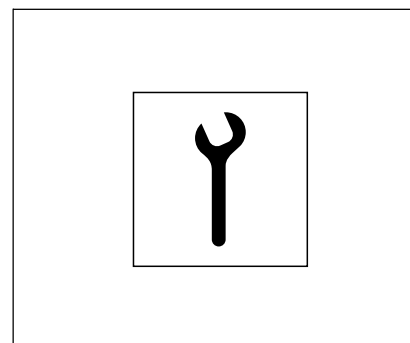
Diesel fuel

Recommendation

Use diesel fuel with a cetane index of 40 or more and a sulfur content of not more than 500 ppm that is based on American diesel fuel standard ASTM D975/03.

Caution

- **Not for use with bio-diesel fuel. Damage to engine will occur.**
- **In winter, use winter diesel fuel to prevent clogging of the fuel filter caused by paraffin precipitation.**



Diagnosis lamp

If an abnormality is registered by the SAS, OPSS, 4Y engine ECU, mini lever, or truck speed limiter, the respective lamps will light up or blink to inform the operator and the diagnosis error code will be displayed in the hour meter display area.

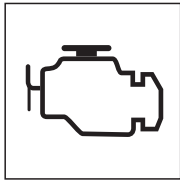
If the following conditions occur to the lamp, there may be a system abnormality. Contact your Toyota dealer to request an inspection.

- The lamp does not light up when the ignition switch is turned ON.
- The lamp lights up when ignition switch is turned ON and stays ON.
- The lamp blinks during travel (operation).

Caution

- **Continuing to use the truck while the diagnosis lamp is lighted or blinking may lead to a breakdown. When the lamp lights up or blinks, halt all operations and park the truck in a safe location. Contact your Toyota dealer to request an inspection. (In the case of diesel engine trucks, the diagnosis lamp may light up during engine warm-up after a cold-start, but does not indicate an abnormality.)**
- **If the operator remains seated for a long period with the ignition switch turned to OFF, the next time the ignition switch is turned to ON, the diagnosis lamp may start blinking. If this occurs, turn the ignition switch to OFF, return to the normal seated position and then turn the ignition switch back to ON. The diagnosis lamp will then turn off.**

III. KNOW YOUR TRUCK

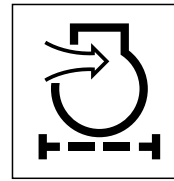


Engine check lamp (Gasoline, LPG and CNG engine models)

1. When an error occurs to the engine controller, the display will light up to inform the operator.
2. When condition is normal, the lamp will light up when the ignition switch is turned to the ON position. The lamp will turn off when the engine is started.

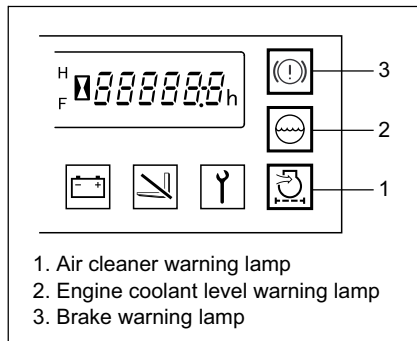
Caution

Once the engine check lamp has come on during the operation, discontinue the operation in progress and park the truck at a safe location. Then, ask a Toyota dealer for an inspection.



1. Air cleaner warning lamp

1. This lamp comes on when the air cleaner element gets clogged while engine is running.
2. If normal, the lamp comes on when the ignition switch is turned to ON and goes off when the engine starts.
3. If the lamp comes on while the engine is running, stop operations and park the truck in a safe location, stop the engine and clean the element and dust cup. For the cleaning method, refer to the Weekly Inspection Section in this manual.

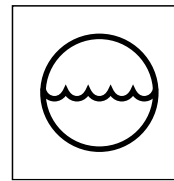


OK monitor (Option)

Senses the fuel level, engine coolant level, battery fluid level, and parking brake status. The lamp comes on to indicate a problem. If the lamp comes on when the ignition switch is on (irrespective of the engine speed), the corresponding part may be abnormal. Contact your Toyota dealer to request an inspection.

Caution

Always perform pre-operation checks. Do not rely on the OK monitor, even if it is not lit.



2. Engine coolant level warning lamp

1. Comes on when the engine coolant level in the radiator reservoir tank is too low.
2. When the lamp comes on, add engine coolant to the upper level in the reservoir tank.
(See the Engine Coolant Inspection Section for the method of adding coolant.)

Note

Even if the coolant level warning lamp is not lighted, always inspect the coolant level before starting operations.

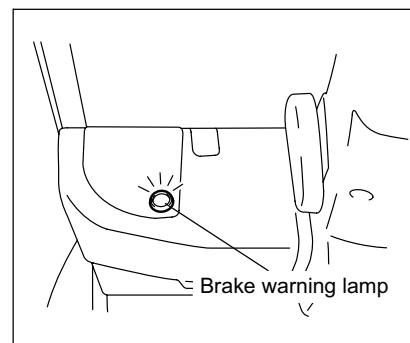


3. Brake warning lamp

1. Comes on in any of the following cases:
 - When the parking brake is applied. (It goes off when the brake is released.)
 - When the brake fluid is below the specified level.
2. If the lamp remains on even after the parking brake is released, inspect the brake fluid level. (See the Brake Fluid Inspection section for the method of adding brake fluid.)
3. If the lamp comes on while the brake fluid level is proper, ask your Toyota dealer to inspect.

Caution

Before starting the truck, always release the parking brake lever and check that the lamp goes off.



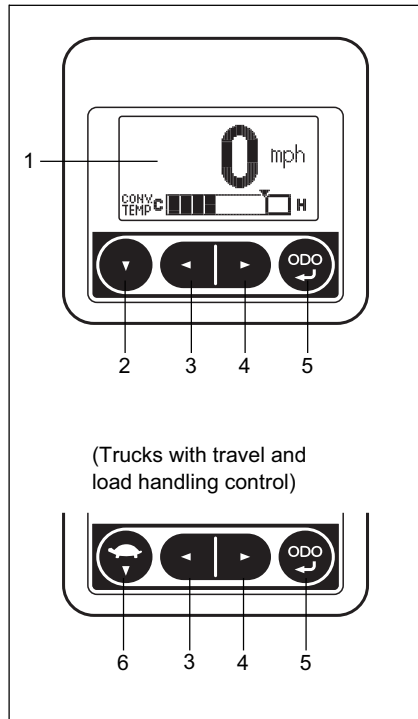
Brake warning lamp (Option)

If the optional wet brake function is abnormal, the warning lamp will light and warning buzzer will sound to inform operator. When the warning lamp lights and buzzer sounds, cease work operation and stop the truck in a safe place. Securely set the parking brake and fully depress the brake pedal until buzzer stops with the engine still running. After confirming that buzzer is turned off and warning lamp goes off, restart the work operation.

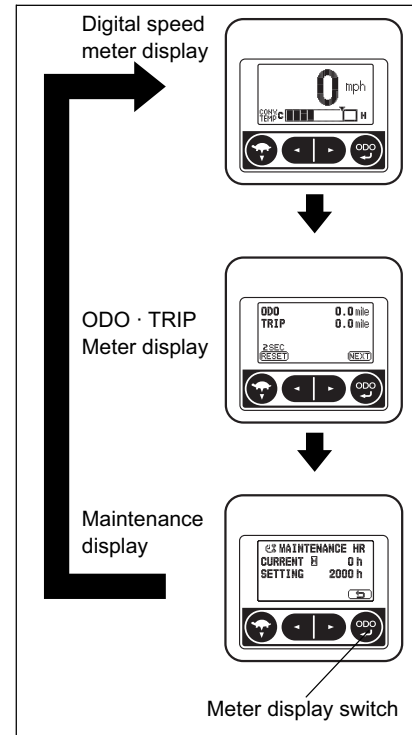
Caution

If the warning lamp does not go off or buzzer does not stop even when the brake pedal is depressed, ask a Toyota dealer for an inspection.

MULTIFUNCTION DISPLAY DX (Option)



1. Multifunction display area
2. Down switch
3. Left switch
4. Right switch
5. Meter display switch
6. Low speed setting switch (only for trucks with travel and load handling control)

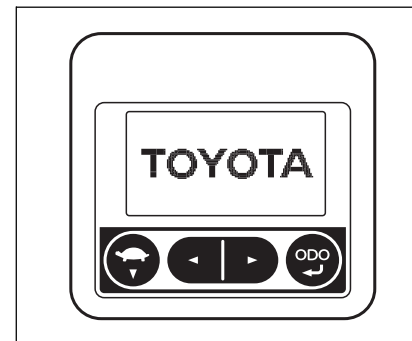


DISPLAY SCREEN TABLE

From the standard screen display, pressing the meter display switch will change the display between ODO, TRIP, and Maintenance Hour Meter display screens.

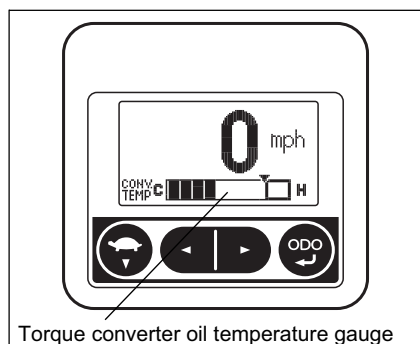
Note

Always operate the switch panel with your finger tips and when the truck is stopped.

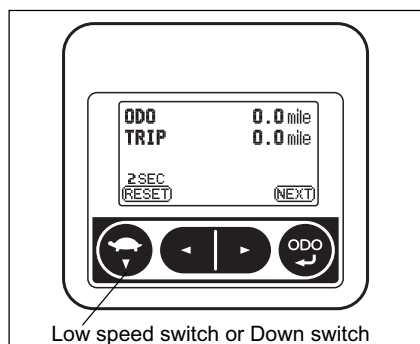


Start screen

When the ignition switch is turned ON, the start screen will appear for 1 second.



Torque converter oil temperature gauge



Low speed switch or Down switch

Standard screen

Truck speed is digitally indicated in mph at the top of the screen. At the bottom of the screen, torque converter oil temperature is indicated in a 10-stage level.

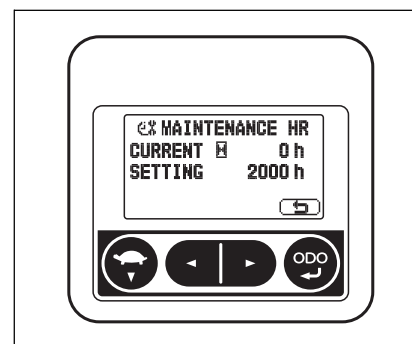
ODO - TRIP Meter

ODO Displays the total distance traveled.

TRIP After resetting this function, displays the total distance traveled until now.

Note

- Pressing the low speed setting switch (truck with travel and load handling control) or the down switch for more than 2 seconds will reset the total travel distance.
- Always operate the switch panel with your finger tips and when the truck is stopped.



Maintenance Hour Meter

Displays the set value and current value of the maintenance hour meter.

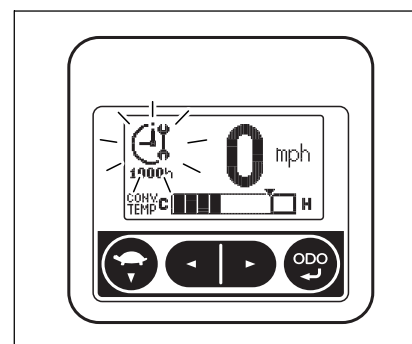
Current Displays the current time.

Setting Displays the maintenance hour time setting.

The maintenance hour meter time setting can be set between 10-2000 hours. 10-200 hours setting can be set in 10 hours intervals, and 200-2000 hours setting can be set in 50 hours intervals.

Note

To change the time setting, consult your supervisor or Toyota dealer.

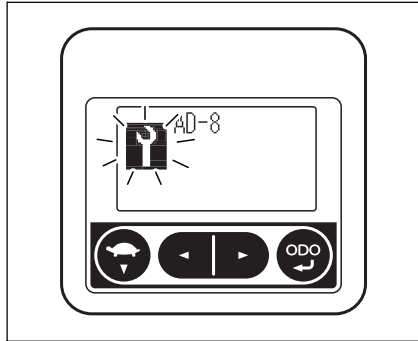


Maintenance indicator

When the reset maintenance time occurs, the operator will be notified through the display and a warning sound. Conduct necessary maintenance.

Note

The maintenance time setting should be determined by the supervisor. For information on setting the maintenance time, contact your supervisor or Toyota dealer.

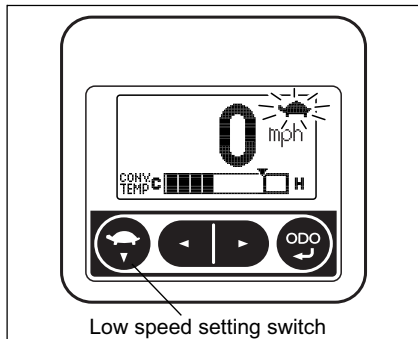


Diagnosis indicator

If an abnormality is registered within the multifunction display, the operator is notified through a warning sound together with diagnosis error content display.

Note

- The error code display will be different depending on the abnormal area and nature of the abnormality. There are also occasions when the error is not indicated depending on the area of abnormality.
- When the diagnosis indicator is displayed, contact your Toyota dealer to request an inspection.



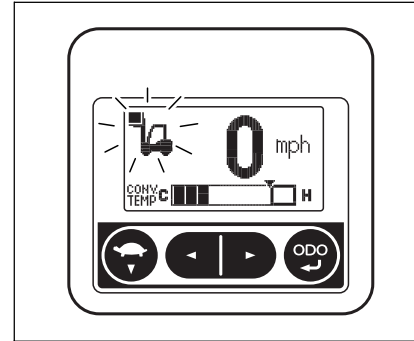
Low speed setting indicator (Only for trucks with travel and load handling control)

When low speed settings have been set, the turtle mark is displayed.

Each time the low speed setting switch is pressed, the turtle indicator will blink. When the indicator is lighted, the low speed setting control is active.

Note

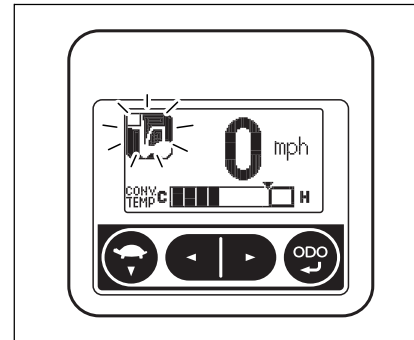
Always operate the switch panel with your finger tips.



Travel and load handling control indicator

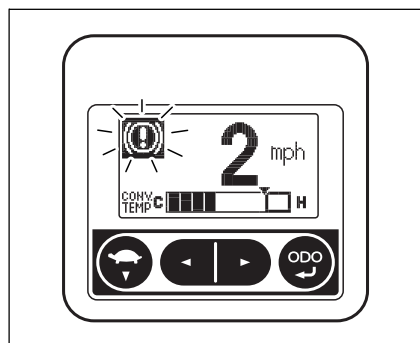
(Only for trucks with travel and load handling control)

When the speed limiting sensor or the load height sensor, to prevent sudden acceleration, is activated, the indicator will warn the operator.



Function On/Off indicator (Only for trucks with travel and load handling control)

Even when the speed limiting load height sensor is deactivated, the operator is notified via display indication that the preventing sudden starts function is active.



Parking brake on warning

When the parking brake is engaged while the lift truck is traveling, the warning indicator will blink and a warning will sound to notify the operator.

Caution

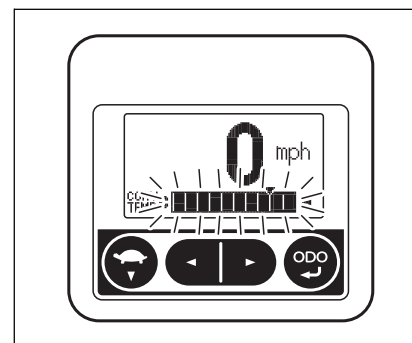
- If the truck is operated without disengaging the parking brake, the brake will lose effectiveness. Contact your Toyota dealer to request an inspection.
- If the indicator lamp does not turn off even when the parking brake is disengaged, stop operations and contact your Toyota dealer to request inspection.

Parking brake off warning

When the ignition switch is turned to OFF or the operator leaves the seat while the parking brake is disengaged, a warning sound will warn the operator to apply the park brake. The warning will also be sounded if the operator returns to the normal seated position and turns the ignition switch to ON while the parking brake is disengaged.

Note

When leaving the driver's seat, always shift to NEUTRAL, set the parking brake, lower the forks, and tilt forward so the tips will not trip pedestrians, and turn off the ignition switch.

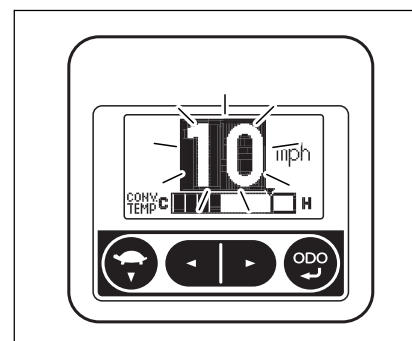


Torque converter oil temperature overheating warning

When the torque converter oil temperature reaches level 9 on the indicator [120°C (248°F) or over], the indicator will blink to notify the operator. When the indicator reaches level 10 [140°C (284°F) or over], the entire indicator will blink to notify the operator.

Note

When the warning lamp blinks, park the truck in a safe location, engage the parking brake, open the engine hood with the motor idling, and cool the torque converter oil.



Over-speed alarm

When the set traveling speed is exceeded, the speed meter will blink and a warning will sound to notify the operator.

Note

This function does not limit traveling speed. Pay attention to your speed when operating the truck.

III. KNOW YOUR TRUCK

Setting menu screen

With the multifunction display, pressing the low speed setting switch or the down switch in the standard screen for more than 2 seconds will display the setting menu screen.

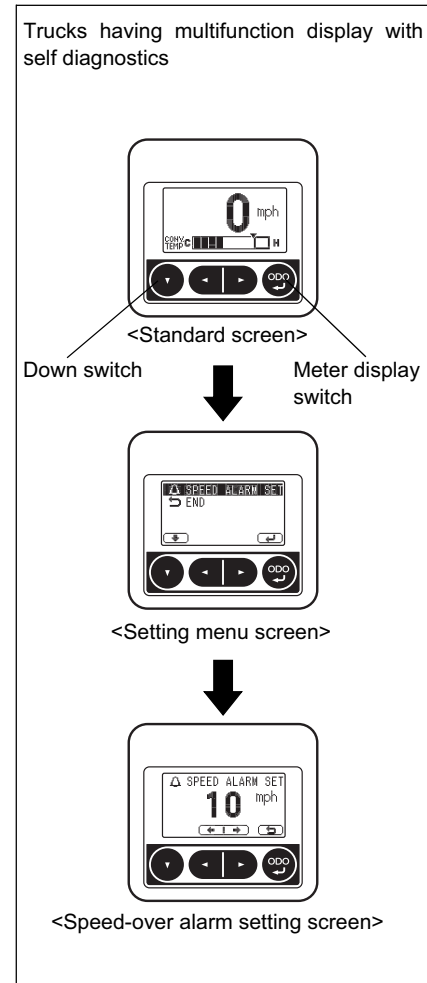
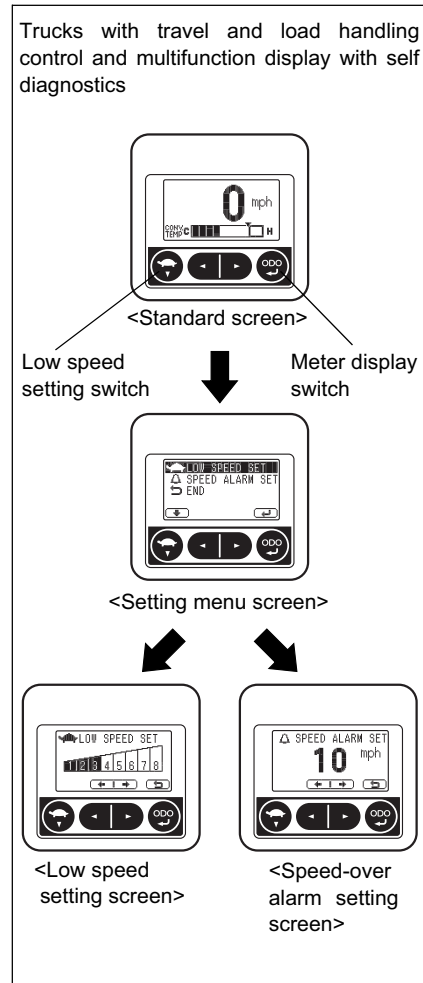
Note

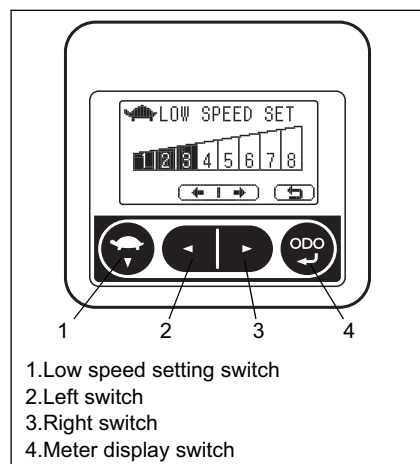
When the supervisor unlocks the menu, these screen settings can be displayed.

Selecting the options using the low speed setting switch or the down switch and then pressing the meter display switch will display each screen settings. Select [END] in the setting menu screen, and then pressing the meter display switch will return to the standard screen.

Note

Always operate the switch panel using your finger tips.





Low speed setting screen

When the low speed settings are activated, a range of 8 maximum speed settings can be established.

Selecting the level 8 setting will turn off the function.

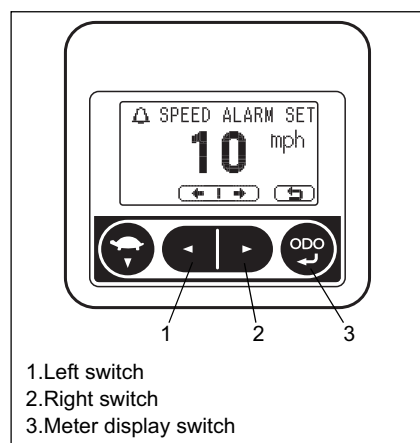
Left switch.....Decreases the setting level

Right switchIncreases the setting level

Meter display switch.....Switches to the menu screen

Note

- If level 8 is selected, the setting cannot be changed by pressing the low speed setting switch or the down switch in the standard screen.
- Always operate the switch panel using your finger tips.



Speed-over alarm setting screen

This function allows you to set the traveling speed which will set off the alarm.

Left switch.....Reduces travel speed

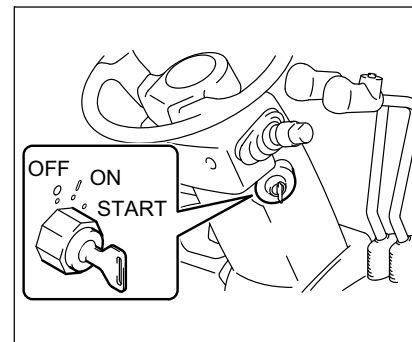
Right switch Increases travel speed

Meter display switch.....Switches to the menu screen

Note

Always operate the switch panel using your finger tips.

SWITCHES AND LEVERS



Ignition switch

OFF --- Engine stop position. Key insertion and withdrawal are performed in this position.

ON --- Engine operation position. Located one position clockwise from the OFF position.

The glow plug light must go off before starting the diesel model.

START --- Engine start position. Located one position clockwise from the ON position.

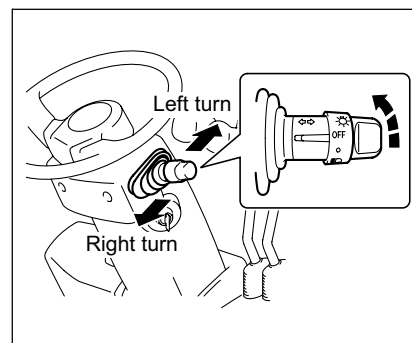
After engine start, release the key and it will return to the ON position automatically.

The engine does not start unless the control lever is in the NEUTRAL position.

WARNING

- Always start the engine with the control lever in the NEUTRAL position.
- If the OPSS lamp is on, return the direction control and load handling control levers to their neutral positions and sit in the seat. Confirm that the OPSS lamp turns off.
- Do not leave the switch in the ON position when the engine is stopped. It will cause the battery to discharge.
- Do not turn the switch to the START position while the engine is running. This may damage the starter motor.

- Do not operate the starter motor for more than 30 seconds continuously. Return the switch to the OFF position and wait at least 20 seconds prior to attempting restart.
- Your truck has an anti-restart ignition switch to protect the engine. You must turn the switch from the ON position to the OFF position before attempting to start the engine again.
- With the ignition switch OFF (engine stopped), the forks will not move down even if the lift lever is so operated. Do not operate the lift lever before getting on the truck and starting up the engine. (key-lift interlock)
- If the diagnosis lamp does not turn off when you sit on the seat, battery voltage may be low. Do not operate the truck until the lamp turns off, as low voltage may inhibit normal operation. If the diagnosis lamp does not turn off approximately 1 to 2 minutes after engine starts or when the engine speed is increased, stop operation and have the truck inspected by your Toyota dealer. (On diesel engine models, the diagnosis lamp may remain on during warm-up after a cold engine start. This does not indicate a OPSS malfunction.)



This switch serves as both two-position light control and turn signal switch.

Irrespective of a key switch position, this switch allows you to turn on and off lighting.

This switch has two positions. With the switch at each position, the lamp comes on as shown below.

Lamp name	Step1	Step2
Head lamps	—	○
Side clearance lamps, tail lamps (Option)	○	○
Meter illumination lamp	○	○

Caution

Do not keep lamps such as head lamps kept on for a long time when the engine is stopped. It may cause overdischarge of the battery to make engine starting impossible.

Turn signal switch

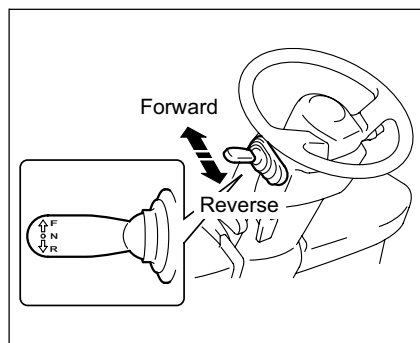
Makes the turn signal lamps blink

Left turn..... Push forward

Right turn Pull backward

The signal switch will operate when the ignition switch is ON.

The turn signal lever returns automatically to the original position after making a direction change.



Control lever

Lever for shifting between forward and reverse.

Forward Push forward

Reverse Pull backward

The neutral position is halfway between the forward and reverse position.

Note

To resume powered travel after activation of the OPSS, return the accelerator pedal and the direction control lever to their neutral positions and sit in the seat so that powered traveling can be resumed. (Sitting in the seat without returning the direction control lever to its neutral position will not allow powered traveling.)

Caution

The engine cannot be started unless the direction control lever is at the neutral position.

Stop the truck before shifting between forward and reverse.

Torque converter interlock function (Intelli-shift option)

If you switch the control lever direction to something other than the current travel direction while moving at high speed, this function electrically disengages the drive and sets the torque converter to neutral. Once the speed drops below the set speed while running in neutral, the travel direction is automatically switched.

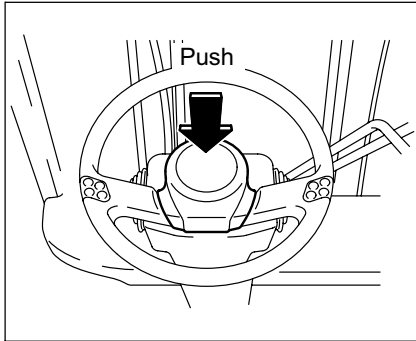
To change travel direction, operate the control lever after travel speed is reduced sufficiently.

Ask your Toyota dealer for changing speed settings.

Caution

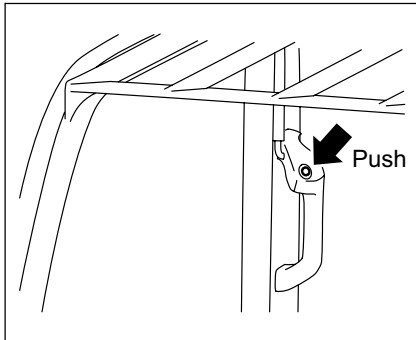
- When the interlock has engaged, release the accelerator pedal and use the brake pedal to reduce speed. After the truck has stopped moving, slowly press the accelerator pedal down to start moving again. Disengaging the interlock while the accelerator pedal is pressed down could result in wheel spin.
- Do not perform Forward or Backward operation on slopes. If control lever is operated on a down slope, torque converter interlock function may not operate correctly.

III. KNOW YOUR TRUCK



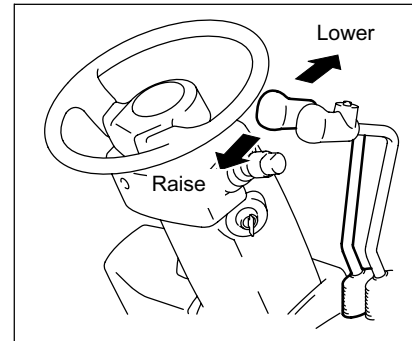
Horn button

Make sure your truck is seen. Press the button in the center of the steering wheel to sound the horn. The horn will sound even when the ignition switch is off.



Horn button (Option)

The horn will sound when the button at the top of the rear pillar assist grip is pressed. Use this horn when backing up. The horn will sound even when the ignition switch is off.



Lift lever

Raises and lowers the forks.

Raise..... Pull backward

Lower..... Push forward

The lifting speed can be adjusted by varying the engine speed and the distance the lever is moved.

The lowering speed can be adjusted only by the distance the lever is moved.

Note

- As soon as the engine is stopped, the forks do not move down even if so operated. However, if you sit in the seat and turn on the ignition switch, you can lower the forks.
- The idling lifting speed increase function (option) automatically increases lifting speed by engaging the lift lever.

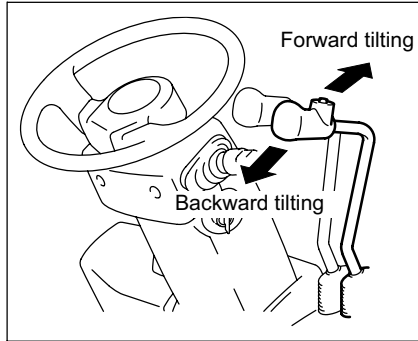
⚠ WARNING

- |||||



When the ignition switch is OFF, the lift will not descend even if the lift lever is lowered. However, if the operator sits properly in the seat and turns the ignition switch ON, the forks can be lowered even if the engine is off (except mini lever / joy stick operated trucks). If the ignition switch cannot be turned ON for whatever reason, loosen the manual lowering valve located on the oil control valve beneath the floor board, and operate the lift lever in the downward direction.

Once the forks have been lowered with the manual down valve, close and lock the valve.



Tilt lever

Tilts the mast forward and backward.

Forward Push forward

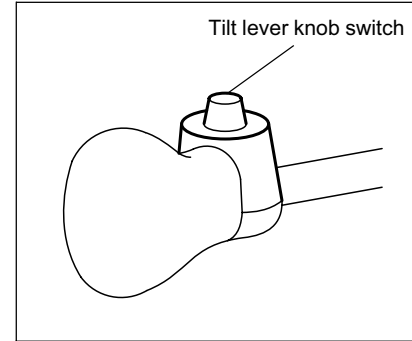
Backward Pull backward

The forward or backward tilting speed can be adjusted by varying the engine speed and distance the lever is moved.



⚠ WARNING

- Insure the load-handling control levers are in their neutral positions before returning to the operator's seat, if not load-handling functions will start movement 1 second after operator returns to the seat.
- Always operate the tilt lever from a seated position.
- After the OPSS operates, return the accelerator pedal to its fully released position and the tilt lever to the neutral position, and sit on the seat to restart the operation. (If you sit on the seat without returning the tilt lever to the neutral position, the forks may resume moving.)



Tilt lever knob switch

Depressing the tilt lever knob switch while changing from backward to forward tilt will automatically stop forks at their horizontal position.

It is also possible to slow down the backward tilt speed at a low lift height by depressing knob switch.

Automatic forks leveling control

With the forks positioned at backward tilt, use the lever to tilt the forks forward while depressing the tilt lever knob. Then, the mast will be automatically stopped with the forks horizontally positioned.

Motion upon change of tilt from backward to forward, with tilt lever knob switch pressed:

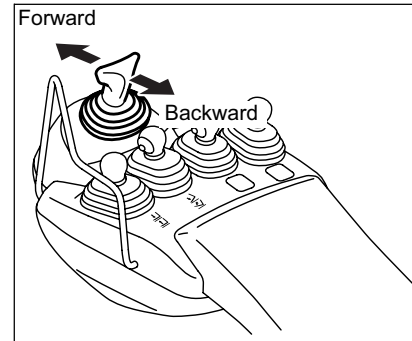
	Not Loaded	Loaded
High Lift	Forks stopped at its horizontal position (with mast vertically positioned)	No forward tilt
Low Lift	Forks stopped at its horizontal position (with mast vertically positioned)	horizontal

Active mast rear tilt speed control

At a high lift height, the mast has a backward tilt speed controller (slow down) irrespective of the load. When lowering from a high lift height to a lower lift height while tilting the mast backward, the control speed will not change.

At a low lift height, the mast can be tilted at full speed even if there is a load. If the mast is tilted backward at a low lift height with the tilt knob switch pressed, the mast has the backward tilt speed controlled (slowed down) as long as the tilt lever knob switch is pressed.

If the low lift height is then changed to a high lift height while tilting the mast backward, the control speed will not change as long as the tilt lever knob switch is pressed. The mast may be tilted backward at the full speed so long as the tilt lever knob switch is not pressed.



Mini levers (Option)

Control lever

Lever for shifting between forward and reverse.

Forward Push forward

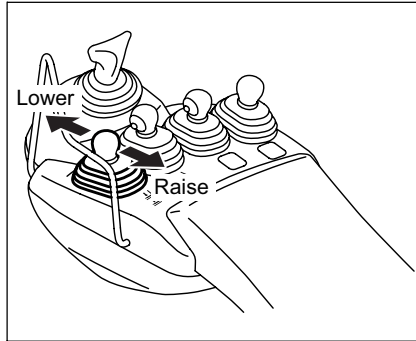
Reverse Pull backward

The speed of forward and backward traveling can be adjusted by the extent of pressing the accelerator pedal.

Note

- Stop the truck before shifting between forward and backward traveling.
- After the OPSS has been activated, return the accelerator pedal and control lever to their neutral positions and return to the seat before recommencing operations.
- Always operate the control lever from a properly seated position.
- Depending on the truck specifications, the position of the control lever may vary.

III. KNOW YOUR TRUCK



Lift lever

Raise and lower the forks for loading.

Raise Pull backward

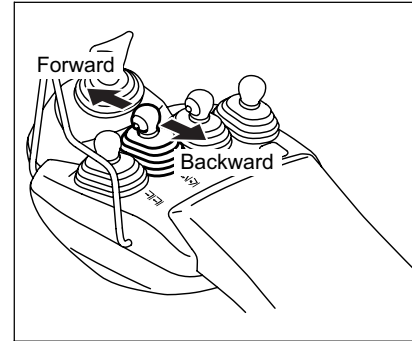
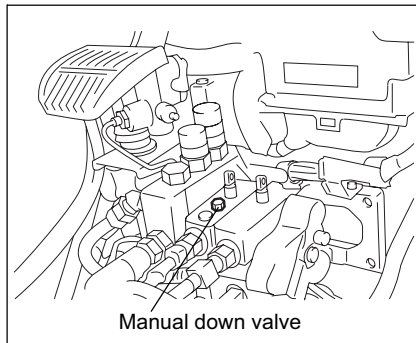
Lower Push forward

Raising speed can be adjusted by the extent of pressing the accelerator pedal and pulling the lift lever.

Lowering speed can be adjusted by the extent of pushing the lift lever.

Note

- After the OPSS has been activated, return the accelerator pedal and all of the levers to their neutral positions and return to the seat before recommencing operations.
- If you return to the seat while lowering the lift lever, the forks will not descend due to the return to neutral function.
- Always operate the lift lever from a properly seated position.
- If you use the idling lifting speed increase function (option), pulling the lift lever will automatically increase the engine speed and raise the forks at a constant speed without having to step on the accelerator pedal.
- When ignition switch is turned OFF, the forks will not descend even if the lift lever is lowered. (key-lift interlock)
- When the forks will not lower due to system malfunction or other reasons, they can be lowered by opening the manual down valve.
- If you lower the forks by opening the manual down valve, close and lock the valve.



Tilt lever

Tilt the mast forward and backward.

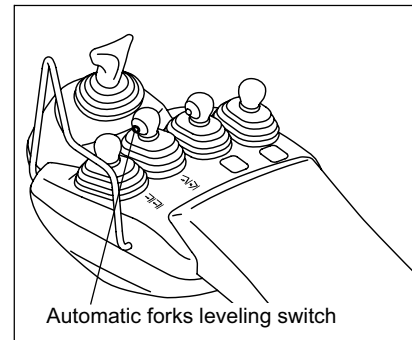
Forward Push forward

Backward Pull backward

Forward or backward tilting speed can be adjusted by the extent of pressing the accelerator pedal and operating the lever.

Note

- After the OPSS has been activated, return the accelerator pedal and all of the levers to their neutral positions and return to the seat before recommencing operations.
- Always operate the tilt lever from a properly seated position.



Automatic fork leveling switch

(Works same as the tilt lever knob switch.)

While pressing the switch when the forks are tilted forward from a backward-tilted position, or backward from a forward-tilted position, the forks will automatically stop at a horizontal position. Releasing the switch will allow for normal tilt lever operation.

Forks forward-tilt automatic leveling control

When the forks are in a backward-tilted position, operating the lever forward while depressing the automatic fork leveling switch will automatically stop the forks at a horizontal position. This function is convenient when handling loads or attaching/detaching forks.

While depressing the automatic fork leveling switch, operating the forks forward from a backward-tilted position:

	Not Loaded	Loaded
High Lift	Forks stopped at their horizontal position (with mast vertically positioned)	No forward tilt
Low Lift	Forks stopped at their horizontal position (with mast vertically positioned)	
Highest position	No forward tilt	

Forks backward-tilt automatic leveling control

When the forks are in a forward-tilted position, operating the lever backward while depressing the automatic fork leveling switch will automatically stop the forks at a horizontal position. This function is convenient when using clamp-type attachments while the mast is tilted forward from the vertical position.

While pressing the automatic forks leveling switch, operating the forks backward from a forward-tilted position:

	Not Loaded	Loaded
High Lift	Forks stopped at their horizontal position (with mast vertically positioned)	
Low Lift		

Active mast rear tilt speed control

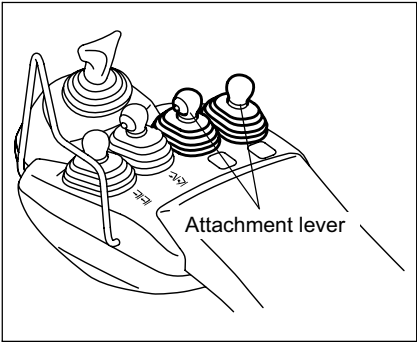
When the mast is raised to a high position, backward-tilt speed will automatically slow down.

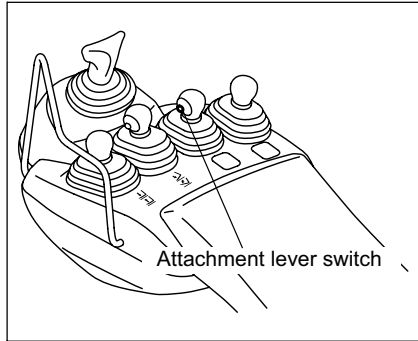
Attachment lever

Operates the attachment. Attachment speed can be adjusted by the extent of pressing the accelerator pedal and operating the lever.

Note

- After the OPSS has been activated, return the accelerator pedal and all of the levers to their neutral positions and return to the seat before recommencing operations.
- Always operate the attachment lever from a properly seated position.



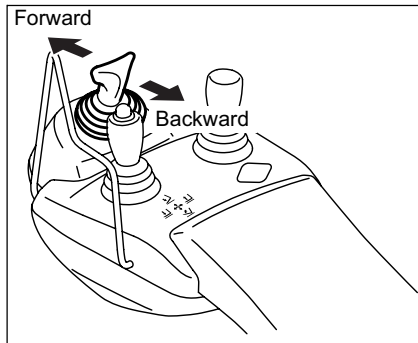


Attachment lever switch (only for 5 ways series)

This switch allows attachment lever operations to be switched between the 3rd and 5th ways. It is for 3rd operation when the changing-over switch is not pushed. And it switches to 5th operation while pushing it.

Caution

Operate the attachment lever switch when attachment operations are stopped.



Joy stick (Option) Control lever

Lever for shifting between forward and reverse.

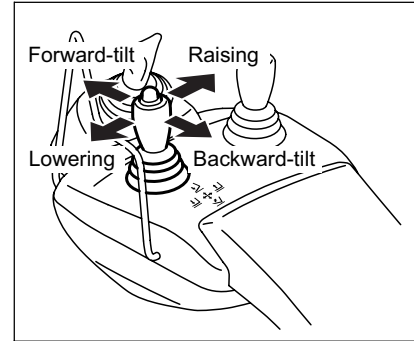
Forward Push forward

Reverse Pull backward

The speed of forward and backward traveling can be adjusted by the extent of pressing the accelerator pedal.

Note

- Stop the truck before shifting between forward and backward traveling.
- After the OPSS has been activated, return the accelerator pedal and control lever to their neutral positions and return to the seat recommencing operations.
- Always operate the control lever from a properly seated position.
- Depending on the truck specifications, the position of the control lever may vary.



Load handling lever

Operation to the left and right controls lift, and forward and backward operation controls tilt.

Raising Operate the lever to the right

Lowering Operate the lever to the left

Forward-tilt Operate the lever forward

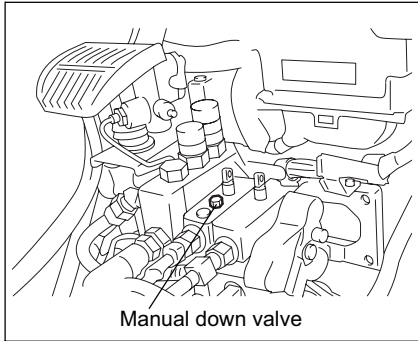
Backward-tilt Operate the lever backward

Raising speed and forward and backward-tilt speed can be adjusted by the extent of pressing down on the accelerator pedal and operating the lever.

Lowering speed can be adjusted by operating the lever.

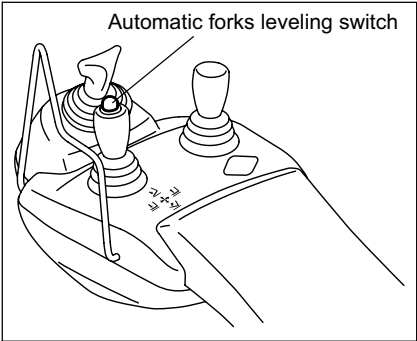
Note

Using the idling lifting speed increase function (option), when the lift tilt lever is raised, the engine's speed is automatically increased without having to press the accelerator pedal, allowing the forks to be raised at a constant speed.



Note

- After the OPSS has been activated, return the accelerator pedal and all of the levers to their neutral positions and return to the seat before recommencing operations.
- If you return to seated position while lowering the lift lever, the lift will not descend due to the return to neutral warning function.
- Always operate the load handling lever from a properly seated position.
- When the ignition switch is turned OFF, the forks will not descend even if the load handling lever is lowered. (key-lift interlock)
- When the forks will not lower due to system malfunction or other reasons, they can be lowered by opening the manual down valve.
- If you lower the forks by opening the manual down valve, close and lock the valve.



Forks automatic leveling switch

While pressing the switch and the forks are operated forward from a backward-tilted position, or backward from a forward-tilted position, the forks will automatically stop at a horizontal position. Releasing the switch will allow for normal tilt lever operation.

Forks forward-tilt automatic leveling control

When the forks are in a backward-tilted position, operating the lever forward while pressing the automatic forks leveling switch will automatically stop the forks at a horizontal position. This function is convenient when handling loads or attaching/detaching forks.

While pressing the automatic forks leveling switch, operating the forks forward from a backward-tilted position:

	Not Loaded	Loaded
High Lift	Forks stopped at their horizontal position (with mast vertically positioned)	No forward tilt
Low Lift	Forks stopped at their horizontal position (with mast vertically positioned)	
Highest position	No forward tilt	

Forks backward-tilt automatic leveling control

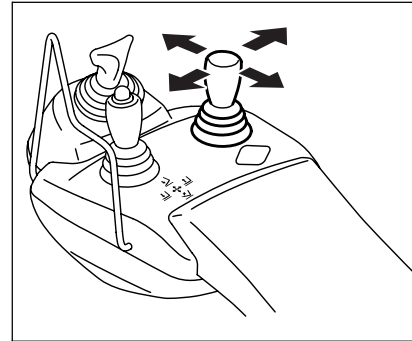
When the forks are in a forward-tilted position, operating the lever backward while depressing the automatic fork leveling switch will automatically stop the forks at a horizontal position. This function is convenient when using clamp-type attachments while the mast is tilted forward from the vertical position.

While pressing the automatic forks leveling switch, operating the forks backward from a forward-tilted position:

	Not Loaded	Loaded
High Lift	Forks stopped at their horizontal position (with mast vertically positioned)	
Low Lift		

Active mast rear tilt speed control

When the mast is raised to a high position, backward-tilt speed will automatically slow down.

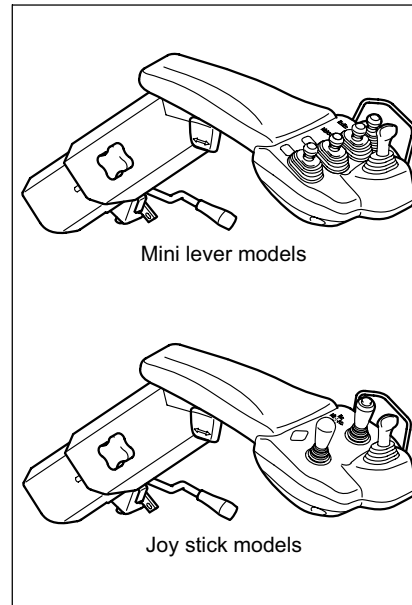


Attachment lever

Operates attachment. Attachment speed can be adjusted by the extent of pressing the accelerator pedal and operating the lever.

Note

- After the OPSS has been activated, return the accelerator pedal and all of the levers to their neutral positions and return to the seat before recommencing operations.
- Operate the attachment lever after you are seated properly in the truck.



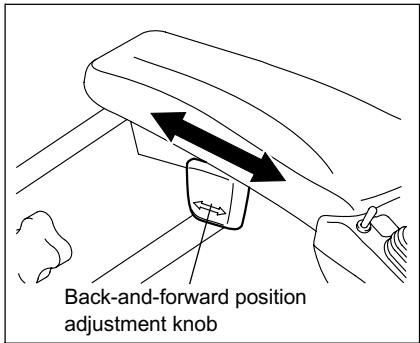
Arm rest

(mini lever and joy stick models)

Before starting up the engine, adjust the arm rest to set the optimum driving position.

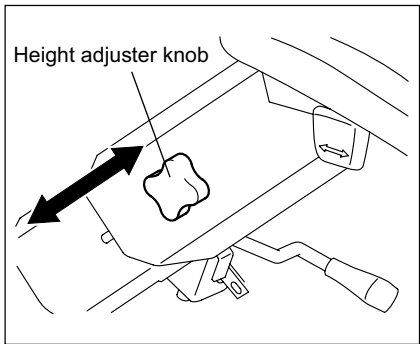
Caution

- After you have finished adjusting the arm rest forward-backward position, height and tilt, confirm that the knob and the lever have been secured in place. A loose knob or lever may cause an accident.
- Do not adjust the position of the arm rest while operating the truck.
- To operate the truck safely, lock the arm rest securely. Before operating the truck, always confirm that the lever for turning and securing the arm rest is locked.



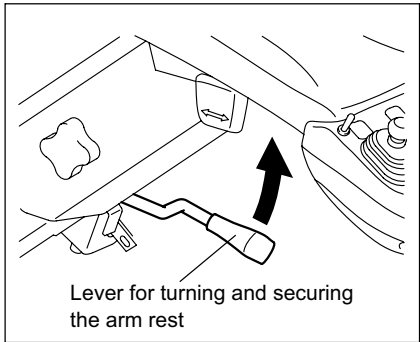
Adjusting the forward-and-backward position

Pull up and loosen the forward-backward position adjusting knob. Adjust the arm rest forward-backward position. Then press the knob, securing it in place.



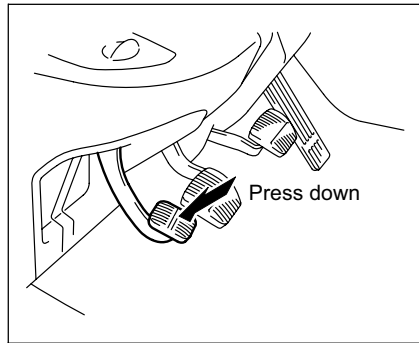
Adjusting the height position

Turn the knob counterclockwise to release the lock. Then, move the arm rest up-and-down to place it to an appropriate position. Turn knob clockwise to lock.



Tilt adjustment

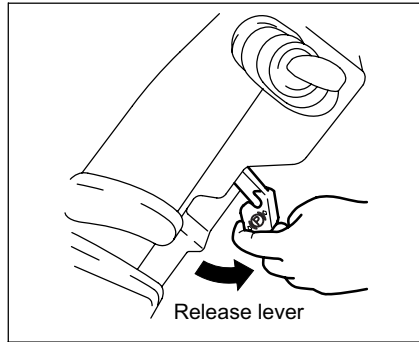
Lift and loosen the lever for turning and securing the arm rest. Adjust the tilt of the arm rest. Then push the lever down, securing it in place. This lever is used in order to turn the arm rest when you open and close the engine hood.



Parking brake pedal

Use the parking brake pedal when parking or stopping.

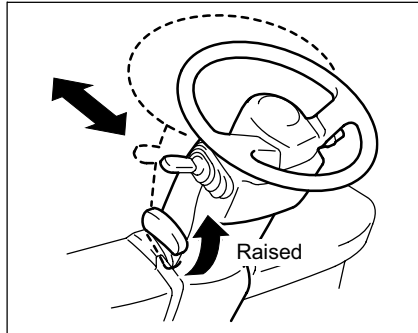
1. When engaging the parking brake, while stepping on the brake pedal, fully press down on the parking brake pedal.
2. To disengage the parking brake pedal, while stepping on the brake pedal, pull the release lever toward you.



⚠ WARNING

- Before operating the parking brake pedal, step on the brake pedal and always confirm that the truck has come to a stop.
- When parking on a slope, apply wheel chocks to the wheels.
- Traveling without releasing the brake will decrease the brake performance.

III. KNOW YOUR TRUCK



Tilt steering adjustment

1. The steering wheel position can be adjusted backward and forward by holding the tilt steering adjust lever up.
2. Lowering the lever at the desired position locks the steering wheel in that position.
3. After adjustment, try to move the steering wheel backward and forward to make sure it is locked.



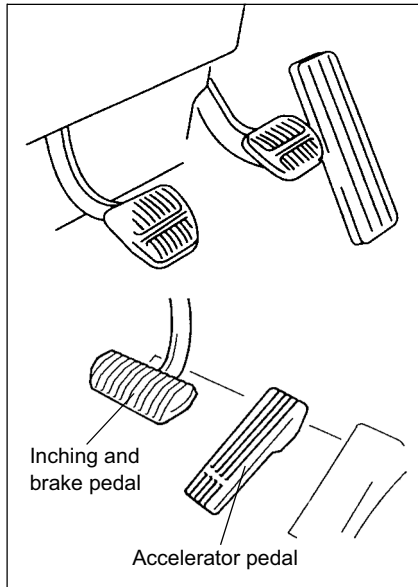
⚠ WARNING

Do not adjust the steering wheel position while the truck is moving.

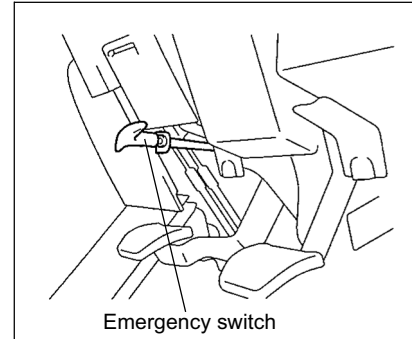


Pedals

From the right: accelerator pedal, brake pedal and inching pedal.



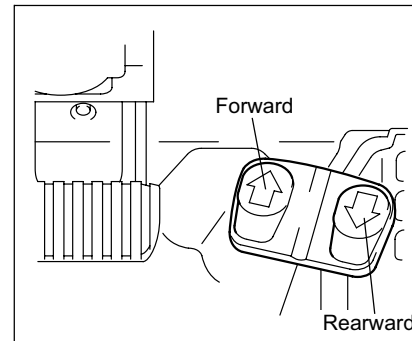
From the right, accelerator pedal and inching and brake pedal.



Emergency switch

(Used only on UL-GS, LPS, and DS types)

This switch cuts off electric current from the battery. Pull this switch in the event of electric circuit problems, short circuiting, or other dangerous conditions, to prevent fire.



EZ pedals (Option)

These pedals are used for foot pedal for travel shifting, and acceleration.
Forward..... Step on the Left Side of the pedal.

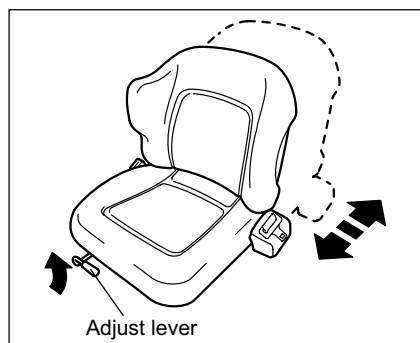
Rearward..... Step on the Right Side of the pedal.

Each pedal functions as accelerator pedal. Speed is adjustable by depth of depression.

Note

When OPSS has been activated, it can be released by sitting on the operator's seat correctly and depressing EZ pedal or by depressing parking brake pedal once and then releasing it.

BODY COMPONENTS



Operator's seat

The seat position can be adjusted back and forth by pulling the adjusting lever to the right.

After adjustment, lightly shake the seat back and forth to make sure it is locked.



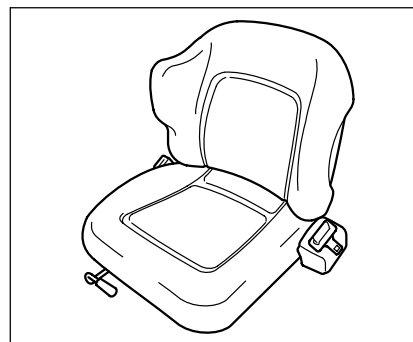
⚠ WARNING

- Never adjust your seating position while the truck is moving.
- Always move the seat to the front position before opening the engine hood to prevent interference with the radiator cover.



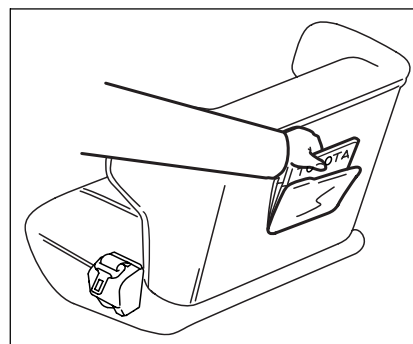
Caution

- The OPSS seat switch prevents powered traveling and load handling operations when the operator is not seated in the seat. Be sure to remain seated in the seat while in operation. Do not operate the truck with any objects placed on the seat.
- The OPSS seat switch should be activated only by the operator sitting on the seat.



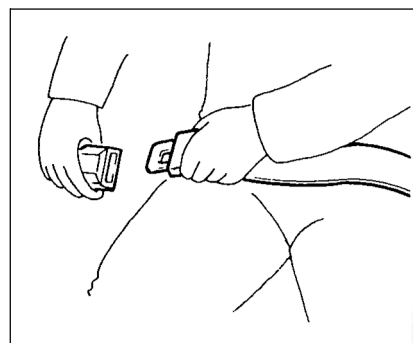
Operator Restraint System

A specially designed operator's seat and seat belt are provided for your safety. Get in the habit of using the seat belt whenever you sit on the trucks.



An Operator's and Owner's Manual is located behind the seat.

If your truck does not have a manual, contact your Toyota dealer to obtain a copy.



Seat belt

To fasten your seat belt, pull it out of the retractor and insert the tab into the buckle.

You will hear a click when the tab locks into the buckle. Pull on the belt to make sure the buckle is securely latched.

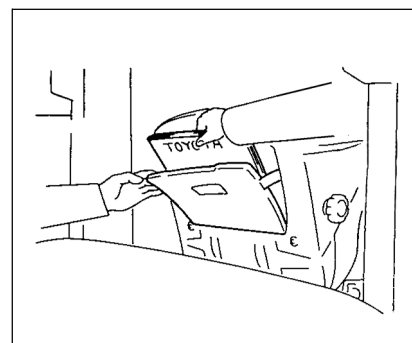
The seat belt length automatically adjusts to your size.

4. Lumbar adjust knob

Use this adjust knob to adjust the lumbar support. Turning the adjust knob clockwise ((+) direction) increases the supporting power and counterclockwise ((-) direction) decreases it.

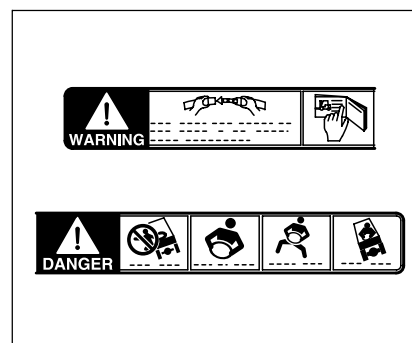
Caution

- The OPSS seat switch prevents powered traveling and load handling operations when the operator is not seated in the seat. Be sure to remain seated on the seat while in operation. Do not operate the truck with any objects placed on the seat.
- The OPSS seat switch should be activated only by the operator sitting on the seat.



An Operator's and Owner's Manual is located behind the seat.

If your truck does not have a manual, contact your Toyota dealer to obtain a copy.



⚠ WARNING

- Never adjust your seating position while the truck is moving.
- Always move the seat to the front position before opening the engine hood to prevent interference with the radiator cover.

⚠ WARNING

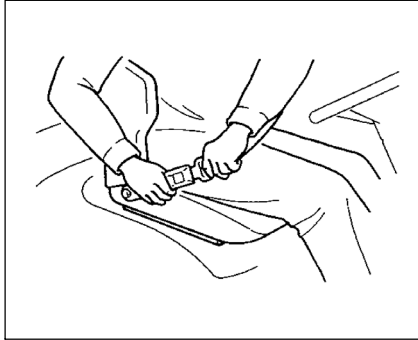
Always wear your seat belt when driving the truck. The truck can tip over if operated improperly.

To protect operators from the risk of serious injury or death in the event of a tipover, it is best to be held securely in the seat.

The seat and seat belt will help to keep you safely within the truck and operator's compartment.

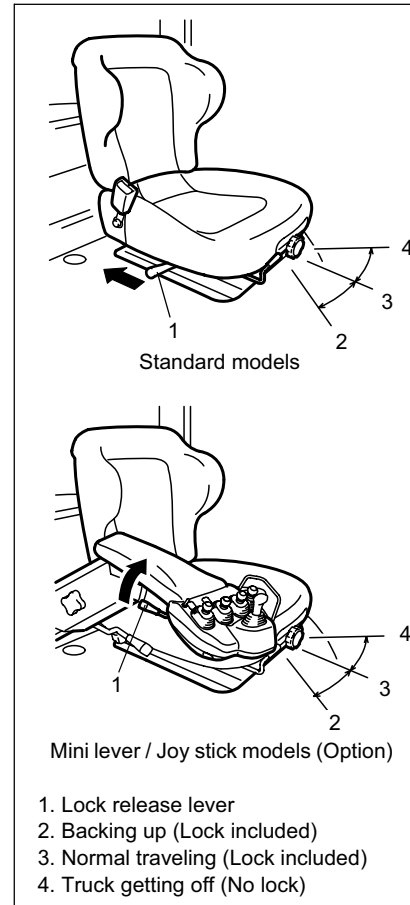
In the event of a tipover, don't jump, grip the steering wheel, brace your feet, lean away from the direction of tipover, and stay with the truck

III. KNOW YOUR TRUCK



Disconnecting method

Push the release button and allow the belt to retract.



Swivel seat (Option)

This rotating seat is useful when backing up over long distances or when getting off from the truck.

Backing up (rotation to the right)

1. For standard models, pull the release lever backward to release the lock.
For mini lever and joy stick models (option) pull the lock release lever upward to release the lock.

Note

Let go of the lock release lever once the seat starts to rotate.

2. Rotate the seat to the right and lock the seat.
3. After backing up, return the seat to normal position.

Getting off from the truck (rotation to the left)

1. For standard models, pull the lock release lever backward to release the lock.
For mini lever and joy stick models (option), pull the lock release lever upward to release the lock.

Note

Let go of the lock release lever once the seat starts to rotate.

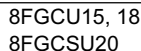
2. Rotate the seat to the left when getting off the truck. The seat will not lock into place when dismounting truck.

- When rotating the seat, be careful not to get your hand caught between the seat and cab.
- After using this function, return the seat to normal position and confirm that the seat is locked in place.
- While operating the truck forward or backward, make sure that the seat is securely locked in a normal operating position.
- To prevent accidents, do not rotate the seat while operating the truck.
- The seat will not lock into place when dismounting the truck.

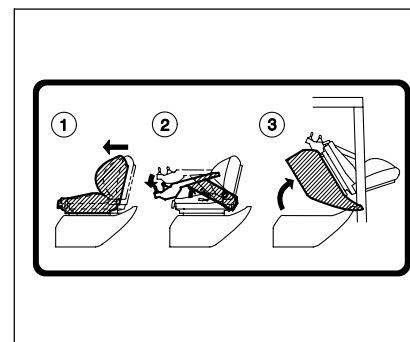
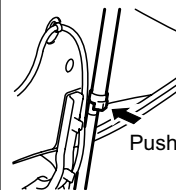


Opening

1. Pulling up on the engine hood lock release lever will release the engine hood lock, and the engine hood will pop up slightly.
2. Lift the engine hood.
3. Open the engine hood all the way, then shake the hood slightly to check that the hood damper has been securely fastened before letting go.



Except
8FGCU15, 18
8FGCSU20



⚠ WARNING

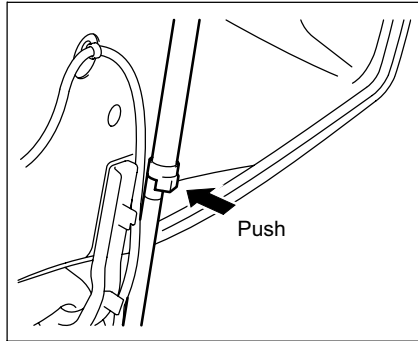
[illegible]

1. Lift up the engine hood and press the hood damper lock to release the lock.
2. Close the engine hood quietly, and press down on the hood until you hear a clicking sound.

(Mini lever / Joy stick models:
Option)

1. Pull up the slide lever and slide the seat to the forward-most position.
2. Pull up on the arm rest forward-backward angular adjustment lever, and after tilting the arm rests downward, lower the forward-backward angular adjustment lever and again lock it in place.
3. Pull up on the engine hood lock release lever to release the lock. The engine hood will slightly pop up and lift up the engine hood.
4. Open the engine hood all the way, shake the hood to check that the hood damper is engaged before letting go.

III. KNOW YOUR TRUCK

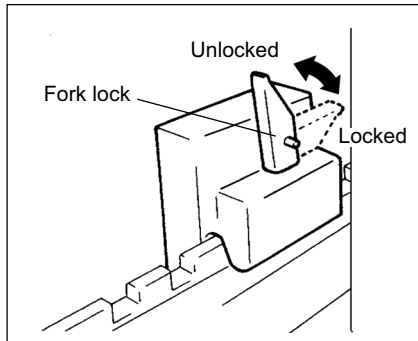


⚠ WARNING

Working on the engine without firm locking of the hood may be hazardous.

Closing

1. Lift up the engine hood, press the lock release button to release the lock.
2. Close the hood quietly, and press down on the hood until you hear a clicking sound.
3. Return the seat and arm rests to their normal position.

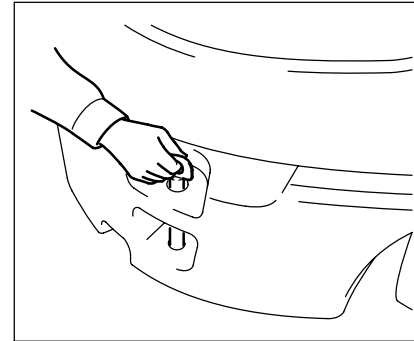


Forks

Lift each fork lock so that the forks can be shifted left and right. Adjust the forks in the position most appropriate for the load and as wide as possible. When adjusting the forks, make sure that the weight of the load is centered on the truck. After adjustment, set the fork locks to keep the forks in place.

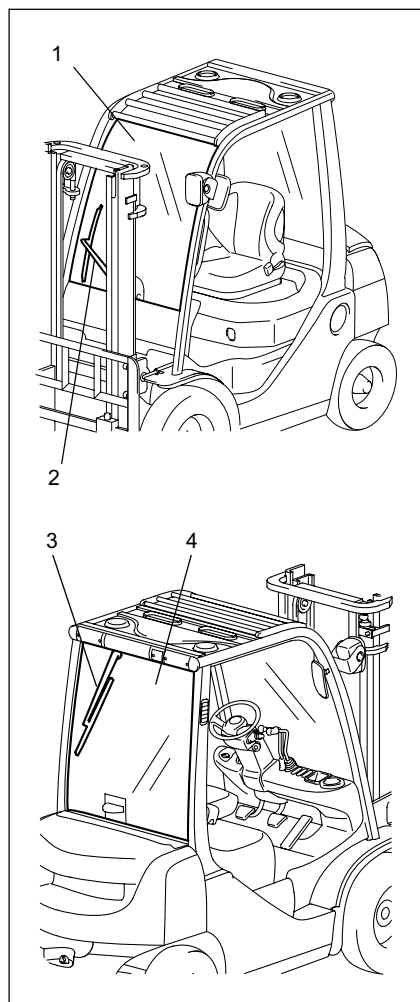
⚠ WARNING

Make sure the forks are locked before carrying a load.



Draw bar

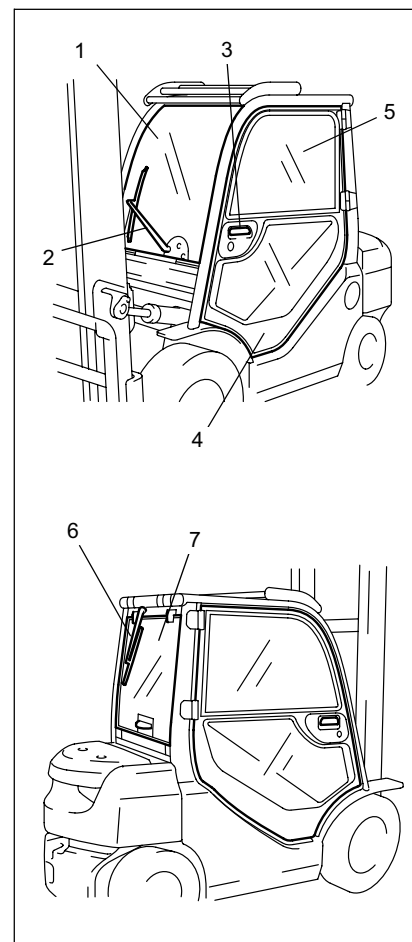
The draw bar is located at the back of the counterweight, and is used to pull the truck should its tires drop into a gutter or become stuck in mud. It can also be used for loading the forklift onto a truck or another truck. The draw bar should not be used for towing the forklift or for towing another truck using the forklift.



Using the cabin (Option)

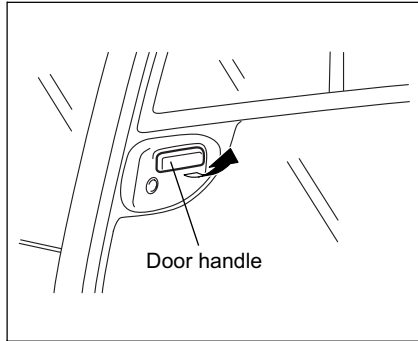
Steel cabin models

1. Front glass
2. Front wiper
3. Rear wiper
4. Rear window



Premium cabin models

1. Front glass
2. Front wiper
3. Door handle
4. Front door
5. Side door window
6. Rear wiper
7. Rear window



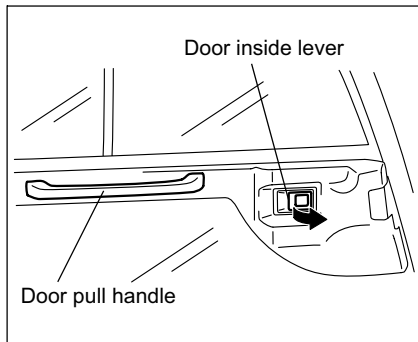
Opening/closing doors (Premium cabin models)

Outside the truck

1. Grasp the door handle and pull toward you to release the lock and open the door.
2. When closing the door, press until the door lock catches.

Note

To open the engine hood for cabin type models, first open the cabin doors to the right and left.

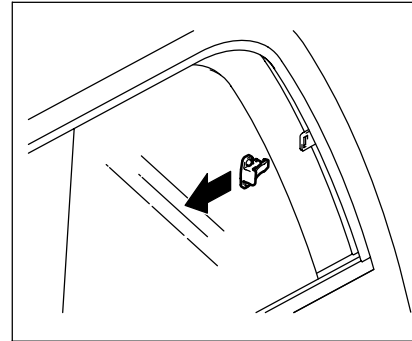


Inside the truck

1. Pulling the door inside lever toward you will release the lock and open the door.
2. Open the door using the door pull handle.

Caution

- When opening doors, be aware of pedestrians or other trucks.
- When closing the door, make sure to use the door pull handle. Before operating the truck, confirm that the doors are securely shut.

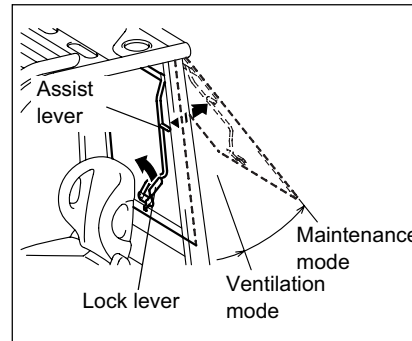


Opening/closing side door window (Premium cabin models)

(Premium cabin models)

The side windows can be opened in the right-left direction.

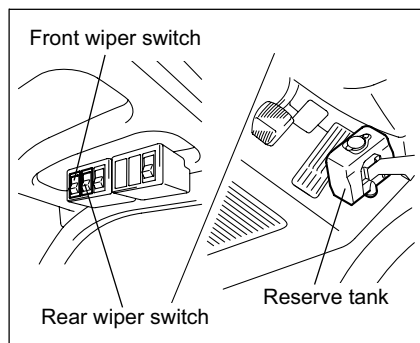
1. Grasping the knob at the center of the window will release the lock. Open the window to the left or right.
2. To close the side windows, use the knobs to slide the window to the right or left.



Opening/closing rear window

The rear window is a flip-up style window which can be set in two stages - to ventilation mode or maintenance mode.

1. Releasing the lock lever at the bottom of the window will open the window. Gripping the assist lever and pushing the window back will open the window to ventilation mode.
2. Pushing the assist lever further back to extend the damper, the window will be opened to maintenance mode.
3. To close the rear window, grip the assist lever and pull on the rear window until it closes completely, then operate the bottom lock lever to lock position.

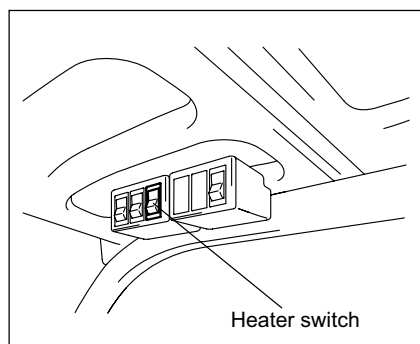


Operating the wiper

The wiper will operate by turning the front or rear wiper switch to the right of the head guard to the ON position. Press the washer button at the bottom of the front wiper switch to release wiper fluid.

Note

To inspect or replenish washer fluid, use the reserve tank located to the right of the driver's seat.



Using the heater

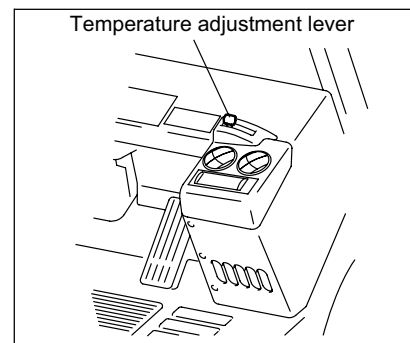
(Premium cabin models)

The heater is attached to the step region on the right-side of the driver's seat. The heater switch is located at the right-side of the head guard.

The heater switch can be set to Hi or Low which will operate the heater in two air volumes. The air outlet can be opened or closed and the entry of debris and dust into the heater unit can be prevented.

Note

- Use the heater after sufficiently warming up the engine.
- Running the heater fan for extended periods of time while the engine is stopped or while idling the motor may cause the battery to run out.
- Using the heater for extended periods of time will cause the air inside the cabin to become stale and the glass to fog, so take care to open windows and ventilate the interior.



Temperature adjustment lever

(Premium cabin models)

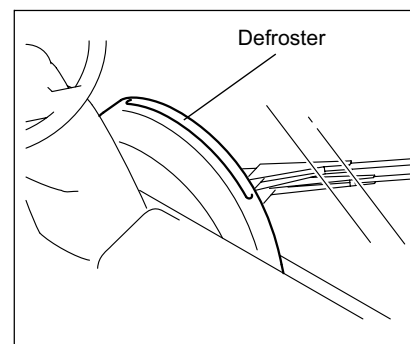
This lever adjusts the temperature of the heater. Adjust the temperature to your preference.

Raise the temperature

.....Shift the lever to the left

Lower the temperature

.....Shift the lever to the right.



Using the defroster

(Premium cabin models)

The defroster is attached at the base of the front windshield.

Closing the air outlet of the heater will switch function to the defroster. Use the heater switches to operate and stop the defroster. It will allow you to quickly defog the front windshield.

III. KNOW YOUR TRUCK

IV. DAILY OPERATION
CONTENTS

●	Daily Inspection (Pre-Operation Checks)
●	Starting Engine (Gas, LP, Diesel)
●	Brakes
●	Lifting and Lowering Controls
●	Direction Controls
●	Lifting Loads
●	Traveling
●	Stacking Loads
●	Attachments
●	Parking
●	Unexpected Emergencies

PRE-OPERATION CHECK	67
OPERATING YOUR TRUCK	76
UNEXPECTED EMERGENCIES	79

PRE-OPERATION CHECK

=====

 WARNING

OSHA and ANSI require users to inspect trucks at the start of every shift. These and other periodic inspections are your responsibility and these guidelines are to assist you in your job.

=====

Item	Inspection
Previously detected malfunctions	Correct.
Exterior	Body, oil leakage, water leakage, loose parts and exterior damage.
Wheels	Tire pressure, wear or damage, rims and hub nuts.
Lamps	Lamp condition and damaged lamps.
Hydraulic oil	Oil level, contamination and consistency.
Radiator	Coolant level and antifreeze requirement.
Engine	Oil level, contamination, consistency, noise and exhaust.
Inching and brake pedal	Pedal play and braking effect.
Brake fluid	Fluid level.
Parking brake	Operating force
Steering wheel	Looseness, play, vibration and pulling.
Horn	Sound.
Instruments	Functioning.
Load handling system	Parts, oil leakage, cracking and looseness. Make certain that the SAS is functioning.
Fuel	Amount.
Overhead guard	Bends, cracks and looseness.
Load back rest	Bends, cracks and looseness.
Operator restraint	Cut, frays and deck latch OK, seat belt & buckle.

Alignment

Check for any oil or coolant leakage on the ground or floor where the truck was parked. Check for loose parts or damage.

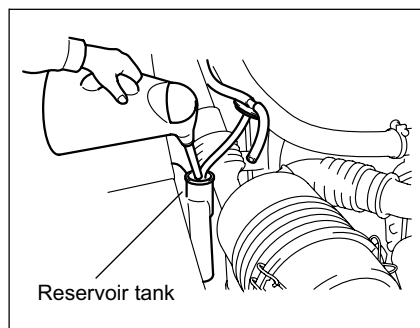
Damage, cracking and wear of tires and rims

Use a tire pressure gauge and measure the inflation pressure. Adjust it to the proper pressure.

Check the tightness of the hub nuts. Avoid uneven torque and tighten all of the nuts uniformly. Refer to service data for proper torque and tightening method.

Are the filaments intact? Is there any lens damage? Always keep the lenses clean to insure proper forward vision. (Rearview mirror and turn signal lamps are optional)

Engine coolant level check and supply



1. With the engine off, open the engine hood and check the engine coolant level in the reservoir tank.

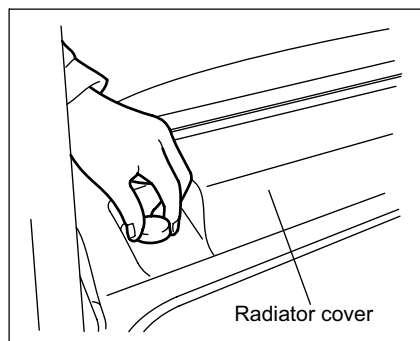
Note

The reservoir tank automatically supplies the engine coolant when the level in the radiator is low.

2. The coolant level is proper if it is between the upper and lower limits. If the level is below the lower limit, supply coolant to the upper limit.
3. The concentration of the long-life coolant (LLC) in the engine coolant must be 30% (or 50% in a frigid zone).

Note

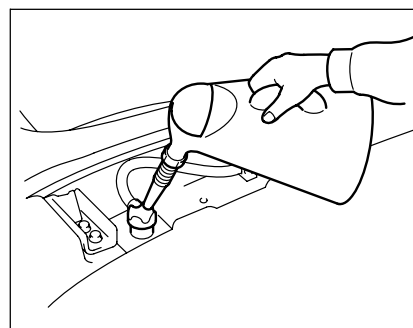
If no engine coolant remains in the reservoir tank, be sure to check the coolant level in the radiator, too, but only when cool.



Checking the engine coolant level in radiator



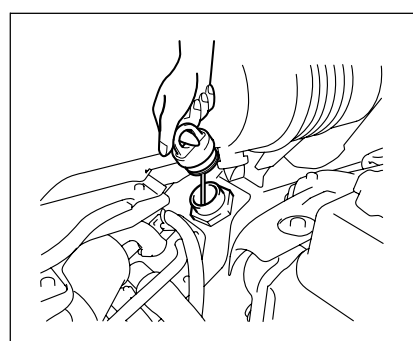
Coolant level checks musts always be performed when the radiator is cold. Removing the cap when engine is hot can cause serious injury.



1. Remove the radiator cover.
2. Remove the cap and check the coolant level from the filler port.
3. If the engine coolant is not visible through the filler port, add an appropriately amount of diluted coolant (LLC).

Note

To close and tighten the radiator cap, match the pawl on the reverse side of the cap with the notch in the filler port and turn the cap fully clockwise while pressing it down.

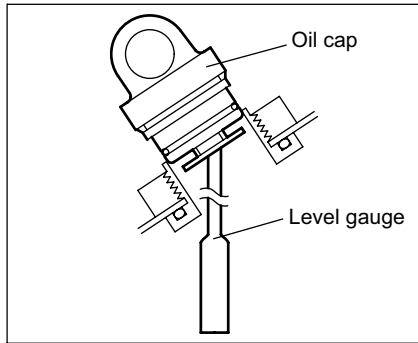


Checking hydraulic oil level

Always stop the engine and lower the forks to the ground before checking the level of the hydraulic oil while the truck is on level ground.

1. Open the engine hood and remove the oil cap.
2. Wipe the level gauge attached to the oil cap with a clean cloth, and insert it again into the tank.

IV. DAILY OPERATION



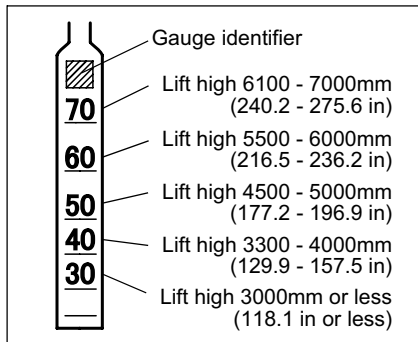
Note

Inspect the oil level by placing the level gauge on the opening of the oil supply inlet, without pushing the oil cap in.

3. Remove the level gauge gently and check if the oil is up to the level line.

Note

The oil level varies with the maximum forks height.

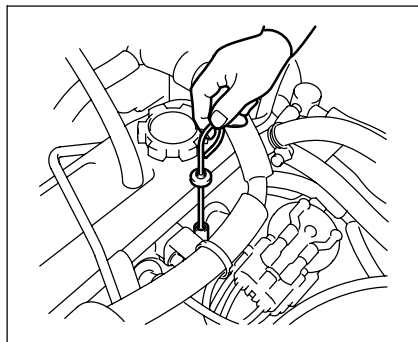


4. If the oil level is insufficient, add oil. Spilled and splashed oil must be wiped off thoroughly.

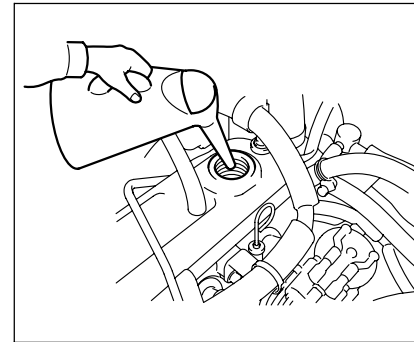
Gauge identifier	Applicable models
10, 18, K2, k3	8FGU/8FDU15, 18 series
C2, C3	8FGCU20, 25 ,30 ,35 series
20, 25	8FGU/8FDU20, 25 series
28, 30, 35	8FGU/8FDU30, 32 series

Engine oil inspection

1. Park the truck on flat ground. If the truck is inclined, the indicated level may be incorrect.
2. The oil level must be checked before starting the engine or at least 3 minutes after the engine is stopped.
3. Remove the oil level gauge and wipe it with a clean cloth. Insert it again and check if the oil level is between the F and L levels.



4. If the oil level is below the L line, add oil to the F line.



Adding engine oil

1. To supply oil, remove the filler cap and pour oil through the filler port. Never let the oil level exceed the F line.
2. The oil to be supplied must be appropriate for the season.
SAE40 Ambient temperature higher than 30°C (86°F)
SAE30 Ambient temperature 0°C to 30°C (32°F - 86°F)
SAE40 Ambient temperature 0°C to -10°C (32°F - 50°F)

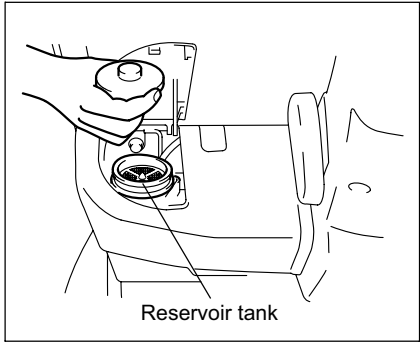
Caution

Always use the same brand of oil.

Leakage inspection

Check the engine compartment for any oil or fuel leakage.

Clean the radiator if it is clogged and check if there are any foreign objects, such as paper, on the radiator grill.



On board Truck Inspection Brake fluid inspection

With the engine off, check the level of the brake fluid in the reservoir tank. The level should be within the range shown in the figure. If the level is below the lower limit, add brake fluid up to the proper level.



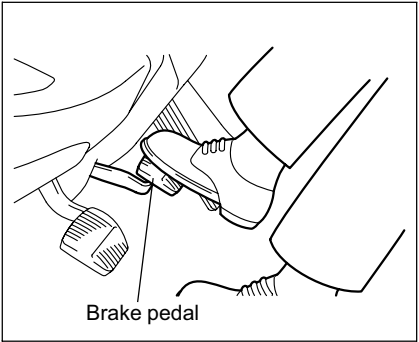
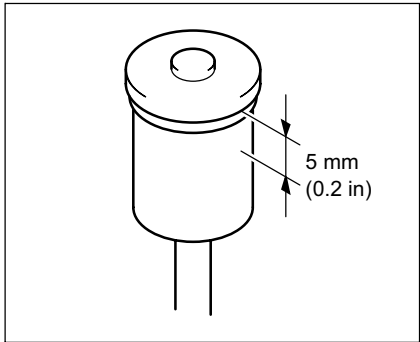
⚠ WARNING

If the level of brake fluid is low, the brake system may be leaky. Conduct an inspection immediately. Call your Toyota dealer for professional help.



⚠ WARNING

- Never use any oil other than brake fluid.
- Do not allow dirt to get into the reservoir tank. Even a small amount of dirt in the brake fluid can prevent proper braking.
- Check the small vent hole in the reservoir tank cap frequently to make sure that it is not clogged with dirt.



Brake pedal inspection

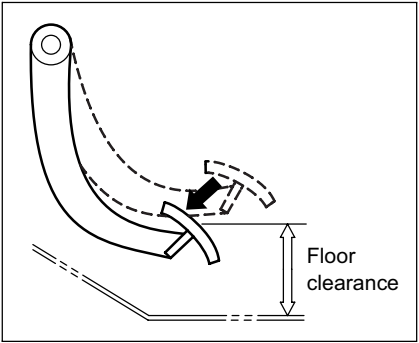
Note

The brake pedal must be inspected after starting the engine.

1. Depress the brake pedal fully, and check the floor clearance (clearance between the pedal and floor).

Note

See the Service Data Section for the floor clearance.



2. Make sure that the pedal does not go any further when it is kept depressed.
3. Also check that no problem is observed with pedal depression and return.
4. Manually depress the brake pedal to check the play until a resistance is felt.

Note

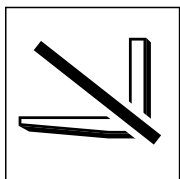
See the Service Data Section for the amount of brake pedal play.



⚠ WARNING

Ask your Toyota dealer to inspect if play is excessive, pedal movement is a problem or brake performance is improper.



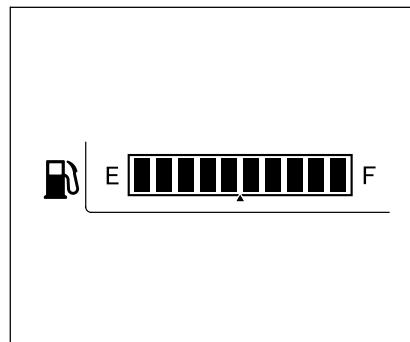


OPSS lamp inspection

Sit on the seat, start the engine, and check that the OPSS lamp is not lighted.

In the following circumstances, a malfunction to the OPSS lamp may have occurred. Park the truck at a safe location and contact your Toyota dealer.

- The OPSS lamp does not light up when the operator leaves the seat.
- The OPSS lamp does not turn off when the operator returns to the seat.



Fuel level check and supply (Gasoline and diesel)

⚠ DANGER

Follow these rules when refueling to avoid serious injury or death:

- Refuel only in authorized areas
- Turn ignition off
- No smoking or naked flames allowed
- Clean up spills before restarting

1. Check the gauge to see if the fuel is sufficient to finish your shift.

Note

After the end of daily operation, fill the tank with fuel to prevent the moisture in the air in the tank from mixing with the fuel.

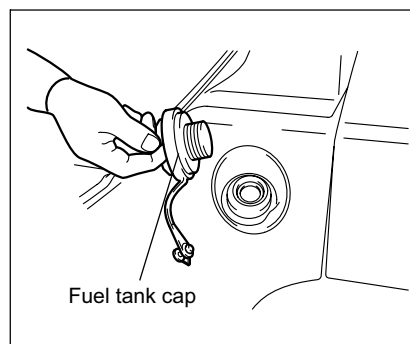
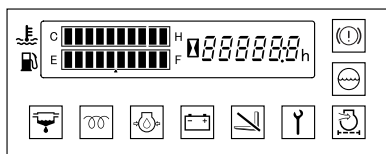
2. When supplying fuel, stop the engine, remove the fuel tank cap by turning it counterclockwise, and pour fuel through the fuel filler neck.
3. After fueling, be sure to tighten the fuel tank cap.

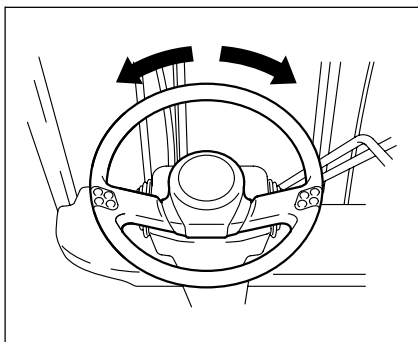
Caution

Carefully prevent entrance of water and dirt into the tank during fueling, use only clean, properly marked containers.

Instrument inspection

Start the engine and see that they operate properly.





Steering wheel inspection

Note

Perform the inspection after starting the engine.

1. Check steering wheel play with the rear wheels set for straight travel.

Note

See the Service Data Section for the standard steering wheel play.

2. Rotate the steering wheel and also move it up and down to check that there is no looseness.
3. Push the horn button to check if the horn sounds normally.
4. If any problems are found, ask your Toyota dealer to inspect.

While moving slowly

Torque converter clutch disengagement

Press the inching pedal and check clutch engagement while moving.

Caution

Insure that the control lever operates properly in each gear and then make the above checks while moving slowly.

Steering inspection

While moving the truck slowly in a safe location, turn the steering wheel to the left and right and check for any unusual movement.

Brake effectiveness

Inspect to see if there is anything unusual when the inching and brake pedal is pressed or if the brakes only work on one side.
Pull the parking brake on and insure that the truck can be stopped and held against moderate engine speed.

|||||

⚠ WARNING

If anything feels even slightly unusual, stop truck operation immediately and have the truck inspected by your Toyota dealer.

|||||

Inspecting SAS system

Check the mast to make certain that it can be properly tilted either forward or backward and moved up. Besides, make certain that the mast can automatically stop at its horizontal position.

Caution

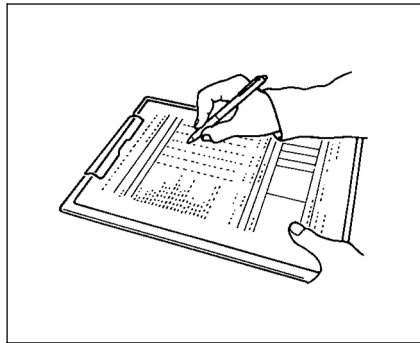
If you should feel something abnormal even slightly, or when the diagnosis lamp blinks, or once an error code has appeared on the hour-meter display, immediately stop operating the truck and request an inspection by a Toyota dealer.

IV. DAILY OPERATION

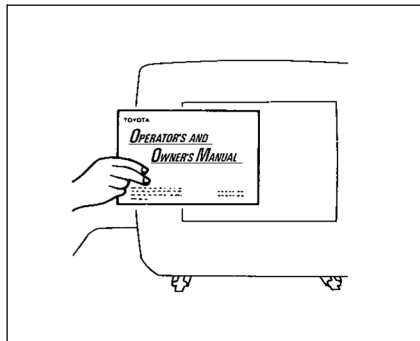
OPERATING YOUR TRUCK

The skills needed to operate a lift truck must be developed through training and experience. Both the truck and jobs to be done change from one work area to another.

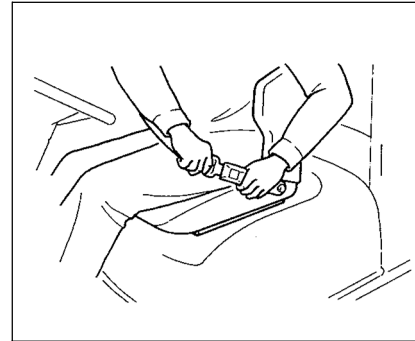
This review will quickly describe use of the basic operating controls. When you get into new jobs or unfamiliar areas, ask your supervisor for help.



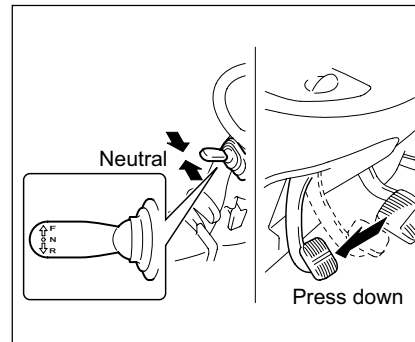
Make sure your truck is safe. Check over the truck before every shift, using the information at the beginning of this section.



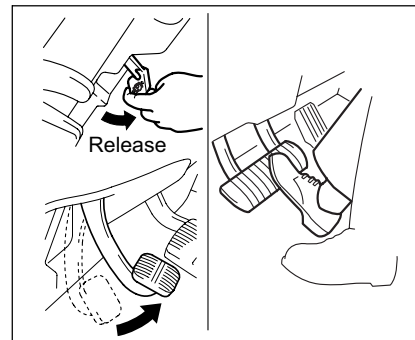
Read the Operator's and Owner's Manual and the warning decals on your truck. Stow the manual behind the seat when you finish. Ask questions if you don't understand.



Use the seat belt to help keep yourself safely inside the truck in case of an overturn or tip up.

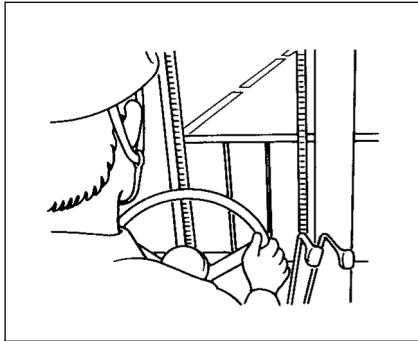


Start the engine only when you are sure the direction control is in NEUTRAL and the parking brake is ON.

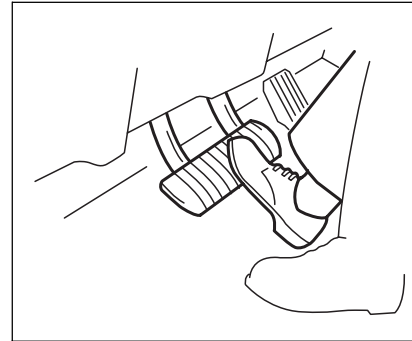


Before you release the parking brake and shift to FORWARD or REVERSE, hold the truck in place with the foot brake.

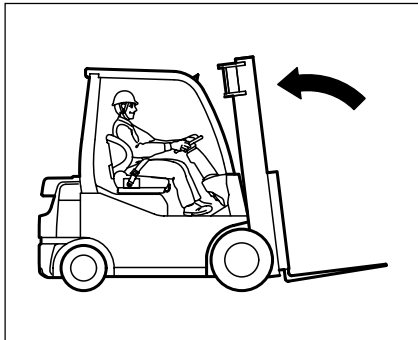
IV. DAILY OPERATION



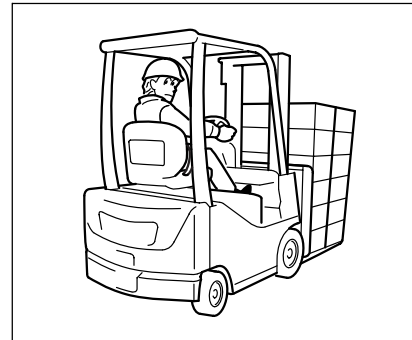
Place the direction control lever in position for travel, then check in that direction to make sure the path you will take is clear of all pedestrians and obstructions.



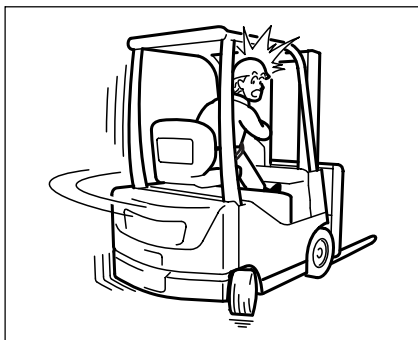
Stopping can be done with the inching and brake pedal. Always use them smoothly to avoid losing a load.



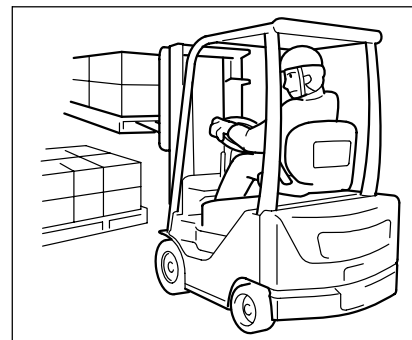
Lift your forks or attachments only high enough to clear the floor and tilt them back.



Pick up loads with forks spread as wide as possible and travel with the load centered and forks tilted slightly back. Travel in REVERSE if the load blocks your view.

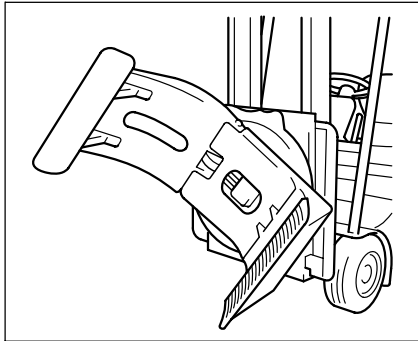


Release brake and push accelerator to change your speed. When steering, remember about tail swing with rear steered wheels.

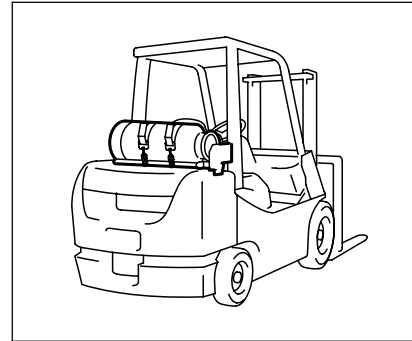


Stacking loads requires patience and practice. Put the truck in position before raising the mast, and lower loads only when stopped and as close to the shelves as possible.

IV. DAILY OPERATION



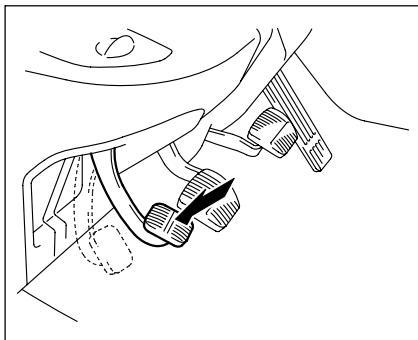
Your employer may use special attachments to replace forks on the truck. There are many different kinds and you must get special training in their operation.



When parking an LP-powered truck overnight, close the shut-off valve and wait until the engine stops before turning off the ignition. Park an LP-powered trucks only in well ventilated area.

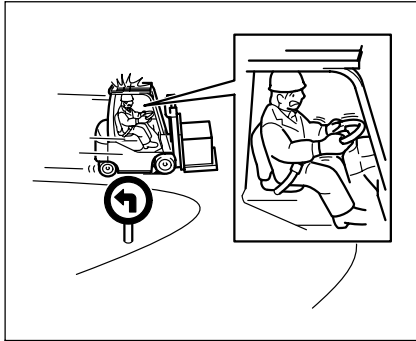


When you leave your truck, park it in an approved area, not on a grade or in a fire lane, or blocking pedestrian walkways.



Be sure your truck will stay parked. Shift to NEUTRAL, set the parking brake, lower the forks, and tilt forward so the tips will not trip pedestrians, and turn off the ignition switch.

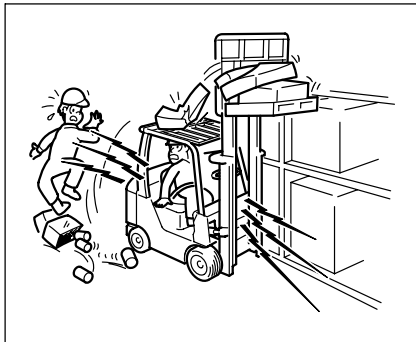
UNEXPECTED EMERGENCIES



Stay alert for unexpected problems. We can't know what might surprise you in your work area, but here are some things to remember.

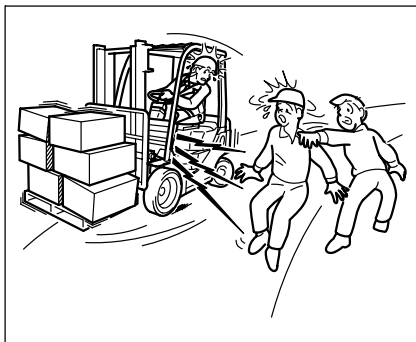
Engine stops while traveling:

Stop the truck as soon as possible. Steering will take much more force to operate than normal.



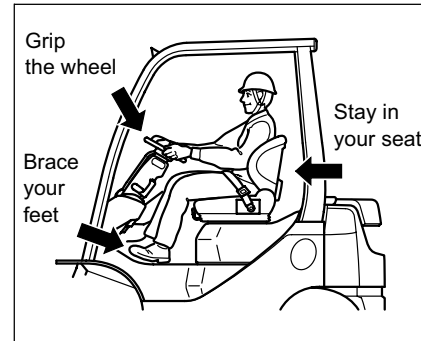
Load starts to shift or fall while stacking:

Lower load if possible. The overhead guard and load backrest should help protect you from most falling loads. Sound horn or shout to warn any others in area.



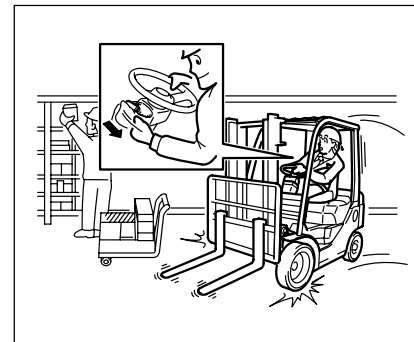
Pedestrians or other machines come into the path of your truck:

Steer or brake to avoid them. Sound horn to alert people who may not be looking at you.



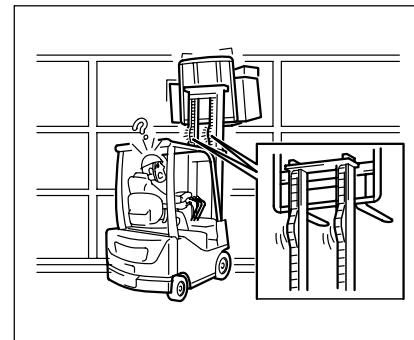
Overturning or tipping forward of the truck:

Stay with the truck. Lean away from the fall and brace your arms and legs to keep them inside the overhead guard.



Loss of brakes while traveling:

Try the inching and brake pedal. Steer around hazards and apply the parking brake. Shift direction control lever to the opposite direction.



Forks, mast or attachments hang up:

If you see a slack chain or the mast stages lower unevenly, raise the forks immediately and find out why. This will avoid a sudden fall.

IV. DAILY OPERATION

V. OPERATOR MAINTENANCE CONTENTS

- Fueling (Gas, LP, Diesel)
- Tires /Wheels
- Antifreeze
- Fuses
- Battery
- Parking Brake Adjustment
- Tag Out

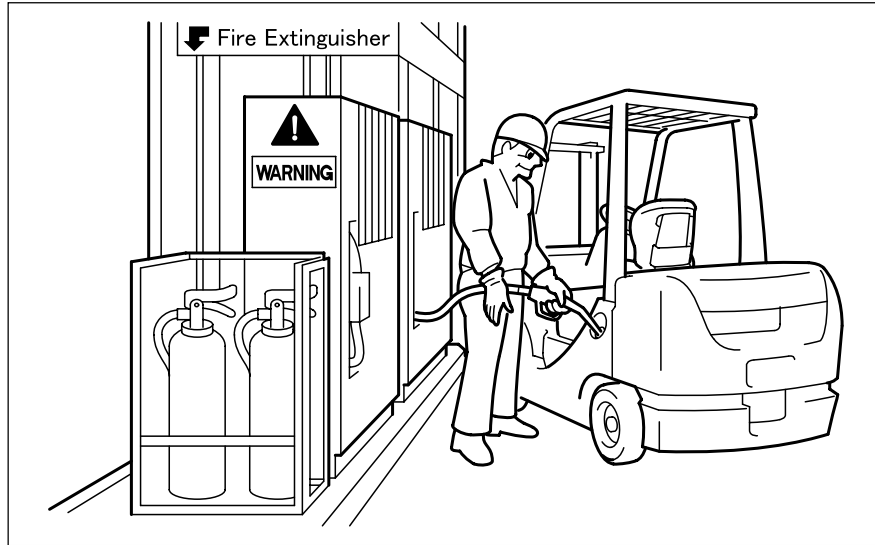
OPERATOR MAINTENANCE	81
REFUELING YOUR TRUCK	82
CHANGING TIRES.....	83
ADDING ANTIFREEZE	85
FUSE REPLACEMENT	86
AIR PURGING OF THE FUEL SYSTEM.....	87
DRAINING THE SEDIMENTER	87
MAINTAINING THE BATTERY TERMINALS.....	87
BATTERY CHARGING.....	88
TAG IT OUT ... PARK IT	88

OPERATOR MAINTENANCE

⚠ WARNING

OSHA requires training for lift truck mechanics. Some employers authorize operators to perform basic maintenance work. This section gives help in a few of these areas, but it is not a substitute for comprehensive training and knowing your employer's work rules.

REFUELING YOUR TRUCK



Don't get caught with an empty tank. Make sure your truck has enough fuel to finish the job you start.

Refuel only in a safe, well ventilated place authorized by your employer. Find out his rules for carrying and storing fuel and follow them.

Gasoline engine models: Use only un-leaded gasoline of 87 octane rating or higher.

LPG fuel is a gas that is heavier than air. It will settle in low spots and on your clothes. Any flame or spark could cause a fire that could cause serious injury.

When changing tanks, check all connections for damage or missing parts. If you have a leak or the truck will not restart, ask a mechanic for help. Never try to restart a truck if you still smell gas.

LPG FUEL

Check the gauge on the tank to make sure there is enough fuel to finish your work.

CHANGING OR REFILLING TANKS

⚠ WARNING

To avoid serious injury from fire or explosion, you must follow these rules:

- Switch ignition and lights off.
- Change tanks only in well ventilated, approved areas.
- No fire or flames allowed.
- Check all connections for damage or missing parts.
- Check for leaks.
- Do not restart until all smell of gas is gone.
- If truck will not restart, get a mechanic to inspect it.
- Filling tanks requires special procedures. Make sure someone explains them all to you.

DIESEL FUEL

Recommendation

Use diesel fuel with a cetane index of 40 or more and a sulfur content of not more than 500 ppm that is based on American diesel fuel standard ASTM D975/03.

Caution

- Not for use with bio-diesel fuel. Damage to engine will occur.
- In winter, use winter diesel fuel to prevent clogging of the fuel filter caused by paraffin precipitation.

CHANGING TIRES

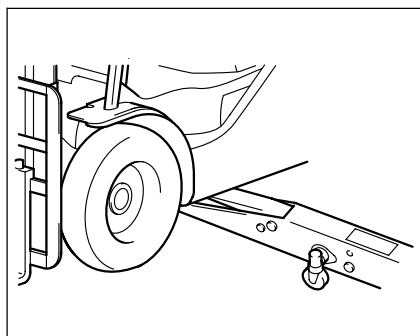
⚠ WARNING

You must use a jack that has a rated capacity of 8,000 lbs or more. Make sure the counterweight mounting bolts are fully tightened before lifting the truck, and follow these instructions to avoid injury:

- Truck must be unloaded and parked on a smooth level floor.
- Make sure the parking brake is properly adjusted and fully applied and opposite end wheels securely chocked.

⚠ DANGER

You can be killed or seriously injured if a lift truck falls on you. After jacking, use blocking or proper stands to hold the truck. Never get under a truck supported by any hydraulic jack.

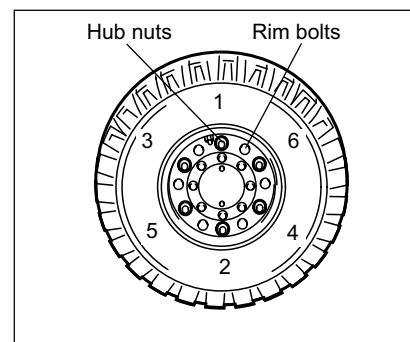
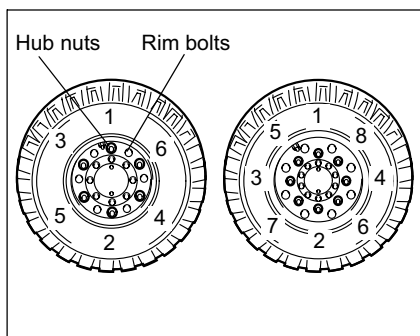


Front wheels

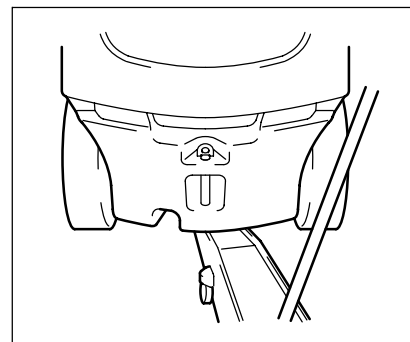
⚠ WARNING

To raise the front, lift the forks about 3 feet with the mast vertical and place a jack under the front frame. Make sure that the jack is properly positioned.

1. Jack up until the wheels are about to leave the ground and then, loosen the hub nuts.
2. Jack up until the wheels come off the ground. Completely remove the air pressure from the tire then remove the hub nuts and remove the wheel.
3. To reinstall the wheel after changing a tire, perform the steps for removal in reverse order. The hub nuts should be tightened evenly and in the sequence shown in the figure.



4. After replacing the wheel, check and adjust the tire air pressure.

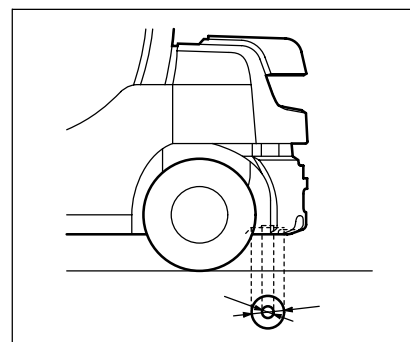


Rear wheels

⚠ WARNING

To raise the rear, use either a floor jack or a bottle jack of the right capacity and place it as shown.

1. Jack up until the wheels are about to leave the ground and then, loosen the hub nuts.
2. Jack up until the wheels come off the ground then remove the hub nuts and remove the wheel.
3. To reinstall the wheel after changing a tire, perform the steps for removal in reverse order.
4. The hub nuts should be tightened evenly and in the sequence shown in the figure
5. After replacing the wheel, check and adjust the tire air pressure.

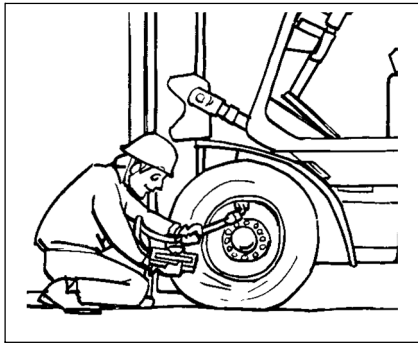


V. OPERATOR MAINTENANCE

Jack setting position

- (1) Hydraulic garage jack
(unavailable in 1-ton models)
- (2) Portable hydraulic jack

Set the jack in the jack point under the counter weight.

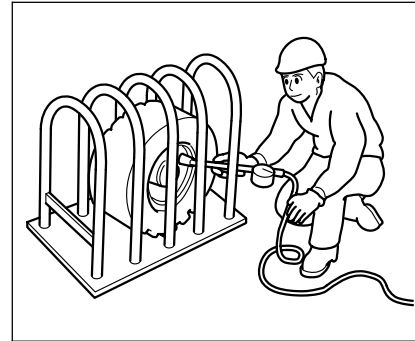
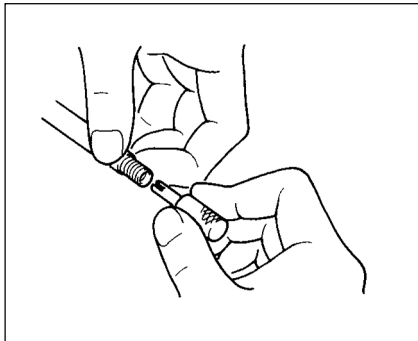


Wheel and tire servicing

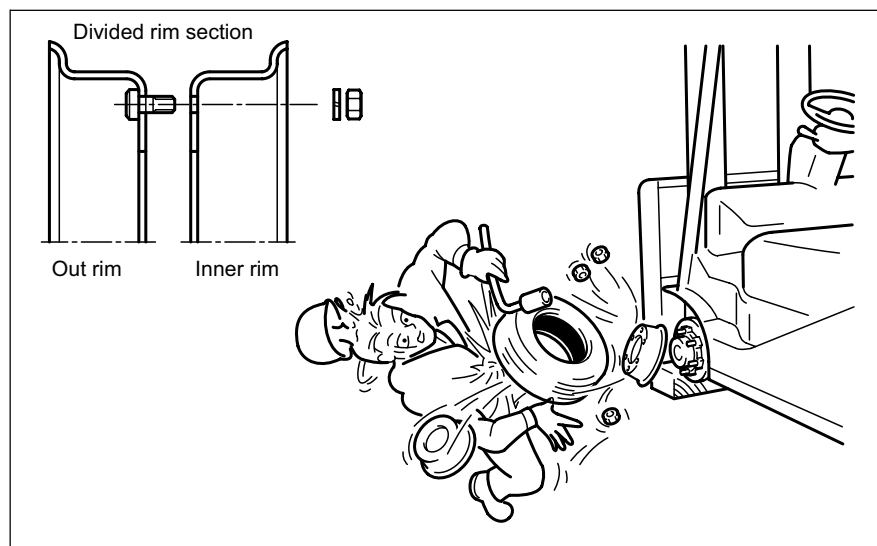
Federal and State laws require you to be fully trained and qualified before doing maintenance on wheels and tires. Injury or death can result from the exploding rim components if servicing is not done correctly.

Check tire pressure facing the tread of the tire, not the side. Use a long handled gauge to keep your body away from rims and wheels.

Remove air from the tire before removing rim locking rings on multi-piece rims.



Always use a safety cage to inflate tires after servicing.



⚠ DANGER

- Tire air pressure is very high, so pay attention to rim deformation, cracks, etc.
Never exceed proper air pressure.
- Never loosen the nuts that hold the inner and outer wheels halves together when there is air pressure in the tire.
- Completely remove the air pressure from the tire before it is removed from the lift truck. Air Pressure in the tires can cause the tire and wheel parts to explode, causing serious injury or death. Tires and wheels must be changed and repaired only by trained service persons.
- If a tire has less than 85% of the correct air pressure, completely remove the air pressure from the tire. Remove the tire from the lift truck. Add air pressure to the tires only in a safety cage.

ADDING ANTIFREEZE

If the truck is left in an area where the temperature is less than 0°C(32°F), the cooling water will freeze and may damage the radiator and/or cylinder block. In such cases, antifreeze coolant must be used.

When long-life coolant (LLC) is used, it must be changed once every two years. The freezing temperature varies depending on the amount of antifreeze added.

Antifreeze mixture (%)

Freeze protection temperature °C (°F)	-12 (10.4)	-15 (5)	-24 (-11.2)	-35 (-31)
Mixture (%)	25	30	40	50

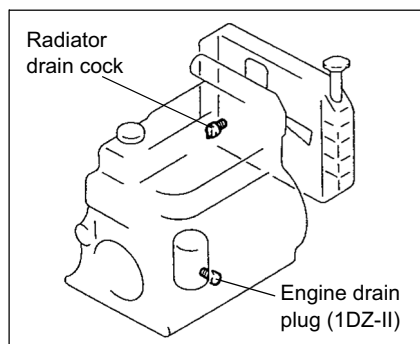
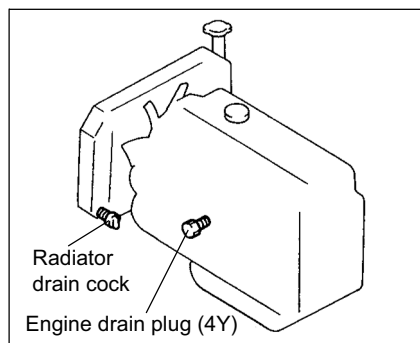
⚠ WARNING

The antifreeze fluid is flammable, so no smoking and avoid open flame.

Prior to adding antifreeze, inspect the radiator, water pump, piping and cylinder block for leaks.

The procedures for adding antifreeze is as follows:

V. OPERATOR MAINTENANCE



⚠ WARNING

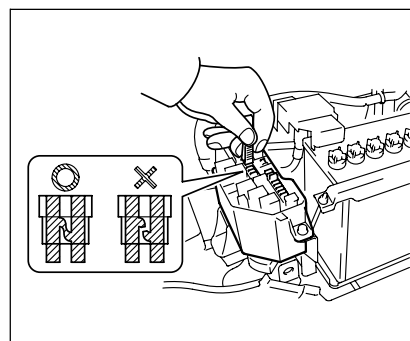
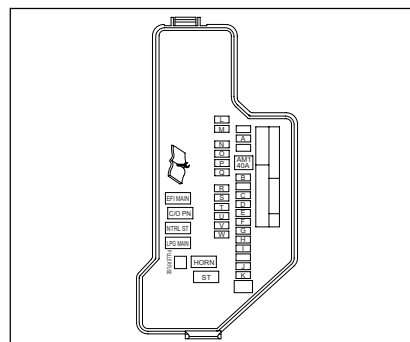
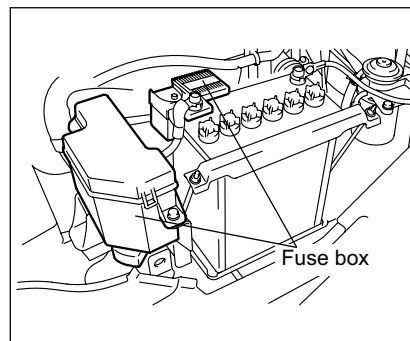
To avoid burns, only open cap when engine is cold.

1. Remove the radiator cap. Loosen the drain cock on the radiator and cylinder block and drain the coolant.
2. Flush out the radiator and cylinder block by adding clean water through the radiator inlet.
3. After the water has drained out of the radiator and cylinder block, tighten the radiator and engine drain cocks.
4. Add the proper amount of antifreeze through the radiator inlet and fill up the remaining space with clean water.

Note

Use of Toyota Long Life Coolant (LLC) is recommend for all season operations.

FUSE REPLACEMENT



If a lamp does not come on or an electrical device does not function, the corresponding fuse may have blown. Check the fuse for each device. The fuse box is located in the front left as seen from the opened engine hood.

Note

See the table below for the device corresponding to each fuse.

Fuse assignment

A	30A	Starter	M	7.5A	Spare
B	7.5A	Horn	N	7.5A	Hour MET
C	15A	EFI	O	7.5A	Starter
D	7.5A	ALT-S	P	10A	Gauge
E	7.5A	Stop	Q	10A	Back LP
F	7.5A	ACC-B	R	7.5A	Shift
G	7.5A	Tail	S	7.5A	Turn
H	7.5A	ECU-B	T	15A	Ignition
I	15A	E-THRO	U	15A	Spare
J	10A	Work LP	V	10A	Spare
K	15A	Head	W	10A	ECU-IG
L	30A	Spare			

Including optional accessories

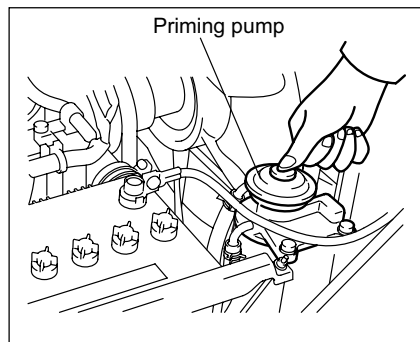
The fuse check and replacement procedure is as follows:

1. Set the ignition switch to the OFF position.
2. Remove the fuse box cover and take off the fuse clip attached to the fuse box.
3. Apply the fuse clip to the fuse to be removed.
4. The fuse is blown if its state is as shown at left in the illustration. Replace it with a spare fuse.

Caution

- Use a fuse having the same capacity as the one removed.
- If the replaced fuse blows again, ask your Toyota dealer to inspect.

AIR PURGING OF THE FUEL SYSTEM

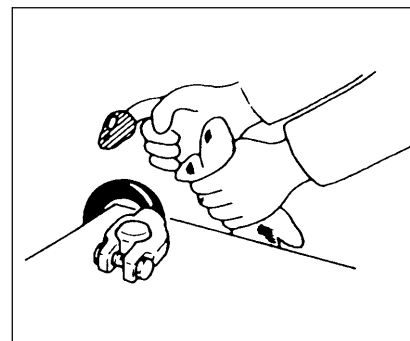


(Diesel engine models)

When fuel has been completely depleted or when maintenance has been performed on the fuel system, be sure to perform air purging in the following sequence.

1. Open the engine hood.
2. Operate the priming pump up and down to perform air bleeding.

MAINTAINING THE BATTERY TERMINALS

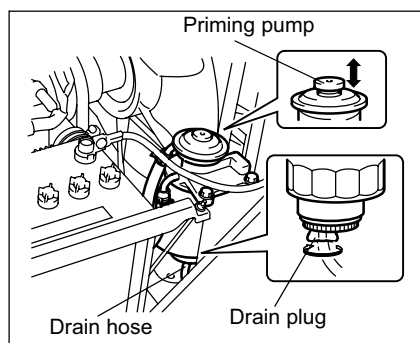


▲ WARNING

Always stop the engine before working on the battery or terminals. Remove the negative terminal (-) first, but replace it last.

1. A loose or corroded terminal causes failure in connection. Remove white powder, if any, from the terminal by pouring warm water over it to dissolve it and then grease the terminal.
2. If the terminal is extremely corroded, remove it from the battery and brush off the corrosion using a wire brush or emery paper. Connect the terminal tightly to the battery and grease the terminal.

DRAINING THE SEDIMENTER



(Diesel engine models)

The sedimenter separates the water contained in the fuel. It is integrated with the fuel filter. If the sedimenter warning lamp comes on, immediately drain water according to the following procedure because the accumulated water in the sedimenter is above the specified level:

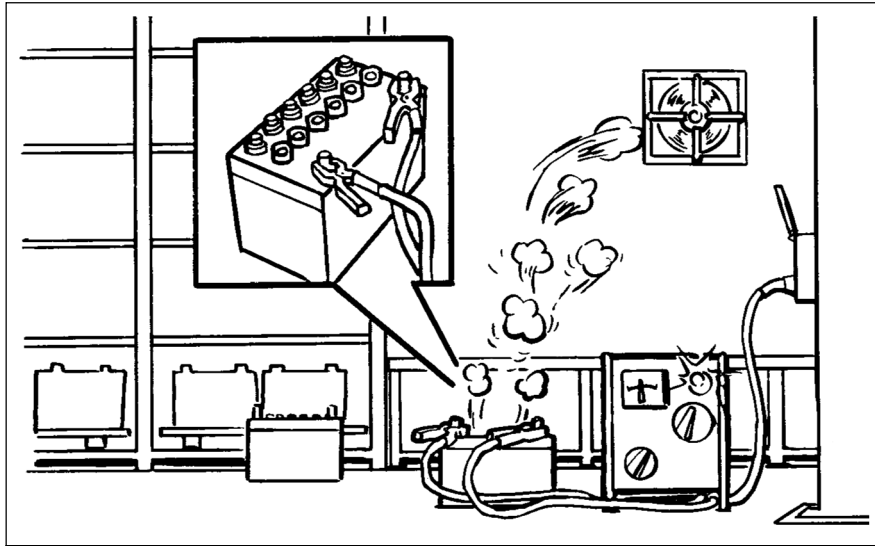
1. Place a container to receive the water under the open end of the drain hose under the fuel filter.
2. Turn the drain cock once or twice to loosen it and operate the priming pump up and down to drain the water from the sedimenter.
3. When light oil starts to flow out after the water has drained, firmly tighten the drain cock.

Caution

Wipe the light oil cleanly from the adjustment area.

V. OPERATOR MAINTENANCE

BATTERY CHARGING



- Charge batteries only in approved, well ventilated areas.
- Remove caps and negative (-) battery cable.
- Connect charger leads positive (red or +) then negative (black or -)
- Charge at the slowest rate and with charger set to rated voltage.
- Make sure electrolyte stays above minimum level. Use distilled water only.
- Turn charger off before disconnecting leads.
- Always replace caps tightly and allow battery to cool before using.

TAG IT OUT ... PARK IT



If any problem starts while you are working, stop and tell your supervisor. When you sideline a truck, tag it so others will not be injured by trying to use it.

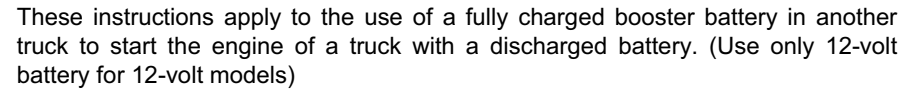
If you find a problem in your daily inspection, get it fixed before starting to work or tag it out and use another truck.

VI. EMERGENCY PROCEDURES

CONTENTS

- Dead Battery, Jump Starting
- Lifting
- Transporting
- Towing
- Jacking

USING JUMPER CABLES	90
TRANSPORTING	91
TOWING	92
RAISING THE TRUCK	93
UL-SPECIFICATION PARTS	93
MODIFICATIONS	94



 WARNING
 To avoid damage to your truck, or the possibility of harm to yourself, follow these instructions and warnings. If you have any questions, consult your Toyota dealer.

1. Use only a 12-volt negative ground battery to start your truck. (Use 12 V battery x 1 for 12 V models) If you are not sure of the voltage, or if the ground is different, do not try to jump start.
Personal injury or damage to the electrical system can result. Damage to the electrical system will not be covered by the manufacturer's warranty.
2. Check fluid level of the discharged battery. If low, add distilled water to the correct level.
Be sure to install the caps before jump starting.
3. Put the booster battery as near to the truck as necessary for the jumper cables to reach both batteries. Check and make sure that the trucks do not touch each other.

Use particular care when connecting a booster battery to prevent sparks.

4. On both trucks:
 - (a) Apply the parking brake.
 - (b) Put the gear shift lever or directional control lever in the NEUTRAL position.
 - (c) Turn the ignition key switch to the OFF position.
 - (d) Turn all lights and accessories to the OFF position and leave them off until after the engine has been started and the jumper cables removed.

5. Connect the jumper cable in the following sequence:
 - (a) Connect the red jumper cable from the positive (+) terminal on one battery to the positive (+) terminal on the other battery. Never connect (+) to (-) or (-) to (+) as this may damage the alternator. Make sure the clamps do not touch any other metal.
 - (b) Next, connect one end of the black cable to the ground (-) terminal of the "helper truck" battery.
 - (c) Last, connect the other end of the black jumper cable to a stationary, solid metal point on the engine of the truck you are starting.
(NOT TO NEGATIVE (-) TERMINAL OF THE BATTERY.)

Make this connection at a point at least 18 inches (450 mm) away from the battery, if possible. Do not connect it to pulleys, fans, or other parts that move.

6. Start the engine on the helper truck, and run the engine at a moderate speed.
7. Start the engine of the truck with the dead battery.
Follow the starting instructions in the Starting and Operating Procedures section of this manual.
Be sure that the engine is at idle speed before disconnecting the jumper cables.
8. Remove the jumper cables by reversing the above sequence exactly. Start by removing the black jumper cable from the engine block of the truck with the discharged battery. Then remove the other end of the negative (-) cable from the "helper truck".
9. Remove both ends of the red cable.

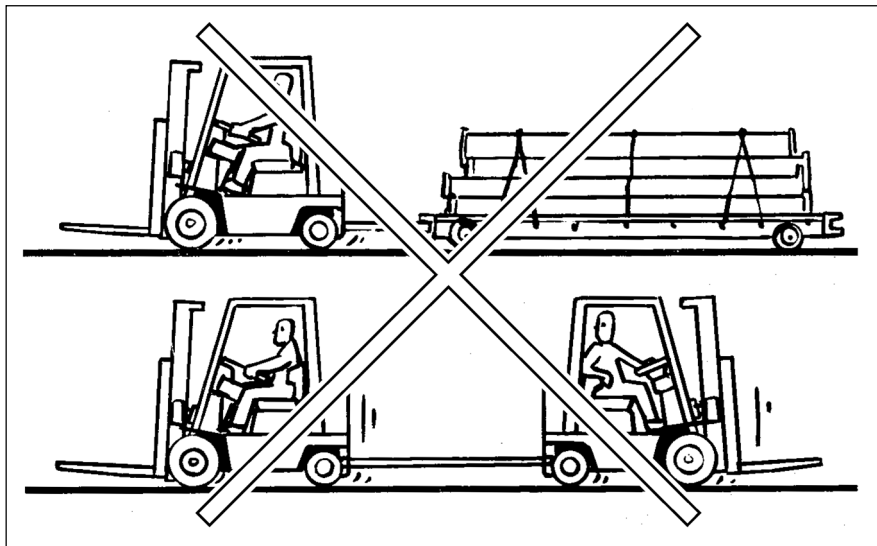
TRANSPORTING

WARNING

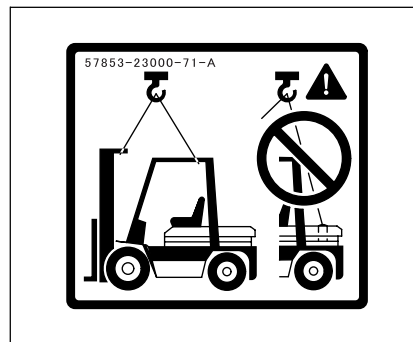
To avoid injury before loading this truck onto a trailer or highway truck, make sure the floor has a capacity strong enough to hold the weight, and that it has tie down points that will keep the truck from moving. Be sure to follow these instructions:

- Stay away from edges and remember that lift trucks steer from the rear.
- Use the draw bar to load the truck on a tilt trailer, or make sure that dock levelers or plates of the proper capacity are used if loading from a dock.
- Center the truck left and right, lower the forks or attachments, apply the parking brake, and stop the engine.
- Install chains and chain tensioners from the draw bar in the rear to the left and right sides and rearward to the transporting truck.
- Install chains on the bar under the front frame or the integral tie down bracket on each fender from left and right with tensioners to a forward position on the transporting truck.
- Operate the transporting truck slowly and smoothly and check the tie downs periodically.

TOWING



The draw bar is located at the back of the counterweight, and is used to pull the truck should its tires drop into a gutter or become stuck in mud. It can also be used for loading the forklift onto a transport truck. The draw bar should not be used for towing the forklift or for towing another truck using the forklift.



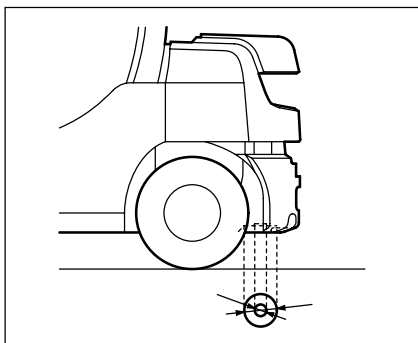
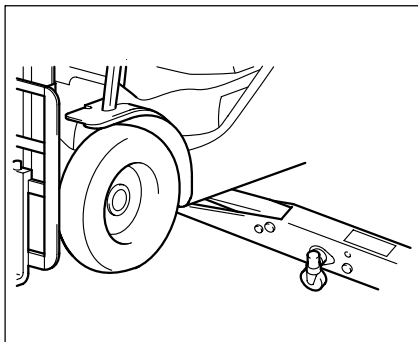
Truck hoisting method

When hoisting the truck, use the lifting holes near the top of the mast and overhead guard as shown in the illustration.

⚠ WARNING

- Use wire cable which is sufficiently strong.
- Never use the holes on the upper part of the counterweight to hoist the truck.

RAISING THE TRUCK



⚠ WARNING

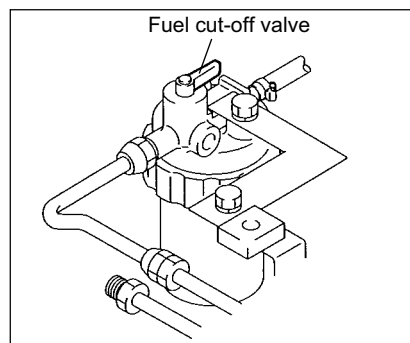
You must use a jack that has a rated capacity of 8,000 lbs or greater. Make sure the counterweight mounting bolts are fully tightened before lifting the truck, and follow these instructions to avoid injury:

- Truck must be unloaded and parked on a smooth level floor.
- Make sure the parking brake is properly adjusted and fully applied and opposite end wheels securely chocked.
- To raise the front, lift the forks about 3 feet with the mast vertical and place a jack under the outer mast rail as shown.
- To raise the rear, use either a floor jack or a bottle jack of the right capacity and place them as shown.

⚠ DANGER

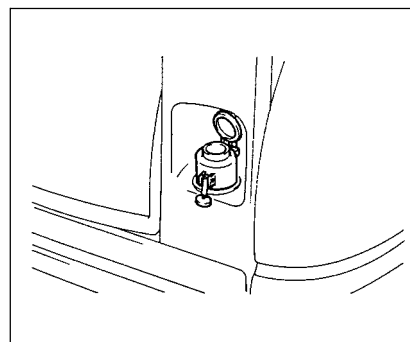
You can be killed or seriously injured if a lift truck falls on you. After jacking, use blocking or proper stands to hold the truck. Never get under a truck supported by any hydraulic jack.

UL-SPECIFICATION PARTS



Fuel Cut-Off Valve

This valve serves to cut off fuel from the fuel tank. It should generally be set to the ON position. Set it to the OFF position in an emergency such as when trouble occurs in the fuel system.



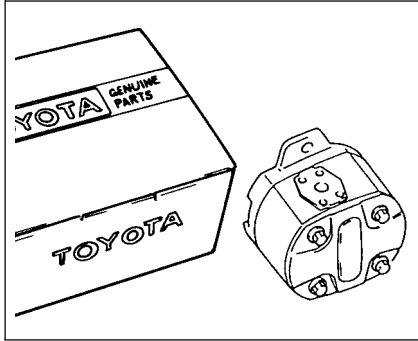
Fuel Tank Cap (Used only on UL-GS, DS types)

The cap cover can be opened by pulling the knob on top of the cap and sliding it sideways. After adding fuel, close the cap cover and tighten the knob securely.

The entire cap can also be removed easily by turning it counterclockwise.

VI. EMERGENCY PROCEDURES

MODIFICATIONS



OSHA and ANSI Standards require users to use repair or replacement parts of the same design and quality as those originally supplied. Your Toyota dealer is a reliable source for these parts and the service to keep your truck running safely and efficiently.

VII. PLANNED MAINTENANCE
CONTENTS

- Training
 - Manuals, Parts
 - Planned Maintenance, Dealer Service
 - Storage
 - Weekly Maintenance
 - Maintenance Schedules
 - Replacement Schedules
 - Service Data
 - Lubrication Chart

PLANNED MAINTENANCE95

STORAGE96

WEEKLY MAINTENANCE96

PERIODIC MAINTENANCE99

PERIODIC REPLACEMENT TABLE99

PERIODIC MAINTENANCE TABLE.....100

SERVICE DATA104

LUBRICATION CHART106

FRAME SERIAL NUMBER.....108

PLANNED MAINTENANCE

Daily inspections are important (and OSHA requires them), but planned maintenance is the least expensive way to keep your Toyota truck safe and productive.

Toyota dealers have trained mechanics and special tools to provide these services for you.

This section of the manual will remind you of the areas that must be inspected and the replacement intervals for various components. It is not meant to be a substitute for the mechanic training OSHA requires.

If you decide to perform any maintenance and have a question or problem, contact your Toyota dealer immediately for the proper repair manuals and professional advice.

OSHA and ANSI require all lift truck mechanics to be trained and authorized to repair and operate this truck.

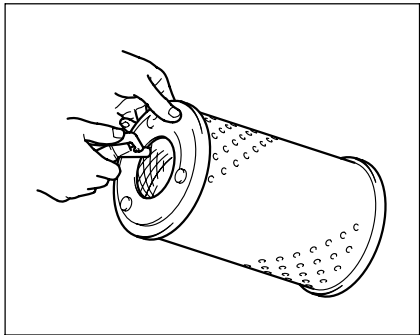
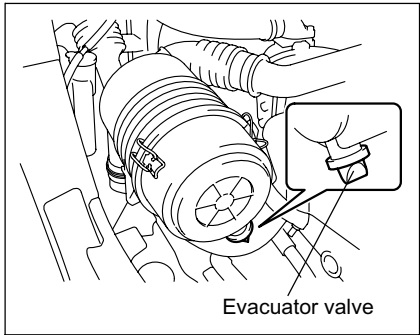
STORAGE

1. Inspect for oil and water leakage.
2. Inspect each component for warping, scratches, dents or cracks.
3. Clean the air filter element and lubricate parts as required.
4. Move the forks all the way up and down to lubricate the inside of the lift cylinder.
5. If you sensed anything unusual during operation, notify the supervisor.

**Even a small malfunction can cause a serious accident.
Do not operate the truck until repairs have been completed.**

Perform the following inspection together with before-operation inspection.

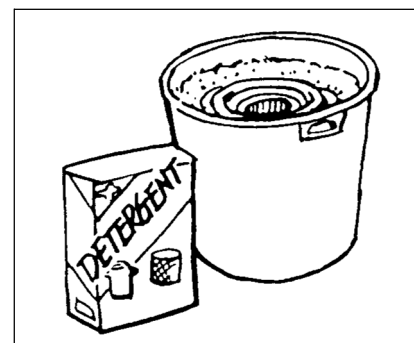
Weekly (40-Hour) Inspection Item	
Air cleaner	Clean
Fan belt	Inspect
Torque converter oil level	Check
Battery electrolyte level	Check
Bolts and nuts	Tighten
Mast and steering linkage	Grease
Chain	Engine oil



The element can be taken out after removing the three catches fixing the element.

1. Tap the element filter paper lightly without causing any damage or blow dust off with compressed air (7 kg/cm² (99.4 psi) or less) from inside.

2. After element cleaning, remove any dust in the evacuator valve.

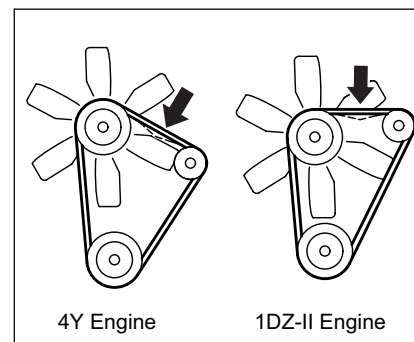


1. Soak the element in water containing mild detergent for approximately 30 minutes and then wash. Use care not to scratch the filter paper.
2. After washing, rinse the element with clean water (water pressure less than 2.8 kg/cm^2 (40 psi)).
3. Allow to dry naturally or use a dryer (cold air). Never use compressed air or flame.

1. The element should be replaced after washing six times or after it has been used for one year.
2. It is unnecessary to clean the inside element when cleaning the double cyclone air cleaner (Option). Only clean the outside element.
It is essential to replace both outside and inside elements, at time of replacement.

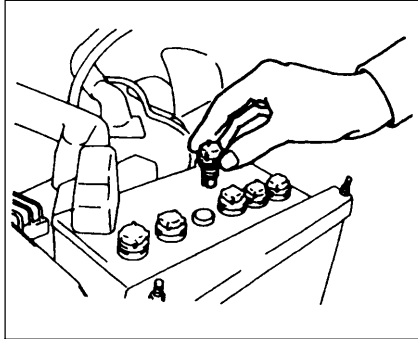
⚠ WARNING
Always use eye protection when cleaning with air.

1. Always replace the element if the filter paper is torn or damaged.
2. Wash the element if heavily contaminated.



Inspect the fan belt for cracks, fraying and tension.
If any problems are found, have the belt replaced or adjusted by a Toyota dealer.
Refer to service data for proper tension.

VII. PLANNED MAINTENANCE



Battery electrolyte check

1. The battery electrolyte should be between the upper and lower levels.

Note

For a battery without level indication on the casing, make sure that the electrolyte level is above the electrodes.

2. If the electrolyte level is below the lower level, remove the cap and add distilled water to the upper level through the water inlet port.



WARNING

See Battery Maintenance Section for other important rules.

Be sure to use distilled water for battery electrolyte. Also, wear protective glasses when working on the battery.

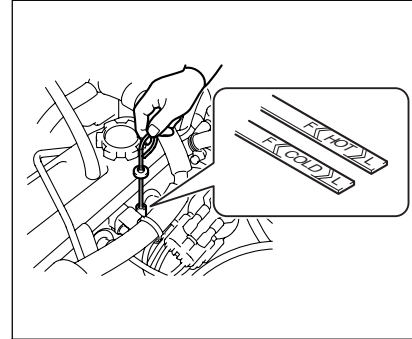
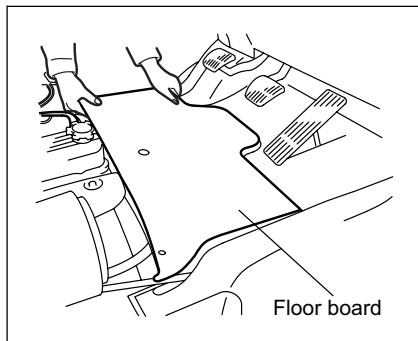


Torque converter oil inspection

1. Park the truck at a safe and level ground, and stop the engine.

Caution

Inspect with the parking brake pedal engaged and the forks lowered to the ground.



2. Open the engine hood and remove the toe board.
3. Remove the level gauge and wipe it with clean cloth.
4. Insert the level gauge back to the hole from which it is removed, and extract it again to check if the oil level is between the F and L lines on the level gauge.

Note

- Perform inspections using the COLD side of the level gauge before operating the truck.
 - The level gauge contains the inscriptions "COLD" and "HOT" on either side. Conduct inspections using the "COLD" side before operating the truck and when the oil temperature is 40°C or under. If you have operated the truck and the oil temperature is 60°C or over, use the "HOT" side to conduct inspections after 30 seconds and within five minutes after the engine is stopped.
5. If the level is near or below the L line, add oil to the F line. Use only oil specified in the lubrication table.

Retightening of nuts and bolts

Retighten each nut and bolt on the chassis and load handling system.

Greasing mast and steering linkage

Grease in accordance with the lubrication table.

Caution

- Clean the grease nipples thoroughly prior to greasing.
- After greasing, wipe off excess grease.

PERIODIC MAINTENANCE

Periodic inspection and maintenance are necessary to keep your Toyota industrial truck running safely and smoothly. The maintenance experts at your Toyota dealer are specially trained to use the right tools and approved procedures to protect the investment you have made in this high quality truck. Do not entrust it to amateurs.

The designated numbers of hours in the inspection cycle are as follow:

Daily (pre-operation check)	Every 8 hours
Weekly	Every 40 hours
6 weeks	Every 250 hours
3 - Month	Every 500 hours
6 - Month	Every 1000 hours
Annual	Every 2000 hours

If the operation time exceeds 250 hours within 6 weeks, use the number of hours as a guide for performing periodic inspection. Pre-operation checks and weekly inspections should preferably be performed by the user. 6-week, 3-month, 6-month and annual inspections should be performed by a Toyota dealer.

Refer to the Periodic Maintenance Table to determine the inspection and maintenance items and inspection cycles.

Use only genuine Toyota parts for replacements, and always use the recommended types of lubricants.

PERIODIC REPLACEMENT TABLE

REPLACEMENT PERIOD (Accumulated hours of operation or monthly periods of operation, whichever comes sooner.)	EVERY	6 WEEKS	3	6	12	MONTHS
	EVERY	250	500	1000	2000	HOURS
Engine oil		●	←	←	←	
Engine oil filter		●*	●	←	←	
Coolant (except LLC, LLC is every 2 years)			●	←	←	
Air cleaner element					●	
Fuel filter				●	←	
Torque converter oil				●	←	
Torque converter oil filter				●	←	
Differential gear oil					●	
Hydraulic oil				●	←	
Hydraulic oil return filter		●*		●	←	
Wheel bearing grease					●	
Spark plugs						(Every 3 years or 2500 hours)
Master cylinder, wheel cylinder cap and seals					●	
Brake fluid				●	←	
Power steering hose						(Every 2 years)
Power steering rubber parts						(Every 2 years)
Hydraulic hose						(Every 2 years)
Reserve tank tube						(Every 2 years)
Fuel hose						(Every 2 years)
Torque converter rubber hose						(Every 2 years)
Chain						(Every 3 years)
Hydraulic oil pump seal						(Every 3 years or 6000 hours)
Swing lock cylinder (SAS models)						(Every 10000 hours)

*For new trucks

Note

In case of the hard operating condition, the service interval of 170 hours or 1 month may be recommended.

VII. PLANNED MAINTENANCE

PERIODIC MAINTENANCE TABLE

Periodic maintenance

INSPECTION METHOD

I: Inspect and correct and replace as required. T: Tighten C: Clean L: Lubricate M: Measure and correct and adjust as required.

INSPECTION PERIOD (Accomplish based on operating hours or months, whichever is soonest.)

EVERY	6 WEEKS	3	6	12	MONTHS
EVERY	250	500	1000	2000	HOURS

ENGINE

Basic components

1. Starting condition and unusual noise.....	I	←	←	←	
2. Rotating condition during idling.....	M	←	←	←	
3. Rotating condition during acceleration.....	M	←	←	←	
4. Exhaust condition	I	←	←	←	
5. Air cleaner element.....	C	←	←	←	
6. Valve clearance	M*			M	
7. Compression.....				M	
8. Cylinder head bolts				T	
9. Muffler rubber mount				I	

Blow by gas reduction device

10. Clogging and damage of PCV valve and piping	I	←	←	←	
--	---	---	---	---	--

Governor

11. Maximum no-load stabilized rotation speed	M	←	←	←	
---	---	---	---	---	--

Lubrication system

12. Oil leakage.....	I	←	←	←	
13. Oil level	I	←	←	←	
14. Clogging and fouling of oil filter.....	I	←	←	←	

Fuel system

15. Fuel leakage	I	←	←	←	
16. Carburetor link mechanism operation	I	←	←	←	
17. Fouling and damage of fuel filter element.....	I	←	←	←	
18. Injection timing.....			M	←	
19. Injection nozzle injection pressure and condition ..				M	
20. Draining of sediment.....			I	←	

INSPECTION PERIOD (Accomplish based on operating hours or months, whichever is soonest.)

EVERY	6 WEEKS	3	6	12	MONTHS
EVERY	250	500	1000	2000	HOURS

Cooling system

21. Radiator coolant level and leakage.....	I	←	←	←	
22. Rubber hose deterioration	I	←	←	←	
23. Radiator cap condition	I	←	←	←	
24. Fan belt tension and damage	I	←	←	←	
25. Radiator rubber mount.....				I	

Three-way exhaust emission control system

26. Exhaust gas (carbon monoxide) concentration measurement	M	←	←	←	
27. Exhaust system piping joint loosening and damage				T	
28. Vacuum piping damage	I	←	←	←	
29. Vacuum sensor damage				I	
30. Injector cleaning and damage.....				I	
31. Register damage.....				I	
32. ABCV damage				I	
33. Water temperature sensor damage				I	
34. Oxygen sensor damage.....				I	

PCV system

35. Clogging and damage in PCV valve and piping....	I	←	←	←	
---	---	---	---	---	--

POWER TRANSMISSION SYSTEM

Differential

1. Oil leakage	I	←	←	←	
2. Oil level	I	←	←	←	
3. Loose bolts.....				T	

Torque converter and transmission

4. Oil leakage	I	←	←	←	
5. Oil level	I	←	←	←	
6. Operating mechanism function and looseness	I	←	←	←	
7. Control valve and clutch function	I	←	←	←	
8. Inching valve function	I	←	←	←	
9. Stall test and oil pressure measurement.....			M	←	

VII. PLANNED MAINTENANCE

INSPECTION PERIOD (Accomplish based on operating hours or months, whichever is soonest.)

	EVERY	6 WEEKS	3	6	12	MONTHS
	EVERY	250	500	1000	2000	HOURS

Propeller shaft and axle shaft

10. Loosening of joint.....			I	←	←	
11. Looseness at spline connection.....					I	
12. Looseness at universal joint					I	
13. Twisting and cracks of axle shaft.....					I	

RUNNING EQUIPMENT

Wheels

1. Tire air pressure.....	M	←	←	←		
2. Tire cuts, damage and uneven treads	I	←	←	←		
3. Loose rim and hub nuts	T	←	←	←		
4. Tread depth	M	←	←	←		
5. Metal fragments, stones or other foreign objects in tires	I	←	←	←		
6. Rim, side ring and disc wheel damage	I	←	←	←		
7. Front wheel bearing unusual noise and looseness	I	←	←	←		
8. Rear wheel bearing unusual noise and looseness	I	←	←	←		

Front axle

9. Housing cracks and damage					I	
------------------------------------	--	--	--	--	---	--

Rear axle

10. Beam cracks, damage and deformation					I	
11. Axle beam forward and backward direction looseness	M*				M	

STEERING SYSTEM

Steering wheel

1. Play and looseness.....	I	←	←	←		
2. Operating condition.....	I	←	←	←		

Steering valve

3. Oil leakage.....	I	←	←	←		
4. Mounting looseness.....	T	←	←	←		

Power steering

5. Oil leakage.....	I	←	←	←		
---------------------	---	---	---	---	--	--

INSPECTION PERIOD (Accomplish based on operating hours or months, whichever is soonest.)

	EVERY	6 WEEKS	3	6	12	MONTHS
	EVERY	250	500	1000	2000	HOURS

6. Mounting and linkage looseness.....	I	←	←	←		
7. Power steering hose damage					I	

Knuckle

8. King pin looseness.....	I	←	←	←		
9. Cracking and deformation.....					I	

BRAKING SYSTEM

Brake pedal

1. Play and reserve	M	←	←	←		
2. Braking effect.....	I	←	←	←		

Parking brake

3. Operating force	I	←	←	←		
4. Braking effect.....	I	←	←	←		
5. Linkage and cable looseness and damage.....	I	←	←	←		

Brake pipe and hose

6. Leakage, damage and mounting condition	I	←	←	←		
---	---	---	---	---	--	--

Brake oil

7. Level	I	←	←	←		
----------------	---	---	---	---	--	--

Master cylinder or wheel cylinder

8. Function, wear, damage and mounting looseness.....					I	
---	--	--	--	--	---	--

Brake drum and brake shoe

9. Clearance between drum and lining	M	←	←	←		
10. Shoe sliding portion and lining wear					I	
11. Drum wear and damage					I	
12. Shoe operating condition					I	
13. Anchor pin rusting					I	
14. Return spring wear, etc.....					M	
15. Automatic adjusting function operation					I	

Backing plate

16. Deformation, cracking and damage					I	
17. MOUNTING looseness					T	

VII. PLANNED MAINTENANCE

INSPECTION PERIOD (Accomplish based on operating hours or months, whichever is soonest.)	EVERY	6 WEEKS	3	6	12	MONTHS
	EVERY	250	500	1000	2000	HOURS

LOAD HANDLING SYSTEM

Forks

1. Forks and stopper pin condition.....	I	←	←	←		
2. Left and right forks uniformity	I	←	←	←		
3. Cracks in forks base and welded portion					I*3	

Mast and lift bracket

4. Deformation, damage and cracks in welded portion	I	←	←	←		
5. Mast and lift bracket looseness	I	←	←	←		
6. Mast support bushing wear and damage.....				I		
7. Roller wear, damage and rotating condition	I	←	←	←		
8. Roller pin wear and damage				I		
9. Mast strip wear and damage	I	←	←	←		

Chain and chain wheel

10. Chain tension, deformation and damage	I	←	←	←		
11. Chain lubrication	I	←	←	←		
12. Chain anchor bolt condition	I	←	←	←		
13. Chain wheel wear, damage and rotating condition	I	←	←	←		

Various attachments (option)

14. Abnormalities and mounting condition	I	←	←	←		
--	---	---	---	---	--	--

HYDRAULIC SYSTEM

Cylinder

1. Cylinder mounting looseness and damage.....	T	←	←	←		
2. Rod and rod screw and rod end deformation and damage.....	I	←	←	←		
3. Cylinder operation.....	I	←	←	←		
4. Natural drop and natural forward tilt	M	←	←	←		
5. Oil leakage and damage	I	←	←	←		
6. Pin and cylinder shaft support wear and damage.....	I	←	←	←		
7. Lifting speed	M	←	←	←		
8. Uneven movement.....	I	←	←	←		

INSPECTION PERIOD (Accomplish based on operating hours or months, whichever is soonest.)	EVERY	6 WEEKS	3	6	12	MONTHS
	EVERY	250	500	1000	2000	HOURS

Oil pump

9. Oil leakage and unusual noise.....	I	←	←	←		
---------------------------------------	---	---	---	---	--	--

Hydraulic oil tank

10. Oil level and contamination	I	←	←	←		
11. Tank and oil strainer			C	←		
12. Oil leakage	I	←	←	←		

Control lever

13. Linkage looseness	I	←	←	←		
14. Operation	I	←	←	←		

Oil control valve

15. Oil leakage.....	I	←	←	←		
16. Relief pressure measurement.....					M	
17. Relief valve and tilt lock valve function	I	←	←	←		

Oil pressure piping

18. Oil leakage	I	←	←	←		
19. Deformation and damage	I	←	←	←		
20. Linkage looseness	T	←	←	←		

ELECTRICAL SYSTEM

Ignition system

1. Distributor cap cracking	I	←	←	←		
2. Spark plug burning and gap	I	←	←	←		
3. Distributor side terminal burning	I	←	←	←		
4. Distributor cap center piece wear and damage	I	←	←	←		
5. Plug cord internal disconnection						I
6. Ignition timing.....			M	←		

Starter

7. Pinion gear meshing	I	←	←	←		
------------------------------	---	---	---	---	--	--

Charger

8. Charging effect.....	I	←	←	←		
-------------------------	---	---	---	---	--	--

Battery

9. Battery electrolyte level.....	I	←	←	←		
10. Specific gravity			M	←		

VII. PLANNED MAINTENANCE

INSPECTION PERIOD (Accomplish based on operating hours or months, whichever is soonest.)	EVERY 6 WEEKS 3 6 12 MONTHS					
	EVERY	250	500	1000	2000	HOURS
	EVERY	250	500	1000	2000	HOURS

Electrical wiring

11. Wiring harness damage.....	I	←	←	←	←	
12. Fuses	I	←	←	←	←	

Preheater

13. Glow plug heat coil breakage				I	←	
--	--	--	--	---	---	--

Engine stopping system

14. Diesel engine key stop device function.....	I	←	←	←	←	
---	---	---	---	---	---	--

SAFETY DEVICES, ETC.

Over head guard

1. Welded portion cracking	I	←	←	←	←	
2. Deformation and damage	I	←	←	←	←	

Back rest

3. Mounting looseness.....	T	←	←	←	←	
4. Deformation, cracking and damage	I	←	←	←	←	

Lighting system

5. Operation and mounting condition.....	I	←	←	←	←	
--	---	---	---	---	---	--

Horn

6. Operation and mounting condition.....	I	←	←	←	←	
--	---	---	---	---	---	--

Direction indicators (option)

7. Operation and mounting condition.....	I	←	←	←	←	
--	---	---	---	---	---	--

Instrument

8. Operation	I	←	←	←	←	
--------------------	---	---	---	---	---	--

Back-up buzzer (option)

9. Operation and mounting condition.....	I	←	←	←	←	
--	---	---	---	---	---	--

SAS

10. Operation	I	←	←	←	←	
11. Looseness at and/or damage to sensor fittings	I	←	←	←	←	
12. Damage to, deformation of and/or oil leakage at functional parts and loosening mounting	I	←	←	←	←	
13. Looseness at and/or damage to wire harnesses ...	I	←	←	←	←	
14. Performance of lock cylinder and/or accumulator					I	

INSPECTION PERIOD (Accomplish based on operating hours or months, whichever is soonest.)	EVERY 6 WEEKS 3 6 12 MONTHS					
	EVERY	250	500	1000	2000	HOURS
	EVERY	250	500	1000	2000	HOURS

15. Rust and/or corrosion in load-handing sensor					I	
--	--	--	--	--	---	--

OPSS

16. Function	I	←	←	←	←	
--------------------	---	---	---	---	---	--

Seat

17. Mounting looseness and damage	I	←	←	←	←	
18. Operating condition of seat switch	I	←	←	←	←	

Seat belt

19. Mounting looseness	I	←	←	←	←	
20. Webbing damage.....	I	←	←	←	←	
21. Plate damage.....	I	←	←	←	←	
22. Buckle damage	I	←	←	←	←	

Body

23. Frame, cross member, etc. damage and cracking.....					I	
24. Bolt looseness.....					T	

Rear-view mirror

25. Dirt, damage	I	←	←	←	←	
26. Rear reflection status	I	←	←	←	←	

Others

27. Lubrication	L	←	←	←	←	
-----------------------	---	---	---	---	---	--

* For new truck

*1 Soap

*2 Leakage detector

*3 Fissure and crack detector

*4 Check voltage of test terminal from controller

Note

In case of the hard operating condition, the service interval of 170 hours or 1 month may be recommended.

VII. PLANNED MAINTENANCE

SERVICE DATA

Adjustment valve table

ENGINE

Engine mode		4Y	1DZ-II
Item			
Fan belt tension [10 kg (22 lb) pressure applied]	mm (in)	8 - 13 (0.31 - 0.51)	8 - 16 (0.31 - 0.63)
Spark plug gap	mm (in)	0.7 - 0.8 (0.028 - 0.031)	—
Spark plug type		WREXR-U	—
Ignition or fuel injection timing (BTDC)	deg	7/Idling speed	O (Static)
Ignition or fuel injection sequence		1-3-4-2	←
Valve clearance (Hot engine)	Intake	O (Self-adjusting)	0.18 - 0.22 (0.007 - 0.009)
	Exhaust	O (Self-adjusting)	0.33 - 0.37 (0.013 - 0.015)
Engine compression kg/cm ² /rpm (psi/rpm)	Standard valve	12.5/250 (178/250)	29/260 (412/260)
	Limit	9.0/250 (128/250)	20/260 (284/260)
No load maximum speed	P1.5-1.75t	2600±50	2600±50
	C1.5-1.75t	2450±50	—
	2.0-2.5t	2600±50	2800±50
	3.0t	2800±50	2800±50
Idling speed	1.5-1.75t	750±30	750±25
	2.0-2.5t	750±30	750±25
	3.0t	750±30	750±25

BATTERY

Battery electrolyte specific gravity [at 20 (68°F)]	1.280
---	-------

STEERING

Steering wheel play (at engine idle speed)	mm (in)	25 - 50 (1.0 - 2.0)
--	---------	---------------------

BRAKE

Model		1 - 2 ton series	3 ton series
Item			
Inching and operating pedal	mm (in)	35 - 40 (0.48 - 0.57)	25 - 30 (0.98 - 1.18)

OIL CONTROL VALVE

Model		1 - 2 ton series	3 ton series
Item			
Set pressure	kg/cm ² (psi)	182 (2580)	191 (2710)
	Lift	120 (1710)	150 (2130)

WHEEL

Model		8FGU15, 18 8FDU15, 18	8FGU20, 25 8FDU20, 25	8FGU30, 32 8FDU30, 32
Item				
Tire air pressure kg/cm ² (lb/in ²)	Front	6.50-10-10PR	8.0 (115)	—
		7.00-12-12PR	—	8.5 (120)
		28 × 9-15-12PR	—	8.5 (120)
	Rear	5.00-8-8PR	8.0 (115)	—
		6.00-9-10PR	—	7.0 (100)
		6.50-10-10PR	—	7.75 (110)

Model		8FGU15, 18 8FDU15, 18	8FGU20, 25 8FDU20, 25	8FGU30, 32 8FDU30, 32	8FGCU15, 18 8FGCSU20	8FGCU20, 25 8FGCU30, 32
Item						
Hub nut tightening torque kg-m (ft-lb)	Font	12-20 (87-145)	18-40 (131-289)	30-60 (217-434)	12-20 (87-144)	30-60 (217-434)
	Rear (Divided)	9-16 (65-116)	12-20 (87-145)	←	—	—
	Rear (Side ring)	9-16 (65-116)	18-40 (131-289)	12-20 (87-145)	9-16 (65-116)	12-20 (87-145)
Divided rim set bolt tightening torque kg-m (ft-lb)		3-4 (21-32)	5-7 (36-50)	←	—	—

TIRE AVAILABILITY

Model	Tire arrangement	Tire size	
		Standard tread	Wide tire & Wide tread
8FGCU15, 18 8FGCSU20	Front	18 × 6 × 12-1/8	18 × 8 × 12-1/8
	Rear	14 × 4-1/2 × 8	—
8FGCU20, 25	Front	21 × 7 × 15	—
	Rear	16 × 5 × 10-1/2	—
8FGCU30 8FGCU32	Front	21 × 8 × 15	—
	Rear	16 × 6 × 10-1/2	—

Lubricant capacities and types

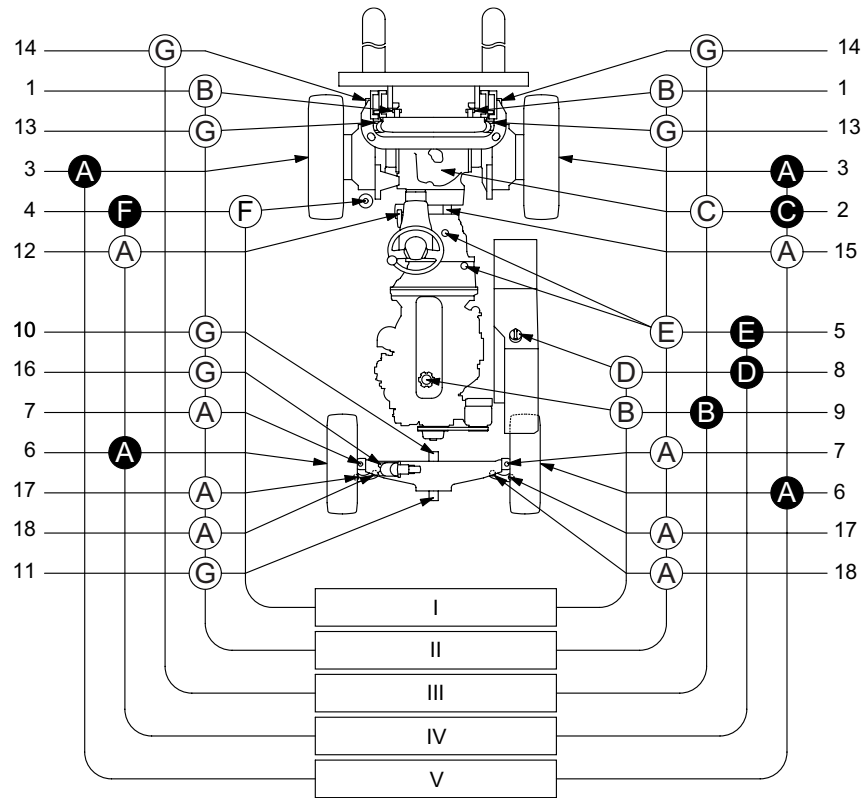
Item				1 ton series	2 ton series	3 ton series	Type
English oil	ℓ (US. gal)	Gasoline	4Y	4.0 (1.06)	←	←	API SL
		Diesel	1DZ-II	7.9 (2.09)	←	←	API CE, CF
Torque converter	ℓ (US. gal)	Pneumatic	1 speed	5.0 (1.32)	←	←	Autofluid DII Kai Exxon 2043K
			2 stator	9.0 (2.38)	←	←	
		Cushion	8.0 (2.11)	9.0 (2.38)	←		
Differential gear	ℓ (US. gal)	Pneumatic		5.8 (1.53)	6.1 (1.61)	8.2 (2.16)	API GL-4, GL-5
		Cushion			6.1 (1.61)	←	Hypoid gear oil SAE85W-90
Fuel tank	ℓ (US. gal)	Pneumatic		45 (11.9)	60 (15.9)	←	
		Cushion		27 (7.1)	45 (11.9)	←	
Wheel bearings, chassis, tilt steering and mast and grease fittings				Appropriate amount			MP Grease
Brake lines	ℓ (US. gal)			0.2 (0.05)	←	←	SAE J-1703, DOT-3
Engine cooling system (excluding reserve tank)	ℓ (US. gal)	Pneumatic	4Y	8.1 (2.14)	9.3 (2.46)	←	L.L.C.*
			1DZ-II	6.8 (1.80)	7.9 (2.09)	←	
		Cushion	4Y	8.3 (2.19)	8.1 (2.14)	←	
Radiator reserve tank (at FULL mark level)	ℓ (US. gal)	Pneumatic		0.48 (0.124)	←	←	
		Cushion		0.6 (0.16)	0.47 (0.124)	←	
Hydraulic oil	ℓ (US. gal)	Pneumatic		30 (7.9)	33 (8.7)	34 (9.0)	Super Hydro 32AT
		Cushion		30 (7.9)	←	←	

*L.L.C. = Long Life Coolant
(Appropriately diluted with fresh water)

VII. PLANNED MAINTENANCE

LUBRICATION CHART

8FGU/8FDU15-32, 8FGCU20-32



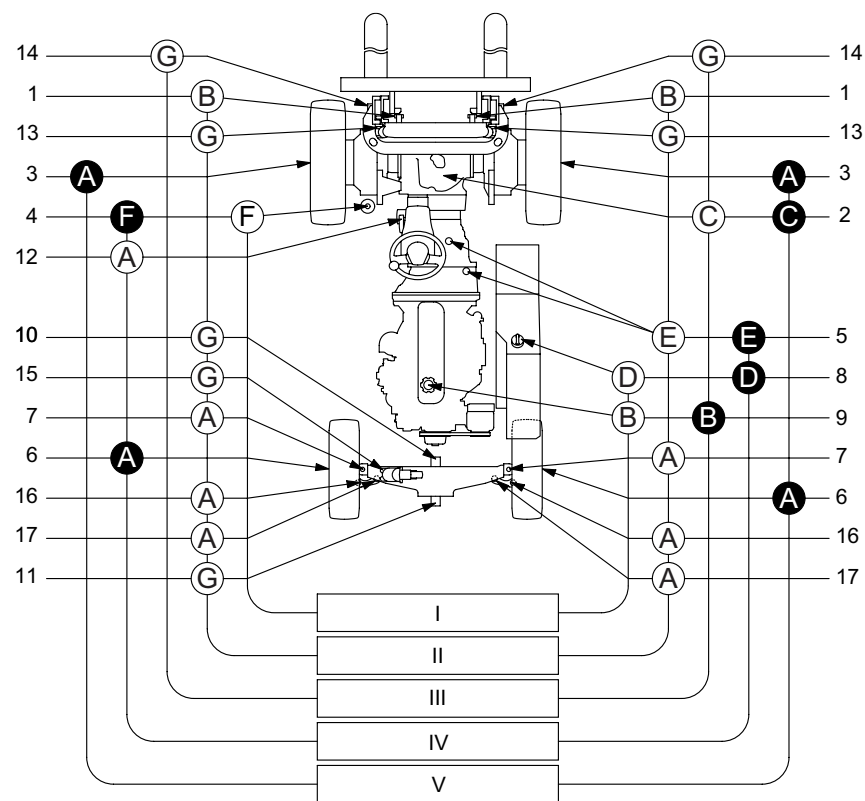
1. Chain
2. Differential gear
3. Front wheel bearing
4. Brake and clutch master cylinder
5. Transmission case
6. Rear wheel bearing
7. Steering knuckle king pin
8. Oil tank
9. Engine crank case
10. Rear axle beam front pin
11. Rear axle beam rear pin
12. Tilt steering locking mechanism
13. Mast support bushing
14. Tire cylinder front pin
15. Propeller shaft
16. Swing lock cylinder
17. Tie rod end pin
18. Rear axle cylinder end pin

- I. Inspect every 8 hours (daily)
- II. Inspect every 40 hours (weekly)
- III. Inspect every 250 hours (6 weeks)
- IV. Inspect every 1000 hours (6 monthly)
- V. Inspect every 2000 hours (annually)
- Inspect and service
- Replace
- A MP grease
- B Engine oil
- C Hypoid gear oil
- D Hydraulic oil
- E Hypoid gear oil
- F Brake fluid
- G Molybdenum disulfide grease

Note

In case of the hard operating condition, the service interval of 170 hours or 1 month may be recommended.

8FGCU15, 18, 8FGCSU20



1. Chain
2. Differential gear
3. Front wheel bearing
4. Brake and clutch master cylinder
5. Transmission case
6. Rear wheel bearing
7. Steering knuckle king pin
8. Oil tank
9. Engine crank case
10. Rear axle beam front pin
11. Rear axle beam rear pin
12. Tilt steering locking mechanism
13. Mast support bushing
14. Tire cylinder front pin
15. Swing lock cylinder
16. Tie rod end pin
17. Rear axle cylinder end pin

- I. Inspect every 8 hours (daily)
- II. Inspect every 40 hours (weekly)
- III. Inspect every 250 hours (6 weeks)
- IV. Inspect every 1000 hours (6 monthly)
- V. Inspect every 2000 hours (annually)

○ Inspect and service

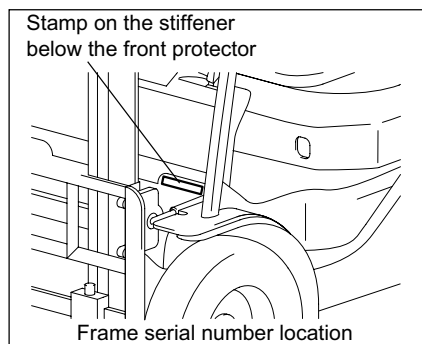
● Replace

- A MP grease
 B Engine oil
 C Hypoid gear oil
 D Hydraulic oil
 E Hypoid gear oil
 F Brake fluid
 G Molybdenum disulfide grease

Note

In case of the hard operating condition, the service interval of 170 hours or 1 month may be recommended.

FRAME SERIAL NUMBER



Frame serial number location

The frame serial number is stamped on the front protector. Refer to the frame serial number when making inquiries about your truck.

WARNING

- *Make sure these important instructions are available for the operator.
Keep this manual in the pocket on the seat back.*
- *Do not start the truck unless you understand these instructions.*



006120-DA

TOYOTA Material Handling Company

A Division of TOYOTA INDUSTRIES CORPORATION

Publication No. AU027-00000 PRINTED IN USA